Concept Note

Curriculum of UG Program

2026

Department of Civil and Infrastructure Engineering

IIT Jodhpur

1. Introduction

The rapidly urbanizing society and increasing quality of life demand reliable and intelligent infrastructure systems which are capable to cater the society at different scales – from an individual to the community level. Therefore, the civil and infrastructure industry has undergone profound changes and is constantly evolving. However, the new-age designs and innovations in the civil and infrastructure industry can only be driven by a group of engineering graduates having multi-disciplianary training and a sound understanding of emerging technologies. Through this undergraduate program in Civil and Infrastructure Engineering, IIT Jodhpur is making a stride to reimagine the course structure that incorporates and integrates the elements of conventional civil engineering with advanced transformative technologies such as artificial intelligence (AI), geographic information system (GIS), intelligent transport system (ITS), automated management and information system. Additionally, a major thrust is also planned on the design, implementation, and maintanace of large scale integrated infrastructure systems across different domains.

2. Objectives of the Program

1. To provide an expanded but holistic understanding of different civil and infrastructure systems and an in-depth understanding of the differences, similarities, and relations between different scales and components of it.
2. To produce graduates with an ability to integrate conventional methods with emerging technologies to design intelligent civil and infrastructure systems from a multi-disciplinary perspective.
3. To provide exposure to techniques such as machine learning, big data etc. in the design and maintenance of different civil and infrastructure systems.

3. Expected Graduate Attributes

1. A strong foundation in the analytical, experimental, and computational methods to study various aspects of civil and infrastructure systems.
2. A thorough understanding of green and sustainable materials, practices, and principles to design resilient infrastructure systems.
3. An ability to understand real-world problems related to air and water pollution, solid waste management, ecological and environmental degradation, and their monitoring and remediation.
4. Ability to adopt safe methodologies coupled with advanced technologies in the design and construction of civil and infrastructure systems.
5. Ability to monitor and evaluate health of old structures and renovating to bring sustainability.

6. An understanding of the use of advanced techniques such as artificial intelligence (AI), geographic information system (GIS), intelligent transport system (ITS) etc. in designing, monitoring, surveillance and security of infrastructures.
7. An innate ability to take multidisciplinary approach to problem solving and willingness to go beyond conventional paths.
8. Ability to engage in ethical and responsible practices while pursuing professional growth.

4. Learning Outcomes

1. Strong understanding of structural engineering and its application in reinforced concrete and steel structure design.
2. In-depth knowledge of geotechnical engineering for ground improvement and foundation engineering.
3. Knowledge of pavement design, traffic planning, and highway design for urban transport management.
4. Knowledge of critical experiments and advanced mathematical models for environmental engineering application.
5. Knowledge of GIS and remote sensing for application in agriculture, environment, geo-resource exploitation, and constructing digital twin of built environment.
6. Knowledge of smart and sustainable material, and best practices for construction and infrastructure development.
7. Strong foundation in computational methods and artificial intelligence (AI) for application in various civil and infrastructure problems.
8. Knowledge of town planning, logistics and layout engineering

5. New Skillsets Targeted

The following advanced and new skillsets have been targeted in the UG programme of Civil and Infrastructure Engineering at IIT Jodhpur:

1. Capability to design and develop smart infrastructure systems integrating cutting-edge intelligent systems, and artificial intelligence (AI).
2. A through understanding of geological and geotechnical engineering as applicable in infrastructure development, natural disaster risk reduction and resilience, and geo-energy extraction.
3. Knowledge of best practices, methods, and materials for sustainable infrastructure planning, construction, and management.
4. Preliminary knowledge of monitoring, evaluation, restoration, and renovation of existing and old infrastructures and maintenance.

6. **Topic Clouds and Mapping of Topic Clouds with Courses [Program Comlusory (PC) and Program Elective (PE)]**

Area	Topics	Courses
Geotechnical Engineering	Soil composition and properties; soil classification; soil compaction; permeability and hydraulic conductivity; seepage and fluid flow; stress principles; capillarity; stress distribution theories; soil compressibility; consolidation testing and theory; shear strength and failure criteria	Soil Mechanics (PC)
	Soil classification; index and engineering properties of soils; compaction and consolidation of soils; shear strength of soils; shallow and deep foundations; earth retaining structures	Geotechnical Engineering (PC)
	In-situ densification methods;preloading and dewateringtechniques; mechanical and chemicalstabilization of soils; grouting and Geosynthetics	Ground Improvement Techniques (PE)
	Vibration theory; wave propagation; cyclic behavior of soils; seismic ground motion; dynamic soil properties; geophysical testing; seismic soil liquefaction; seismic microzonation; seismic slope stability; seismic response of retaining walls	Introduction to Soil Dynamics (PE)
	Stresses and strains in soils; stress path for total and effective stress; critical state soil mechanics; load transfer mechanism of shallow and pile foundations; structural	Advanced Geotechnical Engineering (PE)

	health monitoring of geotechnical structures; introduction to computational geomechanics	
	Structural geology and discontinuities; site investigation and data collection; rock mass strength criteria; discontinuity shear strength; plane failure analysis; wedge failure analysis; circular failure and limit equilibrium; toppling failure kinematics; slope stabilization and hazard rating; slope movement and monitoring	Stability of Slopes (PE)
Water Resources Engineering	Fluid fundamentals and dimensions; fluid statics and pressure; buoyancy, stability and rigid bodies; integral analysis and conservation; kinematics of fluid flow; differential analysis and governing equations; internal incompressible viscous flow; pipe head losses and flow measurement; external flow and boundary layers; pressure gradients, drag and dimensional analysis	Hydraulic Engineering (PC)
	Introduction to hydrology and climate change; precipitation forms and measurement; precipitation analysis; evaporation and evapotranspiration processes; interception, depression storage and infiltration indices; runoff characteristics and yield estimation; hydrograph analysis and unit hydrographs; hydrologic extremes: floods and droughts; flood frequency analysis and routing methods; groundwater hydraulics and aquifer properties	Engineering Hydrology (PC)
	Urban, industrial, and agricultural water demand; surface water resources and	Water Resources Engineering (PC)

	reservoir capacity; groundwater resource occurrence and well hydraulics; dams and spillway design considerations; reservoir storage, sedimentation, and yield; water intake structures and pipe networks; canal system alignment and design theories; integrated watershed management and basin planning; soil erosion mechanics and conservation measures; rainwater harvesting	
	Basic flow concepts; energy and momentum principles; uniform flows; gradually varied flows; rapidly varied flows; unsteady flows; hydraulics of mobile bed channel	Flow in Open Channels (PE)
	Hydropower resource and plant types; tidal barrage and tidal lagoon power systems; hydropower turbines; hydrological analysis and site power assessment; geotechnical and geophysical site investigations; embankment dam design and seismic stability; concrete face rockfill dam construction; concrete gravity dam loading and stability; spillway design and reservoir management; dam foundation preparation and cutoff	Hydropower Engineering (PE)
	Weir and barrage subsurface flow theories; permeable foundation design and exit gradients; gravity and embankment dam stability analysis; reservoir capacity and sediment flushing operations; spillway hydraulic design and control gates; energy dissipation and stilling basin engineering; river morphology and meandering behavior; river training works and control structures; impulse and reaction turbine hydraulics; hydraulic pumps	Hydraulic Structures and Machines (PE)

Construction and Infrastructure	Atomic structure of materials; crystal structure and defects; stress–strain behaviour; elasticity and plasticity; fracture mechanics and toughness; viscoelastic behaviour of materials; surface energy and wetting; thermal and electrical properties; cement; supplementary cementitious materials; fresh and hardened concrete properties; concrete batching and mix design; load bearing and framed structures; shallow and deep foundation systems; formwork, shoring and underpinning; durability and variability of construction materials.	Building Materials and Construction Technology (PC)
	Introduction to estimation and quantity; structural item measurement and bill of quantities (BOQ); rate analysis, costing, and bidding processes; engineering economics and financial alternatives comparison; project resource management, quality, and safety; WBS, scheduling networks (CPM/PERT); project optimization, crashing, and control practices	Quantity Estimation, Costing and Construction Management (PE)
Environmental Engineering	Air pollutant sources, health impacts, and regulations; atmospheric meteorology, stability, and plume dispersion; industrial air pollution control equipment and design; indoor air pollution sources and mitigation technologies; integrated solid waste management legislation and generation; waste collection, transport, and material recovery facilities; biological treatment, thermal conversion, and landfilling	Air Pollution and Solid Waste Management (PC)
	Water demand and quality parameters; physicochemical water treatment design; advanced treatment and disinfection processes; water supply pipeline and network analysis; wastewater	Water and Wastewater Engineering (PC)

	characterization and parameters; suspended and attached growth wastewater treatment; advanced wastewater systems and sludge disposal	
	Wastewater microbiology, metabolism, and reaction kinetics; reactor hydraulics, mass transfer, and temperature effects; primary treatment and flow equalization tank design; activated sludge process operational controls and nutrient removal; attached growth biofilm processes and phosphorus removal; advanced and unit treatment technologies; sludge production, stabilization, and dewatering operations	Wastewater Engineering (PE)
	Mass balance; Reaction kinetics; Micro and macro transport processes; modelling of aquatic systems; Oxygen model; Temperature model; Gaussian plume model and modifications; Dispersion models; Acid rain modeling, dissolved oxygen modelling and add reactor models	Mathematical Modelling in Environmental System (PE)
Transportation Engineering	Introduction to transportation engineering, Geometric design, Pavement materials, Pavement design, Pavement Construction, Pavement Evaluation and Maintenance	Transportation Engineering - I (PC)
	Transportation systems, traffic stream characteristics and speed-flow-density relationships; traffic studies, operations, and intersections control design; capacity and multi-modal facilities; intelligent transportation systems; urban transport planning, survey methods, and transit-oriented development; travel demand estimation and route assignment models; traffic impact assessment and activity-based transport modelling	Transportation Engineering - II (PC)

	Highway cross section elements and geometric design, Horizontal and vertical alignment, Intersection design, Traffic signs and road markings – guidelines and principles, Pedestrian facilities, cycle track, Bus bays, Design layout for On-street and Off-street parking	Highway Geometric Design (PE)
	Urban transportation planning process and concepts, Travel demand estimation, Mode split and route split analysis, Landuse-Transportation models	Urban Transportation Planning (PE)
	Road, user, and vehicle characteristics of traffic system; Traffic control devices; control hierarchy; Intelligent Transport System (ITS); ITS user services; framework for planning, defining, and integrating ITS	Traffic System and ITS (PE)
	Introduction to railways, Components of railway tracks, Design elements, Points, Crossings, Signals, Introduction to air transport, Design of airside facilities, Design of landside facilities	Railway and Airport Engineering (PE)
	Stress and strain, Mechanical behaviour, Static analysis, Elastic bending, Elastic torsion Strain energy, Elastic buckling	Mechanics of Solids (PC)
	Constitutive relations and compatibility conditions; axial force, shear force and bending moment; moment area method; conjugate beam method; principle of virtual work; strain energy method; Maxwell-Betti law; force method; energy method; three moment equation; influence	Structural Analysis (PC)

Structural Engineering	lines; cables, suspension bridges and arches; settlement and temperature effects.	
	Indeterminate structures; force method of analysis; displacement method; stress and strain tensors; plastic analysis; basics of FEM; stability of structure; introduction to structural dynamics.	Advanced Structural Analysis (PC)
	Working stress and limit state method of design of RCC structures; design of beam - singly reinforced, doubly reinforced beam; detailing of reinforcement; serviceability criteria; design of slab - one way and two way slab; design of column - axial loading, uniaxial bending, biaxial bending, slender column; design of foundation - isolated and combined footing.	Reinforced Concrete Design (PC)
	Structural steel sections; design concepts - working stress design, limit state design; design of connections; welded connections; design of tension and compression members, concept of stability and buckling; design of flexural members; design against shear.	Steel Structure Design (PC)
	Mass-spring-damper system; equation of motion; Hamilton's principle; single and multiple degree of freedom system; damped and undamped systems; natural frequencies and modes; damping in structures; basics of earthquake response and vibration control.	Dynamics of Structure (PE)

	Governing equation and its solution approximations; calculus of variations; Lagrangian and serendipity elements; analysis of framed structures; axisymmetric problems; 3D stress analysis; analysis of plate and shell; numerical integration and convergence; application of FEM in dynamics.	Finite Element Method (PE)
	Types of prestressing; pre- and post-tensioning systems; losses in prestress; anchorage slip; analysis and design of members for axial, flexure, shear and torsion loading; deflection, camber and crack width; transmission of prestress - pre- and post-tensioned members; analysis and tendon profile of cantilever and continuous beams; design of composite sections, slabs, compression members etc.	Prestressed Concrete Structure (PE)
	Mix design of concrete; P-M, M-phi relationships; strut-and-tie method; design of deep beam and corbel; design of shear walls; compression field theory; design against torsion; stability design of steel structures; torsional buckling; design of beam-columns; fatigue resistant design; fire rating.	Advanced Structural Design (PE)
	NDT philosophy, concrete defects, and quality control planning; visual inspection procedures and destructive core testing; surface hardness, penetration resistance, and pullout methods; stress wave methods, flaw detection, and tomography; electromagnetic cover measurement and mass transport testing; electrochemical corrosion evaluation and concrete petrography; advanced thermographic, nuclear, and tomographic imaging	Non Destructive Evaluation of Structures (PE)

Smart and Sustainable Infrastructure Systems	Regression and Supervised Classifers; Linear Decision boundaries; Unsupervised Learning; Time Series; Artificial Neural Network	Introduction to Machine Learning (IE)
	Review of artificial neural networks - supervised and unsupervised learning networks; genetic programming; applications in infrastructure engineering – prediction, structural control, damage assessment, ground settlement, uncertainty quantification, surrogate modelling etc.	AI/ML in Civil Engineerinag (PC)
	Design; assessment; sustainable site and landscape; energy and carbon footprint; Built environment hydrological cycle; Green materials; Indoor environmental quality; Green building economics	Green Building and Sustainable Materials (PE)
Common	Photogrammetry; Multispectral remote Sensing; Thermal infrared Remote Sensing; Microwave Remote Sesning; LIDAR; GPS; GIS; Application of GIS, Use of Drones	Geoinformatics (PC)
	Statistics and probability; probability distribution and their applications; concepts of sampling and monitoring; frequency analysis; concept of uncertainty; hypothesis testing and non-parametric test; uni- and multi-variate data analysis; time series analysis; copula	Statistics and Probability in Civil Engineering (PE)

7. Course Categories and Credit Distribution of the B.Tech. Programme

S.N.	Course Type	Course Category	Regular B. Tech.	
			Credit	Total
1	Institute Core (I)	Engineering (IE)	16	**39**
		Science (IS)	17	
		Humanities (IH)	6	
2	Programme Core (P)	Programme Compulsory (PC)	67	**91**
		Programme Electives (PE)	18	
		B.Tech. Project (PP)	6	
3	Open (O)	Open Electives (OE)	12	**12**
Total Graded Credits				**142**
4	Non-Graded (N) / Essential Audit	Humanities (NH)	3	**12**
		Engineering (NE)	3	
		Industry-Academia Summer Internship (ND)	4	
		Design/Practical Experience (ND)	2	
Total Graded and Non-Graded Credits				**154**

8. List of Programme Compulsory (PC) Courses

S. No	Course Names	L-T-P	Contact Hours	Credits
1	Hydraulics Engineering	3-1-0	4	4
2	Mechanics of Solids	3-1-0	4	4
3	Building Materials and Construction Technology	3-0-0	3	3
4	Geoinformatics	3-0-0	3	3
5	Transportation Engineering - I	3-0-0	3	3
6	Structural Analysis	3-0-0	3	3
7	Soil Mechanics	3-1-0	4	4
8	Engineering Hydrology	3-0-0	3	3
9	Air Pollution and Solid Waste Management	3-0-0	3	3
10	Water and Wastewater Engineering	3-0-0	3	3
11	Transportation Engineering – II	3-0-0	3	3
12	Reinforced Concrete Design	3-0-0	3	3
13	Geotechnical Engineering	3-1-0	4	4
14	Steel Structure Design	3-1-0	4	4
15	AI/ML in Civil Engineering	3-0-0	3	3
16	Water Resources Engineering	3-0-0	3	3
17	Advacned Structural Analysis	3-0-0	3	3
18	Civil Engineering Laboratory – I	0-0-4	4	2
19	Civil Engineering Laboratory – II	0-0-6	6	3
20	Civil Engineering Laboratory – III	0-0-6	6	3
21	Civil Engineering Laboratory – IV	0-0-6	6	3
Total				**67**

9. Area-wise Programme Elective (PE) Courses

S. No.	Focus Areas	Course Names	L-T-P	Credits
1	Geotechnical Engineering	Ground Improvement Techniques	3-0-0	3
2		Introduction to Soil Dynamics	3-0-0	3
3		Advanced Geotechnical Engineering	3-0-0	3
4		Stability of Slopes	3-0-0	3
5	Environmental Engineering	Mathematical Modelling in Environmental System	3-0-0	3
6		Wastewater Engineering	3-0-0	3
7	Transportation Engineering	Highway Geometric Design	3-0-0	3
8		Traffic System and ITS	3-0-0	3
9		Urban Transportation Planning	3-0-0	3
10		Railway and Airport Engineering	3-0-0	3
11	Construction and Infrastructure	Quantity Estimation, Costing and Construction Management	3-0-0	3
12	Structural Engineering	Advanced Structural Design	3-0-0	3
13		Dynamics of Structure	3-0-0	3
14		Finite Element Method	3-0-0	3
15		Prestressed Concrete Structures	3-0-0	3
16		Non Destructive Evaluation of Structures	3-0-0	3
17	Water Resources Engineering	Hydraulic Structures and Machines	3-0-0	3
18		Flow in Open Channels	3-0-0	3
19		Hydropower Engineering	3-0-0	3
20	Smart and Sustainable Infrastructure Systems	Green Building and Sustainable Materials	3-0-0	3
21	Common	Statistics and Probability in Civil Engineering	3-0-0	3

10. Semerster-wsie Structure of B.Tech. in Civil and Infrastructure Engineering (Regular)

Cat	Course	LTP	CH	NC	GC	Cat	Course	LTP	CH	NC	GC
	I Semester						**II Semester**				
IE	Engineering Mechanics	3-0-0	3	-	3	IE	Introduction to Machine Learning	3-0-2	5	-	4
IE	Introduction to Computer Science	3-0-2	5	-	4	IS	Chemistry	3-0-0	3	-	3
IS	Physics	3-0-0	3	-	3	IS	Introduction to Bioengineering	2-0-2	4	-	3
IH	HSS I	3-0-0	3		3	IS	Chemistry Lab	0-0-2	2	-	1
IS	Physics Lab	0-0-2	2	-	1						
IS	Mathematics I	2-1-0	3	-	3	IS	Mathematics II	2-1-0	3	-	3
IE	Engineering Drawing	0-0-2	2	-	1	IE	Engineering Workshop	0-0-2	1	-	1
NE	Engineering Design I	0-0-2	2	1	-	NE	Engineering Design II	0-0-2	2	1	-
NH	Communication Skill	0-0-2	2	1	-	IH	HSS II	3-0-0	3	-	3
NH	Social Connect and responsibilities	0-0-1	1	0.5	-	NE	Introduction to Profession	0-0-2	2	1	-
NH	Performing Arts / Sports	0-0-1	1	0.5	-						-
	Total	14-1-12	27	3	**18**		**Total**	13-1-12	26	2	**18**
	III Semester						**IV Semester**				
PC	Hydraulics Engineering	3-1-0	4	-	4	PC	Transportation Engineering – I	3-0-0	3	-	3
PC	Mechanics of Solids	3-1-0	4	-	4	PC	Structural Analysis	3-0-0	3	-	3
PC	Building Materials and Construction Technology	3-0-0	3	-	3	PC	Soil Mechanics	3-1-0	4	-	4
PC	Geoinformatics	3-0-0	3	-	3	PC	Engineering Hydrology	3-0-0	3	-	3
PC	Civil Engineering Laboratory – I	0-0-4	4	-	2	PC	Air Pollution and Solid Waste Management	3-0-0	3	-	3
IE	Environment and Sustainability	3-0-0	3	-	3	PC	Civil Engineering Laboratory – II	0-0-6	6	-	3
	Total	15-2-4	21	-	**19**		**Total**	15-1-6	22	**-**	**19**
Industry-Academia Summer Internship (min. 45 days, outside IIT Jodhpur)											
	V Semester						**VI Semester**				
PC	Water and Wastewater Engineering	3-0-0	3	-	3	PC	Steel Structure Design	3-1-0	4	-	4
PC	Transportation Engineering – II	3-0-0	3	-	3	PC	AI/ML in Civil Engineering	3-0-0	3	-	3

PC	Reinforced Concrete Design	3-0-0	3	-	3	PC	Water Resources Engineering	3-0-0	3	-	3
PC	Geotechnical Engineering	3-1-0	4	-	4	PC	Advanced Structural Analysis	3-0-0	3	-	3
PC	Civil Engineering Laboratory – III	0-0-6	6	-	3	PC	Civil Engineering Laboratory – IV	0-0-6	6	-	3
PE	Programme Elective	3-0-0	3	-	3	PE	Programme Elective	3-0-0	3	-	3
NH	Professional Ethics I	1-0-0	1	1	-						
	Total	**16-1-6**	**23**	**1**	**19**		**Total**	**15-1-6**	**22**	**-**	**19**
Industry-Academia Summer Internship (min. 45 days, outside IIT Jodhpur)											
	VII Semester						**VIII Semester**				
PP	B.Tech. Project 1	0-0-3	-	-	3	PP	B.Tech. Project 2	0-0-3	-	-	3
PE	Programme Electives	2 × (3-0-0)	6	-	6	PE	Programme Electives	2 × (3-0-0)	6	-	6
OE	Open Electives	2 × (3-0-0)	6	-	6	OE	Open Electives	2 × (3-0-0)	6	-	6
	Total	12-0-3	-	-	**15**		**Total**	12-0-3	-	**-**	**15**
										Total of Graded Credits	**-**	**142**
										Non-Graded Credits	**6**	**-**
										Industry-Academia Summer Internship	**4**	**-**
										Non-Graded Design Credits	**2**	**-**
										Grand Total		**154**

11. List of Core Courses for B.Tech. with Minor in Civil Engineering

S. No.	Course Name	L-T-P	Credits
1	Building Materials and Construction Technology	3-0-0	3
2	Transportation Engineering - I	3-0-0	3
3	Structural Analysis	3-0-0	3
4	Water Resources Engineering	3-0-0	3
5	Water and Wastewater Engineering	3-0-0	3
6	Geotechnical Engineering	3-1-0	4
	Total		**19**

12. Detailed Course Contents

Title	**Hydraulic Engineering**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

The instructor will

1. Provide a rigorous and comprehensive foundation in the fundamental laws governing fluid flow
2. Help the students to understand the importance and application of fluid mechanics
3. Exemplify concepts of similarity and model testing

Learning Outcomes :

The students are expected to have the ability to:

1. Understand and apply the fundamental principles of fluid statics, kinematics, and dynamics to analyse forces, pressures, and flow conditions in civil engineering systems
2. Derive and apply the governing equations of fluid motion to solve engineering problems
3. Design physical hydraulic models, interpret experimental results, and extend findings to prototype-scale engineering predictions.

Course Contents:

Introduction: Definition and properties of fluid, Newtonian and Non-Newtonian Fluids, Dimensions and units, Continuum Hypothesis, Knudsen Number, Capillary and surface tension. [3 Lectures]

Pressure and Fluid Statics: Pascal's Law, Manometry, hydrostatic force on submerged planar and curved surfaces, buoyancy and stability, rigid body motion. [5 Lectures]

Fluid Kinematics: Graphical Description of Fluid Flow, Vorticity, Circulation, Stream and Potential Functions, rotational-irrotational flows, Bernoulli's equation and its applications. [5 Lectures]

Fluid Dynamics: Reynold Transport Theorem, Acceleration of fluid, Deformations in fluid particles, derivation of Euler's equation and Navier-Stokes equations. [9 Lectures]

Internal flow: Reynolds number regimes and Reynold experiments, Laminar and Turbulent flow in pipes, Head Losses in pipes, Flow measurement in pipes, Pipe network analysis, Introduction of water hammer effect and cavitation, Moody's chart. [6 Lectures]

External flow: Boundary layer over flat plate, boundary layer thickness, Prandtl-boundary layer equations, momentum integral equation, pressure gradient & separation, drag coefficients. [6 Lecture]

Dimensional analysis and similarity: Dimensionless Numbers, Scale and Similitude, Buckingham Pi theorem and examples. [2 Lectures]

Flow in Open Channel: Discharge in open channel, Concept of Specific Energy and Alternate Depths, Hydraulic Jump. [3 Lecture]

Text Books:

1. Hibbler, R.C., (2022), Fluid Mechanics, 2nd Edition, Pearson.
2. Modi, P.N. and Seth, S.M., Hydraulics and Fluid Mechanics Including Hydraulic Machines. Standard Book House, New Delhi.
3. White, F. M. (2003), Fluid Mechanics, 5th Edition, McGraw-Hill.

Reference Books:

1. Munson, B. R., Rothmayer, A. P., Okiishi, T'H., and Huebsch, W. W. Fundamentals of Fluid Mechanics, 7th Edition, John Wiley and Sons.
2. Fox, R. W., Pitchard, P. J., and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
3. Kundu, P. K, Cohen, I. M and Dowling, D. R, Fluid Mechanics, 6th Edition, Associated Press.
4. Cimbala, J.M. and Cengel, Y.A., (2010), Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

Online Course Material:

1. Prof. S. Dutta, Fluid Mechanics, NPTEL Course Material, Civil Engineering, IIT Guwahati, https://nptel.ac.in/courses/105103192/
2. Prof. M. S. Afzal, Hydraulic Engineering, NPTEL Course Material, Civil Engineering, IIT Kharagpur, https://nptel.ac.in/courses/105105203
3. Prof. T. I. Eldho, Fluid Mechanics, NPTEL Course Material, Civil Engineering, IIT Bombay, https://nptel.ac.in/courses/105101082

Title	**Mechanics of Solids**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisiste	-		
Antirequisite	Mechanics of Solids / Solid Mechanics (by other Academic Units)		

Objectives:

The instructor will

1. To inculcate fundamental concepts of stress, strain and deformation of solids
2. Fundamental of applying equilibrium, compatibility and forced deformation relationships to structural elements

Learning Outcomes :

The students are expected to have the ability to:

1. Apply the fundamentals to analyze structural members.
2. Physical insight into distribution of stresses and strains in structural members.
3. Ability to design structural members using dimensions, force-displacement relationships, boundary conditions, loading and allowable stress

Course Contents:

Stress and strain: Concept of stress at a point, Plane stress case: transformation of stresses at a point, principal stresses and Mohr's circle, Displacement field, Concept of strain at a point, Plane strain case: transformation of strain at a point, principal strains and Mohr's circle, Strain Rosette, Polar coordinates in stress and strain analysis. St. Venant's principles and stress concentrations. [10 Lectures]

Mechanical behaviour: Concepts of elasticity, plasticity, strain hardening, failure (fracture/yielding), Generalized Hooke's law (without and with thermal strains) for isotropic materials, orthotropic and anisotropic materials, Force displacement relationship and geometric compatibility for axially loaded members and thin walled pressure vessels. [7 Lectures]

Static Analysis: Complete equations of elasticity; Force analysis (axial force, shear force, bending moment, and twisting moment diagrams) of slender members. [4 Lectures]

Elastic Bending: Moment curvature relationship for pure bending of beams with symmetric cross section, bending stress, shear stress; Cases of combined stresses, Deflection due to bending, Integration of the moment curvature relationship for simple boundary conditions, Stress concentrations.[5 Lectures]

Elastic torsion: Torsion of circular shafts and thin walled tubes, Stress

concentrations. [4 Lectures]

Strain Energy: Concept of strain energy, strain energy for simple structural elements, Castigliano's theorems for deflection analysis and indeterminate problems. [5 Lectures]

Elastic Buckling: Concept of elastic instability, Introduction to column buckling, Euler's formula. [4 Lectures]

Text Books:

1. Crandall, S.H., Dahl, N.C., and Lardner, T. J., (1978), An Introduction to the Mechanics of Solids, McGraw Hill, Second Ed. with SI Units.
2. Beer, F.P, Johnston, E.R. and De Wolf, J.T., (2004), Mechanics of Materials, Tata McGrawHill.
3. Popov, E.P., (1990),Engineering Mechanics of Solids, First Ed. Prentice Hall
4. Meriam, J.L. and Kraige, L.G., (1980), Engineering Mechanics, Vol. 1: Statics, 2nd Ed. John Wiley
5. Timoshenko, S.P. and Goodier, J.N. ``Theory of Elasticity``, McGraw-Hill, International Edition, 1970.

Online Course Material:

1. https://nptel.ac.in/courses/105/106/105106116
2. https://nptel.ac.in/courses/105/102/105102090
3. https://nptel.ac.in/courses/105/104/105104160

Title	**Building Materials and Construction Technology**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

1. Understand the fundamental behavior of different building materials
2. To introduce the various building materials available for construction
3. Understand the different components of building and their functions

Learning Outcome:

Following this course, the student will have the ability to:

1. Identify and select the right building materials for various applications
2. Identify the correct construction method for different constructions

Course Contents:

Introduction to materials – atomic structure, energy and entropy, structure of solids, Mechanical properties – stress-strain relationships, elastic behavior and elastic constants, failure under different conditions, fracture and toughness, viscoelasticity, surfaces – energy, wetting, electrical and thermal properties, variability in materials and its implications. [9 Lectures]

Classification, physical, mechanical properties and testing of commonly used building materials – Stones, bricks, cement, aggregates, chemical admixtures, supplementary cementitious materials, lime, timber, paints, glass, rebars, polymers etc.

Concrete – Fresh and hardened properties of concrete, test methods, grades of concrete, batching, mixing, placing and curing, Introduction of concrete mix design. [15 Lectures]

Building Construction: Components of building, stone and brick masonry, framed and loading bearing structures, shallow and deep foundations, floors and roofs, lintels, arches and staircases, doors, windows and ventilators, finishing materials, flooring, joints, waterproofing, formworks, scaffolding, shoring and underpinning. [15 Lectures]

Text Books:

1. P.C. Varghese (2008) Building Materials, Prentice-Hall of India, New Delhi.
2. B. C. Punmia (1993) Building Construction, Laxmi Publications, New Delhi.

Reference Books:

1. V. Raghavan (1990) Materials Science and Engineering, Prentice Hall.
2. J.F. Young, S. Mindess, R.J. Gray & A. Bentur (1998) The Science and

Technology of Civil Engineering Materials, Prentice Hall.
3. P.K. Mehta and P.J.M. Monteiro (2006) Concrete: Microstructure, properties and materials, McGraw Hill.
4. Edward Allen and Joseph Iano (2008) Fundamentals of Building Construction, 5th Edition, John Wiley & Sons Inc., New Jersey.
5. M. Soutsos and P. Domone (2018) Construction Materials, Their Nature and Behavior, Fifth Edition, CRC Press.

Online Course Material:

1. NPTEL course - NOC: Modern Construction Materials, Prof. Ravindra Gettu Prof. Aslam Kunhi Mohamed, IIT Madras.
   http://www.nptel.ac.in/courses/105106053/

Title	**Geoinformatics**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

This course aims at:

1. To introduce the basic principles of remote sensing, GPS, Surveying and GIS.
2. To teach application of geo-informatics in geoscience, ecology and environment study, agriculture, and disaster management.

Learning Outcomes:

A graduate of this course will have a thorough idea of:

1. Multispectral, thermal infrared, and microwave remote sensing.
2. Surveying Principles and handling modern surveying equipment
3. Different map projections and use of GPS.
4. Component GIS and applications.
5. Different cartographic models, implementation, validation, and visualization.

Course Contents:

Surveying and Global Positioning System: Surveying Fundamentals: Introduction to modern surveying; Coordinate systems; Measurement of distances and angles; Introduction to Total Station. Basic Geodesy, Map and Projections: Introduction, Basic Geodesy, Shape of Earth, Map and its types, Map Scale and Symbol, Map Projection and its Classification; Georeferencing. Global Positioning System (GPS): Components and operational principles; GPS receivers; Differential GPS (DGPS) concepts and types; Real-time kinematic positioning; limitations, and areas of application. [13 Lectures]

Remote Sensing and Photogrammetry: Physics of RS: Interaction of EMR with atmosphere and Earth's surface; Atmospheric windows. Photogrammetry: Types of aerial photographs and their geometry; Aerial cameras and Use of Drones; Image displacement. Stereoscopic Vision: Binocular observations; Stereoscopic pairs; Types of stereoscope; Measurement of height from photographs; Stereo model. Multispectral and Hyperspectral imaging: Spectral signatures of vegetation, soil, and water, Sensor Utilized; Image correction and interpretation Advanced Remote Sensing: Thermal Infrared (emissivity, characteristics, and applications); Microwave region (SAR characteristics and interpretation); LIDAR principles, LIDAR point cloud and its processing. [15 Lectures]

Geographical Information System and Applications: GIS Concepts Components and spatial data representation; Relationships of spatial objects; GIS functions; Integration of Remote sensing and GIS. Cartographic Modelling: Inductive and

deductive modelling; Model flowcharting; Conflict resolution, verification, and implementation, GIS Model Output. Visualization and Applications: Web mapping and visualization; Virtual and Immersive GIS, Mapping temporal data. Applications in Environment, Agriculture, Public Health, Water Resources, and Disaster Management. [11 Lectures]

Text Books:

1. Satheesh Gopi, R. Sathikumar, and N. Madhu (2025) Advanced Surveying: Total Station, GPS, GIS & Remote Sensing (2nd Ed). Pearson India.
2. T. M. Lillesand, R. W. Kiefer, J. W. Chipman (2004) Remote Sensing and Image Interpretation (5th Ed). Wiley. ISBN: 0-471-15227-7
3. G. S. Srivastava (2014) An Introduction to Geoinformatics. McGraw Hill Education (India) Private Limited. New Delhi. India. ISBN: 9781259058462.
4. Michael N. DeMers (2008) Fundamentals of Geographic Information Systems (4th Ed). Wiley. ISBN: 978-0-470-12906-7.

Reference Books:

1. Basudeb Bhatta (2021) Remote Sensing and GIS (3rd Ed). Oxford University Press.
2. L. E. Johnson (2009) Geographic Information Systems in Water Resources Engineering. CRC Press. Taylor & Francis Group. ISBN 978-1-4200-6913-6.
3. B. Tomaszewski (2020) Geographic Information Systems (GIS) for Disaster Management (2nd Ed). CRC Press.

Online Course Material:

1. Prof. Arun K. Saraf (2016) Introduction to Geographic Information System. NPTEL. https://nptel.ac.in/courses/105/107/105107155/
2. Prof. Arun K. Saraf (2017) Introduction to Remote Sensing. NPTEL. https://nptel.ac.in/courses/121/107/121107009/

Title	**Transportation Engineering - I**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
The course will help students to understand the principles of transportation engineering, including geometric design, pavement materials and design, pavement construction, evaluation, and maintenance.

Learning Outcomes:
At the end of the course, the student should be able to:

1. Design cross section elements, horizontal and vertical alignment.
2. Determine the characteristics of pavement materials.
3. Design flexible and rigid pavements.
4. Explain the construction procedures for different pavement layers and the equipment used in highway construction.
5. Evaluate the functional and structural condition of pavements using appropriate methods and recommend maintenance or rehabilitation measures.

Course Content:
Introduction to Transportation Engineering:Transportation system, Introduction to Transportation Engineering, Functional classification of roads, Terrain classification, Factors affecting transportation, Highway development in India**.** [4 Lectures]

Geometric design: Factors affecting geometric design; Alignment – types, requirements, factors influencing and stages; Highway design process - Highway cross sectional elements, Sight distance; Highway alignment: Horizontal alignment - horizontal curve, Super elevation, extra widening, Transition curve, Setback distance, Visibility, Vertical alignment - gradient, grade compensation, summit curve, valley curve; alignment for hill roads. [10 Lectures]

Pavement materials and design: Pavement characteristics, materials, Types of pavements - Flexible and Rigid pavements, Wheel load calculations, Flexible Pavement materials- soil, aggregate, bitumen; Tests on pavement materials; Bituminous mix design, Marshall stability; Rigid pavement- joints, analysis of stresses; IRC method for design of flexible and rigid pavement; Pavement failures, IIT Pave Software. [11 Lectures]

Pavement Construction: Construction of various layers, equipment used for highway construction and quality control aspects. [6 Lectures]

Pavement Evaluation and Maintenance: Failures of flexible and rigid pavements, condition indicators, various types of functional and structural evaluation techniques, maintenance and rehabilitation alternatives, Design of overlays. [8 Lectures]

Text Books:

1. N. J. Garber, L.A. Hoel. Traffic and Highway Engineering.
2. Saxena S. C. Textbook of Highway and Traffic Engineering, CBS Publishers.
3. Wright Dixon, Highway Engineering, Wiley India.

Reference Books:

1. S. K. Khanna, C. E. G. Justo, and A. Veeraragavan. Highway Engineering, Nem Chand and Bros. Publishers.
2. I.S. specifications on concrete, aggregates and bituminous materials.
3. David Croney, Design and performance of road pavements, McGraw Hill.
4. James H. Banks, Introduction to Transportation Engineering, McGraw Hill.
5. R. Srinivasa Kumar, Textbook of Highway Engineering, University Press.
6. Chakroborty, P., Animesh Das. Principles of Transportation Engineering, PHI Learning Private Limited.

Online Course Material:

1. Introduction to Transportation Engineering by Prof. K.S. Reddy and Prof. Bhargab Maitra. NPTEL. https://nptel.ac.in/courses/105105107

Title	**Structural Analysis**	Number	CI2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

This course aims at:

1. Introduction to fundamental concepts of structural analysis.
2. Development of analytical skills to compute quantities like displacements, stresses etc. in structures
3. subjected to static forces.
4. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
5. Problem solving ability for static loads in Civil Engineering design of structures.

Learning Outcomes:

Students will have the ability to:

1. Establish equilibrium equations of structural systems.
2. Analyze the responses and stresses generated in structures subjected to different types of static loading.
3. Apply the structural analysis methods to solve real-world problems of the building, bridges etc.

Course Content:

Analysis of statically determinate structures: Determination of internal force on a system; internal forces acting on typical structural members - axial force, shear force and bending moment; discussion on sign convention and notations for internal forces; obtaining internal forces in a system - general procedure; internal force diagrams for various systems; analysis of pin jointed determinate truss. [8 Lectures]

Determination of slope and deflection of beams: Solving differential equation of the elastic curve, slope and deflection of beams by method of successive integration; Macaulay's method; moment area method; conjugate beam method; principle of virtual work; strain energy method; bending deflection due to temperature variation; Maxwell-Betti law of reciprocal deflections. [9 Lectures]

Force method: Introduction and applications of analysis of statically indeterminate beams - the force method; analysis of statically indeterminate structures by energy method; three moment equation for shear force, bending moment diagrams, deflection and support settlement. [10 Lectures]

Moving loads and influence lines: Introduction to variable loading, construction of influence lines using equilibrium method, use of influence lines for reaction, shear force and bending moment in simply supported beams; influence lines for forces in trusses; analysis for different types of moving loads - concentrated loads, uniformly distributed load etc. [7 Lectures]

Cables, suspension bridges and arches: Analysis of forces in cables, suspension bridges with three-hinged and two-hinged stiffening girders; theory of arches; Eddy's theorem; analysis of three-hinged and two-hinged arches; settlement and temperature effects. [8 Lectures]

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata
3. McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
4. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
5. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.

Online Course Material:

1. Structural Analysis I (2009) by Prof. Siddhartha Ghosh and Prof. R.S. Jangid on NPTEL. https://nptel.ac.in/courses/105/101/105101085/

Title	**Soil Mechanics**	Number	CIE2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered For	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
To introduce the fundamental engineering principles of soil mechanics including their formation, classification, hydraulic behavior, stress distribution, consolidation and shear strength.

Learning outcomes:
Students will have the ability to:

1. Identify and classify soils based on index properties.
2. Assess soil phase relationships and engineering indices.
3. Analyze seepage and permeability problems.
4. Determine stress distribution in soil due to applied loads.
5. Evaluate consolidation and settlement behaviour.
6. Determine shear strength parameters of soils.

Course Contents:
Soil Formation: Phase Relationships, Index Properties and Compaction: Geological soil formation and classification, Types of soils, Soil particle size and shape, Clay minerals and clay-water interaction, Three phase system of soils, Phase relationships, Index properties of Soils, Grain size distribution, Atterberg limits, Soil classification systems (USCS), Relative density of sands, Mechanism of compaction, Standard and modified Proctor tests. [9 Lectures]

Permeability and Seepage: Darcy's law, Hydraulic gradient and seepage velocity, Coefficient of permeability, Constant head permeability test, Falling head permeability test, Permeability of stratified soils, Field permeability of soils, Seepage through soils, Flow net construction and applications, Quick sand condition and piping. [7 Lectures]

Effective Stress and Stress Distribution: Effective stress principle, Pore water pressure, Capillary rise in soils, Vertical stresses in soil mass, Stress distribution due to surface loads, Boussinesq's theory, Westergaard's theory, Influence charts. [7 Lectures]

Compressibility and Consolidation: Compressibility of soils, One-dimensional consolidation, Terzaghi consolidation theory, Consolidation test (oedometer), Determination of coefficient of consolidation, Primary and secondary consolidation, Consolidation settlement. [8 Lectures]

Shear Strength of Soils: Concept of shear strength, Mohr circle of stress, Mohr–Coulomb failure criterion, Stress conditions in soil mass, Direct shear test, Triaxial compression test, Unconfined compression test, Drained and undrained shear strength. [8 Lectures]

Text books:

1. Budhu, M. (2010). *Soil mechanics and foundations*. John Wiley and Sons.
2. Murthy, V. N. S. (2002). Principles and practices of soil mechanics and foundation engineering. *New York: Marcek Decker INC*.
3. Ranjan, G., & Rao, A. S. R. (2011). *Basic and applied soil mechanics*. New Age International.

Reference Books:

1. Terzaghi, K., Peck, R. B., & Mesri, G. (1996). *Soil mechanics in engineering practice*. John wiley & sons.
2. Lambe, T. W. (1951). *Soil testing for engineers* (Vol. 72, No. 5, p. 406). LWW.

Title	**Engineering Hydrology**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
To provide an understanding of hydrologic processes, analysis methods, and design applications with emphasis on mathematical and statistical analysis in water resources management.

Learning outcomes:

1. Become familiar with engineering aspects of surface and sub-surface hydrology
2. Utilize knowledge of hydrological processes in solving engineering problems within hydrology and water resources
3. Develop the understanding of the major impact of changing climate on the water cycle.

Course Contents:
Introduction: Engineering hydrology and its applications, Hydrological cycle, Concept of Catchment, Water balance and fluxes, World Water Balance, Residence time, Climate Change Impacts. [4 Lectures]

Precipitation: Introduction, Forms of precipitation, Measurement of Precipitation, Mean Precipitation over an Area, Analysis of precipitation data, Precipitation variability and monsoon characteristics in India, Depth–Area–Duration Relationships, Maximum Intensity/Depth–Duration–Frequency Relationship, Probable Maximum Precipitation (PMP). [6 Lectures]

Abstractions from precipitation: Introduction, Evaporation and Evapotranspiration: factors affecting evaporation, measurement and estimation of evaporation, Reservoir Evaporation and Methods for Its Reduction, Transpiration, Measurement of Evapotranspiration, Potential Evapotranspiration and its estimation, Actual Evapotranspiration; Interception; Depression Storage; Infiltration: Infiltration Capacity and factors affecting Infiltration capacity, Measuring and modeling infiltration, Infiltration Indices. [7 Lectures]

Runoff and streamflow: Types of runoff, Runoff characteristics of stream, Catchment factors affecting runoff, Flow–Duration and Flow–Mass Curve, Estimation of Reservoir Volume, Hydrograph, Factors affecting Storm Hydrograph, Components of a Hydrograph, Base Flow Separation, Effective Rainfall, Unit Hydrograph and its derivation, Unit Hydrographs of Different Durations, Synthetic Unit Hydrograph. [8 Lectures]

Droughts, Floods and Flood Routing: Introduction to Hydrologic Extremes, Drought, Drought Indices, Flood Estimation: Rational Formula, Runoff Coefficient and Unit hydrograph method, Flood Frequency Analysis, Design Flood and Design Storm, Risk, reliability and Safety Factor in hydrologic design, Introduction to Flood routing, Level Pool Routing, Hydrological Channel Routing, Hydraulic Methods of Flood Routing, Flood Control, Flood in changing climate. [9 Lectures]

Groundwater: Sources and form of Groundwater, Saturated Formations, Water Table, Aquifer Properties, Porosity and Permeability, Groundwater Movement, Hydraulics of wells, Recharge and its techniques, Groundwater Resources of India. [5 Lectures]

Text Books:

1. Subramanya, K. (2024). Engineering hydrology (6th ed.). McGraw Hill Education (India) Pvt Ltd
2. Ojha, C. S. P., Berndtsson, R., & Bhunya, P. (2008). Engineering hydrology. Oxford University Press.
3. Viessman Jr., W., Lewis, G. (2002). Introduction to Hydrology. 5th Edition, Pearson

Reference Books:

1. Chow, V.T., Applied Hydrology, McGraw-Hill
2. Todd, D.K., & Mays, W.L. (2004) Groundwater Hydrology, 3rd Edition, Wiley

Online Course Material:

1. Margulis, S. (2016). Introduction to hydrology. https://margulis-group.github.io/textbook/
2. Prof. Rajib Maity (2012). Advanced Hydrology. NPTEL. https://nptel.ac.in/courses/105105214

Title	**Air Pollution and Solid Waste Management**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

1. To explain and classify sources, types, impacts of air pollutants, along with relevant standards and legislation.
2. To analyze the role of meteorological parameters in air pollutant dispersion and evaluate control strategies.
3. To explain and characterize solid waste sources, composition, and generation, along with regulatory and management frameworks.
4. To apply and evaluate integrated solid waste management principles, including waste minimization, treatment, and disposal technologies.

Learning Outcomes:

Upon successful completion of the course, students will be able to:

1. Classify and assess air pollutants and interpret their impacts on health, environment, and regulatory compliance.
2. Analyze and predict pollutant dispersion using meteorological concepts such as lapse rate, stability, and plume behavior.
3. Evaluate and justify the selection of air pollution monitoring and control technologies for specific applications.
4. Design and propose integrated solid waste management strategies including collection, processing, treatment, and disposal.

Course Contents:

Air pollution

Sources, Types and classification of air pollutants, air pollution episodes, effect of air pollutants on human health, vegetation and materials, air quality and emission standards and legislation. [7 Lectures]

Effect of meteorology, Lapse rate, Atmospheric Stability, Plume behaviour and dispersion, Stack height. [4 Lectures]

Engineering emission control concepts, Selection and basic design approach for air pollution control equipments, source control technologies, Indoor air pollution sources and control. [9 Lectures]

Municipal Solid Waste Management

Fundamentals of Integrated Solid Waste Management, legislation and regulations. [4 Lectures]

Waste generation, collection, and transportation; Characteristics of solid waste.[6 Lectures]

Transfer stations, material recovery facility; biological treatment methods (such as

composting and anaerobic digestion), incineration, pyrolysis and gasification, and sanitary landfilling. [9 Lectures]

Text Books:

1. Noel De Nevers, Air Pollution Control Engineering, 3rd Ed., Waveland Press, Inc., 2016
2. Tchobanoglous, G., Theisen, H., and Vigil, S.A. (2014) Integrated Solid Waste Management: Engineering Principles and Management Issues, McGraw Hill Book Company, Singapore.
3. Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.

Reference Books:

1. Air Pollution Control: A Design Approach, C. David Cooper & F. C. Alley, 4th Ed., Waveland Press, Inc., 2011
2. CPHEEO Manual of Solid Waste Management, GOI Publication, 2016
3. M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Title	**Transportation Engineering - II**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
The course will introduce students to the principles of traffic engineering, traffic operations, and transportation planning. It aims to develop an understanding of traffic flow characteristics, traffic control systems, travel demand estimation, and urban transportation planning for sustainable mobility systems.

Learning Outcomes:
At the end of the course, the student should be able to:

1. Analyze traffic flow characteristics and interpret speed–flow–density relationships
2. Conduct traffic engineering studies and evaluate traffic operations at different road facilities
3. Understand transportation planning principles and carry out surveys involved in transport planning
4. Apply four-step travel demand forecasting models

Course Content:
Introduction to Transportation systems: Overview of transportation systems and their role in economic and social development, Elements of traffic engineering and transport planning and Current transportation challenges. [2 Lectures]

Traffic Engineering: Transportation system, Types of facilities, Factors affecting traffic performance, Categories of traffic flow, Traffic stream components and characteristics, Analysis of speed, Flow and density relationships, Fundamental diagram of traffic flow, Trajectory diagrams. [9 Lectures]

Traffic Operations: Theories of traffic flow; Traffic studies (volume, speed, delay, parking, accident studies); Design of control strategies for simple systems like intersections, roundabouts, freeways, etc., Introduction to traffic signs and signals, Capacity and level of services of various transportation facilities, Introduction of Intelligent Transportation System (ITS), software tools, AI/ML applications. [11 Lectures]

Transportation Planning: Characteristics and objectives of planning, Urban Transportation Planning- Comprehensive planning process, multimodal mobility, Transit-Oriented Development (TOD), NMT planning, Transportation data needs, and sources, types of trips, Study area and analysis zone, Transportation surveys/data collection methods. [8 Lectures]

Travel demand estimation: Four-step transportation planning process, Trip generation, Trip distribution, Modal split, Route assignment, Traffic impact assessment studies, Activity based models. [9 Lectures]

Text Books:

1. N. J. Garber, L.A. Hoel. Traffic and Highway Engineering, 4th Edition
2. R. P. Roess, E.S. Prassas, W.R. McShane. Traffic Engineering, 4th edition, Pearson.
3. C. S. Papacostas, P. D. Prevedouros. Transportation Engineering and Planning, Third Edition, Pearson

Reference Books:

1. Coleman A. O'Flaherty. Transport Planning and Traffic Engineering, CRC Press
2. K. Khanna, C. E. G. Justo, A. Veeraragavan. Highway Engineering, Tenth Edition, NemChand & Bros.
3. Dr. L. R. Kadiyali, Traffic Engineering and Transport Planning, Khanna Publishers.
4. James H. Banks, Introduction to Transportation Engineering, McGraw Hill
5. Saxena S. C. Textbook of Highway and Traffic Engineering, CBS Publishers.

Online Course Material:

1. 1. Introduction to Transportation Engineering by Prof K.S. Reddy and Prof Bhargab Maitra.NPTEL. https://nptel.ac.in/courses/105105107
2. Urban Transportation Planning by Prof. V. Thamizh Arasan. NPTEL. https://nptel.ac.in/courses/105106058

Title	**Water and Wastewater Engineering**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

The Instructor will:

1. Provide an overview on water demand, water quality and treatment unit operation
2. Provide an overview on wastewater characteristics and treatment unit operation
3. Provide knowledge on water distribution system

Learning Outcomes:

The students are expected to be able to:

1. Understand water treatment unit operation and would be able to design it
2. Understand wastewater treatment unit operation and would be able to design it
3. Analyse water distribution system for further design

Course Contents:

Water demand and Water quality assessment: Water demand; Water quality parameters: Physical, chemical and biological parameters; Water quality standards. [4 Lectures]

Water treatment theory and operation:Theory and design parameters for various water treatment unit processes such as screening; sedimentation tank; coagulation-flocculation; filtration; disinfection; hardness; Exposure to advanced water treatment processes. [12 Lectures]

Water distribution system:Design of water supply pipeline; Pumping; Head loss and water distribution network analysis. [4 Lectures]

Wastewater characterization: Wastewater characterization : physical, chemical and biological parameters. [5 Lectures]

Wastewater treatment and sludge disposal: Introduction to Unit processes in wastewater treatment plant and design criteria; Focus on secondary treatment: suspended growth system (e.g., activated sludge process) and attached growth systems (e.g., trickling filter); nutrient removal; Introduction to natural system; Exposure to advanced treatment process and design software. [14 Lectures]

Text Books:

1. Metcalf & Eddy, Wastewater Engineering- Treatment and Reuse (Revised by G. Tchobanoglous, F. L. Burton and H.D. Stensel), Tata McGraw Hill, 4th Edn., 2004.H.
2. Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
3. M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Reference Books:

1. Nathanson and Jerry, Basic Environmental Technology: Water Supply, Waste Management and Pollution Control, Prentice Hall India Learning Pvt. Ltd., India 2009
2. Garg, Environmental Engineering: Water Supply Engineering, Khanna Publishers, 18th Edn, India 2004
3. Manual on Sewerage and Sewage Treatment Systems, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 2013.
4. Manual on Water Supply and Treatment, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 1999.

Online Course Material:

1. Guidelines for drinking-water quality, World Health Organization 2022, ISBN 978-92-4-004506-4, https://www.who.int/publications/i/item/9789240045064
2. Drinking Water – Specification, Bureau of Indian Standard , IS 10500 : 2012, New Delhi, India, 110002

Title	**Reinforced Concrete Design**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Structural Analysis		

Objectives:

This course aims at:

1. Providing an in-depth knowledge on the behaviour of RCC structures.
2. Highlighting the significance of working stress and limit state design methods.
3. Comprehensive development of the design and reinforcement detailing of beams, columns, slabs, staircases, footings.
4. Develop an understanding with respect to the importance of stiffness, strength, ductility and configuration which is useful for earthquake resistant design.
5. Practicing reinforcement detailing procedures of a RCC building considering wind and seismic forces as per Indian Standard codes and developing design skills.

Learning Outcomes:

Students will have the ability to:

1. Assess the behaviour of RCC structures and apply the appropriate design methodology and reinforcement detailing for different types of RCC structural elements i.e. beams, columns, slabs, staircases, foundations etc.
2. Design and provide reinforcement detailing of multistoried RCC buildings considering dead, wind and seismic forces as per Indian Standard codes.

Course Contents:

Introduction to reinforced concrete structures: Properties of concrete; codes of practices; design philosophy of reinforced concrete structures - working stress and limit state design methods. Design of RCC beams: Singly and doubly reinforced rectangular/flanged sections; design for shear, torsion, bond and anchorage of reinforcement; design of continuous beams. [10 Lectures]

Design of Slabs: One-way and two-way slabs; design of slabs for serviceability; design of continuous slab systems. [7 Lectures]

Design of RCC columns: design of short columns under pure compression, uniaxial and biaxial bending; design of long columns. [8 Lectures]

Design of foundations: Isolated footings; combined and strip footing; raft foundation. Design of retaining walls. [8 Lectures]

Design of RCC structures considering wind and seismic forces: Concept of earthquake resistant design of RCC structures - design philosophy, stiffness, strength, ductility and configurations; estimation of wind and seismic forces as per IS codes; Introduction to prestressed concrete. [6 Lectures]

Text Books:

1. U.S. Pillai, D. Menon (2016) Reinforced Concrete Design (3rd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 9780070141100.
2. S.N. Sinha (2014) Reinforced Concrete Design (3rd Ed) McGraw Hill Education Pvt. Ltd. ISBN 9789351342472.
3. A.K. Jain (2012) Reinforced Concrete - Limit State Design (7th Ed), NemChand Brothers. ISBN 9788185240664 50
4. P.C. Varghese (2010) Limit State Design of Reinforced Concrete (2nd Ed), Prentice Hall of India. ISBN 8120320395.
5. S.K. Mallick, A.K. Gupta (2002) Reinforced Concrete (2nd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 0070473323.

Online Course Material:

1. Design of Reinforced Concrete Structures (2009) by Dr. Nirjhar Dhang on NPTEL. https://nptel.ac.in/courses/105/105/105105105/

Title	**Geotechnical Engineering**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered For	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objective:
To introduce geotechnical investigation methods and the design principles of slopes, retaining structures, and shallow and deep foundations.

Learning outcomes:
Students will be able to:

1. Plan and interpret geotechnical site investigations methods and data
2. Assess slope stability problems
3. Evaluate earth pressure on retaining structures
4. Evaluate bearing capacity and settlement of shallow foundations
5. Understand design principles of deep foundations

Course Contents:
Site Investigation and Data Collection: Objectives of site investigation, Methods of subsurface exploration, Boring, drilling, and logging methods, Sampling techniques (disturbed and undisturbed), Rock Mass Characterization, Rock Mass Classification – RMR, Q-System, and GSI.; Standard Penetration Test (SPT), Cone Penetration Test (CPT), Geophysical exploration methods, Preparation of soil/engineering geology investigation reports. [10 Lectures]

Shallow Foundations: Types of shallow foundations, Bearing capacity concepts, Modes of shear failure, Terzaghi bearing capacity theory, Bearing capacity factors, Effect of water table, Plate load test, Settlement of foundations, Safe bearing capacity. [8 Lectures]

Deep Foundations: Types of piles, Methods of pile installation, Load carrying capacity of piles, shaft resistance and base resistance, Pile load tests, Group action of piles, Settlement of pile groups, Introduction to drilled piers and well foundations. [7 Lectures]

Lateral Earth Pressure and Retaining Structures: Types of earth pressure, Earth pressure at rest, Rankine earth pressure theory, Coulomb earth pressure theory, Retaining wall types, Stability of retaining walls. [7 Lectures]

Stability of Slopes: Types of slopes and slope failures, Factor of safety concept, Stability of infinite slopes, Stability of finite slopes, Plane failure and circular failure, Method of slices, Bishop simplified method. [7 Lectures]

Text Books:

1. Das, Braja M. (2015). Principles of Foundation Engineering, SI (7th Edition). Cengage Learning.
2. Ranjan, Gopal, and A. S. R. Rao. 2023. *Basic and Applied Soil Mechanics*. 5th ed. New Delhi: New Age International Private Limited.
3. Budhu, Muni. (2011). Soil Mechanics and Foundations (3rd Edition). John Wiley & Sons, Hoboken, New Jersey.
4. Murthy, V. N. S. (2002). Geotechnical Engineering: Principles and Practices of Soil Mechanics and Foundation Engineering. New York: Marcel Dekker Inc.
5. Deb, D., Verma AK (2016) Fundamentals and Applications of Rock Mechanics. PHI Learning Pvt. Ltd.

Title	**Water Resources Engineering**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

1. To quantify water demand for varied sectors and estimate available surface and groundwater resources.
2. To master the hydraulic design of storage, diversion, and conveyance structures.
3. To apply computational tools (like EPANET) for the analysis of complex water distribution networks.
4. To integrate watershed management principles for sustainable water resource development.

Learning Outcomes :

A graduate of this course will have a thorough idea of::

1. Predict future water demands and estimate reservoir capacity.
2. Design stable alluvial and non-alluvial canals using Lacey's and Kennedy's theories.
3. Analyze and optimize pipe networks using the Hardy-Cross method and software.
4. Design hydraulic profiles for spillways, energy dissipators, and cross-drainage works.
5. Develop a River Basin Master Plan incorporating soil conservation and rainwater harvesting.

Course Contents:

Water Demand: Introduction, Role of Water in Modern Lifestyle, Estimating Urban Water Demand, Irrigation requirement, Industrial Water Demand, Projecting future demand, Concept of Water Equivalent.. [4 Lectures]

Available Water Resources: Introduction; Overview of Available Water Resources in India; Surface Water Resources: Stream flow Measurement Techniques, Hydrograph, Flow-Duration and Flow-mass Curve, Reservoir Capacity Estimation using Sequent Peak Algorithm; Groundwater Resources: Occurrence, Aquifer Properties, Hydraulics of Wells, Yield, Artificial Recharge. [8 Lectures]

Water Retention: Dam: types and their role in water retention, Suitable Site; Spillways: Design Considerations, Location of Spillway, Spillway Types; Reservoir: Different Storage in Reservoir, Reservoir Losses, Reservoir Sedimentation, Trap Efficiency, Methods to Control Silting, Safe Yield, Multi-Purpose Reservoir

Operation. [10 Lectures]

Water Conveyance: Intake Structures: Introduction; design considerations; Pipe Network: Design of Pipe Network, Turbulent Flow in Pipes, Minor Loss, Hardy-Cross Analysis, Software Analysis (for instance EPANET) of pipe network; Canal System: Alignment, Command Area, Time and Capacity Factors, Water losses in canal, Canal Falls, Canal Regulation, Alluvial vs Non-Alluvial Canal, Regime Theories, Design and economics of Canal Lining, Cross Drainage Works. [8 Lectures]

Integrated Watershed Management (IWM): Watershed Concepts: Definition, types, and characterization, River Basin Master Plan; Soil Erosion and Conservation: Mechanics and estimation of water erosion, Structural and vegetative measures (contouring, terracing, gully plugs); Rainwater Harvesting: Rooftop harvesting, recharge pits, and check dams; Land Use Management: Impact of deforestation and urbanization on runoff, Concept of Low Impact Development; Participatory Management. [9 Lectures]

Text Books:

1. Mays, L.W. *Water Resources Engineering*, John Wiley and sons, Inc., Kundli
2. Todd, D. K.and Mays, L. W. *Ground Water Hydrology,* John Wiley India Edition
3. Garg, S.K. Irrigation Engineering and Hydraulic Structures. Khanna Publishers, Delhi.

References Books:

1. Chow, V.T., Maidment, D.R., and Mays L.W. Applied Hydrology. McGraw-Hill Inc., New York.
2. James, L.D., and Lee, R.R. Economics of water resources planning. McGraw-Hill Book Inc., New York.
3. Todd, D. K.and Mays, L. W. *Ground Water Hydrology,* John Wiley India Edition
4. Linsley, R.K., Franzini, J.B., Freyberg, D.L., and Tchobanoglous, G. Water Resources Engineering, McGraw Hill Inc., Singapore.
5. Rohitashw Kumar, Vijay P. Singh, and Sabah Parvaze. Watershed Planning and Management, CRC Press
6. Subramanya, K. (2024). Engineering hydrology (6th ed.). McGraw-Hill Education (India) Pvt Ltd
7. Rossman, L. et al., (2000) EPANET 2.2 User Manual. U.S. Environmental Protection Agency

Title	**AI/ML in Civil Engineering**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Mathematics I, Mathematics II		
Antirequisite	Any department / academic unit specific AI/ML course		

Objectives:

1. Comprehensive understanding of the problem settings and application of AI/ML in civil and infrastructure engineering.
2. Development of programming skills to execute AI/ML algorithms in civil and infrastructure engineering.
3. The ability to bridge the gap between traditional engineering physics and data-driven discovery.

Learning outcomes:

By the end of this course, students will have the ability to:

1. Analyze engineering problems within both deterministic and random (uncertain) environments.
2. Apply modern architectures (CNNs, Transformers, GNNs) to address challenges such as structural damage assessment, traffic forecasting, and uncertainty quantification.
3. Implement end-to-end ML pipelines, from raw sensor data to deployable infrastructure models.

Course Contents:

Foundations and Information Theory: Review of AI/ML: Fundamentals of data-driven infrastructure engineering; Machine Learning Paradigms: Supervised, unsupervised, and reinforcement, Bias-Variance Tradeoff, probability and information theory basics, including Shannon entropy and KL divergence; modern optimization algorithms, including stochastic gradient descent and Adam; support vector machines and feature embedding spaces; Universal Approximation Theorem; autograd and backpropagation. [12 Lectures]

Vision, Representation, and Sequential Intelligence: Autoencoder; Stacked Autoencoder; Computer vision in infrastructure: CNN architectures, Pooling Layer, Transfer Learning; image classification and semantic segmentation for automated damage assessment; sequential sensor data and time series: Recurrent Neural Networks and LSTMs; attention mechanisms; Input Embedding and Transformer architecture; Transformers for real-time structural health monitoring and traffic flow forecasting, Introduction to Graph Neural Network. [15 Lectures]

Surrogate Modeling and Active Learning: Introduction to surrogate modeling; dimensionality reduction of complex physical fields; Gaussian processes and neural

network surrogates. Physics-aware ML: Brief conceptual introduction to Physics-Informed Neural Networks (PINNs) and embedding governing differential equations into AI loss functions. Active learning: Bayesian optimization, uncertainty quantification, and optimal data sampling strategies for data-scarce engineering environments, Recent advances in the field of AI relevant to Infrastructure Engineering (Sensor Network and Edge Computing, Agentic AI in Infrastructure Engineering) [12 Lectures]

Text Books:

1. J.-A. Goulet (2020) Probabilistic Machine Learning for Civil Engineers, The MIT Press.
2. Goodfellow, I., Bengio, Y., and Courville, A. (2016) Deep Learning, MIT Press.
3. S. Shalev-Shwartz, and S. Ben-David (2014) Understanding Machine Learning From Theory to Algorithms, Cambridge University Press. ISBN 9781107057135.

Reference Books:

1. R. S. Sutton and A. G. Barto (2018) Reinforcement Learning: An Introduction (2nd Edition), MIT Press.
2. A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola (2023) Dive into Deep Learning, Cambridge University Press.
3. Y. Singh et al. (2023). *Machine Learning Applications in Civil Engineering*, Elsevier.
4. E. Stevens, L. Antiga, and T. Viehmann (2020) Deep Learning with PyTorch, Manning Publications.
5. Brunton, S. L., & Kutz, J. N. (2019). *Data-driven science and engineering: Machine learning, dynamical systems, and control*. Cambridge University Press.

Online Course Material:

1. Raissi, M., Perdikaris, P. and Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, pp. 686-707. https://doi.org/10.1016/j.jcp.2018.10.045
2. Torralba, A., Isola, P., & Freeman, W., Foundations of Computer Vision, The MIT Press, https://visionbook.mit.edu/

Title	**Advanced Structural Analysis**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Structural Analysis		

Objectives:

This course aims at:

1. Introduction to advanced concepts of structural analysis and review the concepts of solid mechanics.
2. Further development of analytical skills to compute quantities like displacements, stresses etc. in different structures subjected to static forces.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
4. Gaining the basic understanding of structural stability.
5. Problem solving ability for static loads in structural engineering.

Learning Outcomes:

Students will have the ability to:

1. Analyze statically determinate and indeterminate trusses, beams and frames using matrix formulation.
2. Understand the fundamentals of the matrix formulation of structural analysis through finite element method.
3. Analyze cables, suspension bridges with three hinged and two hinged stiffening girders and statically determinate and indeterminate arches.
4. Determine bending moment, shear forces and axial force in the frames subjected to lateral and vertical loads using approximate method.
5. Perform comprehensive methods of structural analysis on real-world problems of practical interest.
6. Model real life structures for solving them in standard software and examine both local and global stability of the structural system.

Course Contents:

Indeterminate structure: Static and kinematic indeterminacy; Force method of analysis of indeterminate structures: Fixed and continuous beams, frames etc. - analysis by consistent deformation method; application of moment area and conjugate beam methods for fixed beams; theorem of three moments for continuous beams. [7 Lectures]

Displacement method of analysis of indeterminate structures: Slope deflection method: no chord rotation and with chord rotation (derivation); analysis of continuous beams; beams with overhang; analysis of rigid frames; frames with sloping legs; settlement effects; moment distribution method. [5 Lectures]

Matrix Method of Analysis: Flexibility matrix method of analysis; stiffness matrix method of analysis; stiffness matrix for a beam-column element; direct stiffness approach; transformation; static condensation; rigid end offset. [6 Lectures]

Introduction to FEM: Energy methods and variational principles; shape functions; use of virtual work method for generating stiffness matrix of structure. [4 Lectures]

Stability of structure: Concept of structural stability; classical stability analysis of beams and frames; energy method of the stability analysis of beams, frames and plates. [6 Lectures]

Plastic analysis: Introduction to plastic theory; concept of plastic hinge; plastic modulus; shape factor; redistribution of moments; collapse mechanism; plastic analysis of beams and portal frames by equilibrium and mechanism methods; moment-curvature for arbitrary composite section. [5 Lectures]

Introduction to structural dynamics: Basic concepts of dynamic loading and response; types of mass-spring damper system; equation of motion using direct equilibration, Hamilton's principle. [2 Lectures]
Single-degree of freedom system: Development of equation of motion; statement of problem and solution; damped and undamped systems; free vibration response. [4 Lectures]

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
3. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
4. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.
5. J.B. Wilbur, C.H. Norris, S. Utku (2006) Elementary Structural Analysis, McGraw Hill, New York. ISBN 9781258242909.

Online Course Material:

1. Advanced Structural Analysis (2012) by Prof. Devdas Menon on NPTEL. https://nptel.ac.in/courses/105/106/105106050/

Title	**Civil Engineering Laboratory – I**	Number	CIP2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-4 [2]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite / Requisite	Building Materials and Construction Technology; Geoinformatics		

Building Materials and Construction Technology Laboratory

Objectives:
To familiarize the students with testing of building materials

Learning Outcome:
Following this course, the student will have the ability to:

1. Assess and interpret the standard materials testing data as per the relevant standards
2. Prepare reports and communicate the materials testing reports
3. Recommend the right building materials based on the material testing reports

List of Experiments:
Cements

1. Specific Gravity, Consistency, Initial and Final Setting Time
2. Blaine's air permeability test, and Sieve Analysis
3. Mortar Strength Test

Aggregates

4. Specific gravity, bulk density and water absorption of coarse and fine aggregates
5. Sieve analysis of fine and coarse aggregates, and Flakiness and elongation index tests
6. Aggregate crushing strength tests

Concrete

7. Workability Tests on Concrete – Slump, Compaction Factor, and Vebe Tests
8. Mechanical Properties of Concrete – 7 and 28 Days compressive/split tensile/flexural strengths, and Modulus of Elasticity of Concrete
9. Self Compacting Concrete - Slump flow, L-Box, V-funnel and J-Ring tests

Rebar

10. Tests on Rebars - tension, stress-strain, and pullout from concrete for the bond properties

Miscellaneous

11. Bricks – Dimensions check, Compressive strength, water absorption,

importance of frog

Text Books:

1. P.C. Varghese (2008) Building Materials, Prentice-Hall of India, New Delhi.
2. M. Soutsos and P. Domone (2018) Construction Materials, Their Nature and Behavior, Fifth Edition, CRC Press.

Reference Books:

1. Relevant BIS standards

Geoinformatics Laboratory

Objectives:
This course aims at:

1. To introduce the basic principles of remote sensing, GPS, Surveying and GIS.
2. To teach application of geo-informatics in geoscience, ecology and environment study, agriculture, and disaster management.

Learning Outcomes:
A graduate of this course will have a thorough idea of:

1. Multispectral, thermal infrared, and microwave remote sensing.
2. Surveying Principles and handling modern surveying equipment
3. Different map projections and use of GPS.
4. Component GIS and applications.
5. Different cartographic models, implementation, validation, and visualization.

List of laboratory exercises:

1. Establishment of Local Control Network and Vertical Profiling using modern surveying Instruments (Total Station or Theodolite, Digital level and EDM).
2. Field data collection using handheld and Differential GPS.
3. Georeferencing when coordinate values are given/not given in a GIS platform.
4. Creating maps using vector data models and cartographic guidelines.
5. Working with Digital Elevation Model (DEM) and terrain analysis.
6. Drone data acquisition using Multi spectral sensor and Lidar sensor. Image data processing and generation of ortho-rectified Images (ORI).
7. Site suitability analysis for service buildings using vector and raster data.
8. Procurement and preparation of land use and land cover map using satellite images for a specified study area.
9. Developing DEM and CHM using LIDAR point cloud data.
10. Network analysis to locate suitable routes.
11. Creating a Web app serving a GIS model (created using model builder) in a GIS platform.

Note: Lab. will have hands-on use of Google Earth Engine, QGIS, ERDAS IMAGINE, ArcGIS, or other relevant software (any one or two tools will suffice).

A surveying excursion for 1 week (or less period) may be planned for nearby places to make students familiar with practical experience in mapping, and using surveying equipment in real-world environments.

Text Books:

1. Satheesh Gopi, R. Sathikumar, and N. Madhu (2025) Advanced Surveying: Total Station, GPS, GIS & Remote Sensing (2nd Ed). Pearson India.
2. T. M. Lillesand, R. W. Kiefer, J. W. Chipman (2004) Remote Sensing and Image Interpretation (5th Ed). Wiley. ISBN: 0-471-15227-7
3. G. S. Srivastava (2014) An Introduction to Geoinformatics. McGraw Hill Education (India) Private Limited. New Delhi. India. ISBN: 9781259058462.
4. Michael N. DeMers (2008) Fundamentals of Geographic Information Systems (4th Ed). Wiley. ISBN: 978-0-470-12906-7.

Reference Books:

1. Basudeb Bhatta (2021) Remote Sensing and GIS (3rd Ed). Oxford University Press.
2. L. E. Johnson (2009) Geographic Information Systems in Water Resources Engineering. CRC Press. Taylor & Francis Group. ISBN 978-1-4200-6913-6.
3. B. Tomaszewski (2020) Geographic Information Systems (GIS) for Disaster Management (2nd Ed). CRC Press.

Online Course Material:

1. Prof. Arun K. Saraf (2016) Introduction to Geographic Information System. NPTEL. https://nptel.ac.in/courses/105/107/105107155/
2. Prof. Arun K. Saraf (2017) Introduction to Remote Sensing. NPTEL. https://nptel.ac.in/courses/121/107/121107009/

Title	**Civil Engineering Laboratory – II**	Number	CIP2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Hydraulic Engineering		
Requisite	Structural Analysis; Soil Mechanics		

Structural Analysis Laboratory

Objectives:
This course aims at:

1. Introduction to fundamental concepts of structural analysis.
2. Development of analytical skills to compute quantities like displacements, stresses etc. in structures
3. subjected to static forces.
4. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
5. Problem solving ability for static loads in Civil Engineering design of structures.

Learning Outcomes:
Students will have the ability to:

1. Establish equilibrium equations of structural systems.
2. Analyze the responses and stresses generated in structures subjected to different types of static loading.
3. Apply the structural analysis methods to solve real-world problems of the building, bridges etc.

List of Experiments:

1. Verification of moment area theorem for the slope and deflection of beam.
2. Verification of Clerk-Maxwell's reciprocal theorem.
3. Experimentally determining the elastic properties of an elastic beam.
4. Experimental and analytical study of defections in a cantilever beam.
5. To study the behaviour of struts and columns with various support end conditions.
6. Determination of the deflection of a pin connected truss through analytical, graphically and experimental calculations.
7. Determine the horizontal thrust in a three hinged arch for a given system of loads experimentally and verify it with theoretical calculations.
8. Experimental and analytical study of sway in portal frame.

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw

Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata
3. McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
4. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
5. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.

Online Course Material:

1. Structural Analysis I (2009) by Prof. Siddhartha Ghosh and Prof. R.S. Jangid on NPTEL. https://nptel.ac.in/courses/105/101/105101085/

Hydraulic Engineering Laboratory

Objectives:
The instructor will

1. Provide hands-on experience with the fundamental theoretical principles of fluid mechanics
2. Introduce various forms of flow measuring devices and techniques

Learning outcomes:
Upon completion of this laboratory course, students will be able to:

1. verify fundamental theoretical principles of fluid mechanics through hands-on experimental measurement
2. calibrate and operate various instruments and assess their accuracy, key hydraulic parameters and suitability for different flow conditions
3. compare experimental results with theoretical principles, and draw evidence-based engineering conclusions

Contents:

1. Observation of streamline formation and flow behaviour in a transparent flow channel
2. Determination of metacentric height of floating bodies
3. Determination of the Reynolds number for fluid flow through a pipe and thereby identify the nature of the flow regime
4. Verification of Bernoulli's equation experimentally and calculation of the total energy at different points
5. Determination of the coefficient of discharge of a venturi meter
6. Determination of the coefficient of discharge of an orifice meter
7. Study the impact of a jet on different types of vanes:
8. Hemispherical Vane
9. Flat Plate Vane
10. Conical vane

11. Study and compare the water surface profile of a forced vortex
12. Determination of head loss in pipes having different diameters, different materials and different roughness
13. Determination of pressure drop at different types of bends and valves
14. Determination of the friction factor for laminar and turbulent flow through pipes
15. Determine the coefficient of discharge for weirs and notches (rectangular, triangular and trapezoidal)

Text Books:

1. Fox, R. W., Pitchard, P. J., and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
2. White, F. M. (2003), Fluid Mechanics, 5th Edition, McGraw-Hill.
3. Cimbala, J.M. and Cengel, Y.A., (2010), Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

Soil Mechanics Laboratory

Objective:
This course aims at:

1. Familiarizing students with the physical and engineering properties of soil.
2. Introducing the laboratory testing equipment and their functioning principles.
3. Introducing basic field soil testing techniques.

Learning outcomes: Students will be able to:

1. Classify the soils roughly based on visual identification
2. Determine the index and engineering properties of soils
3. Use simple and complex field and laboratory testing techniques to determine the shear strength of soils

Course Contents:

1. Visual classification of soils
2. Specific gravity and moisture content determination
3. Dry and wet sieve analysis of soils
4. Atterberg limits: liquid limit; plastic limit and shrinkage limit
5. Permeability of soils through constant head and falling head tests
6. Field unit weight through sand replacement and core cutter tests
7. Laboratory compaction: light weight and heavy weight
8. Consolidation of soil
9. Unconfined compression and CBR test
10. Triaxial and direct shear test

Text Books:

1. Das B.M. (2009). Soil mechanics laboratory manual (7th Ed). OUP USA International Edition. ISBN 10-0195367596.
2. Fratta D. Aguettant J. Smith L.R. (2007). Introduction to Soil Mechanics Laboratory Testing. CRC Press-Taylor & Francis Group (1st Ed). ISBN-10: 1138430242
3. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.

Online Course Materials:

1. NPTEL Soil Mechanics/Geotechnical Engineering-I, IIT Kharagpur, https://nptel.ac.in/courses/105105168
2. MIT Open Course Ware, Advanced Soil Mechanics, https://ocw.mit.edu/courses/1-361-advanced-soil-mechanics-fall-2004/pages/lecture-notes/

Title	**Civil Engineering Laboratory – III**	Number	CIP3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite / Requisite	Transportation Engineering – I; Water and Wastewater Engineering; Reinforced Concrete Design		

Transportation Engineering Laboratory

Objectives:
The instructor will:

1. Introduce students to laboratory testing procedures related to highway construction materials and pavement engineering.
2. Provide hands-on experience in evaluating the engineering properties of aggregates and bituminous materials used in transportation infrastructure.

Learning Outcomes:
The students will have the ability to:

1. Perform standard laboratory tests on aggregates and bituminous materials as per relevant IS/IRC specifications.
2. Determine engineering properties of pavement materials and assess their suitability for highway construction.

Contents:

1. 10% fine value test
2. Aggregate Impact Test
3. Los Angles Abrasion Test
4. Specific Gravity and Water Absorption Test on Aggregates
5. Shape Tests
6. Penetration Test and Softening Point Test
7. Viscosity Test
8. Ductility Test
9. Flash & Fire Test
10. Specific Gravity Tests on Bitumen
11. Marshall Stability Test

Text Books:

1. Khanna, S.K., Justo, C.E.G. and A Veeraragavan, "Highway Material and Pavement Testing Manual", Nem Chand & Bros
2. IRC and IS Codes related to Highway Materials Testing

Environmental Engineering Laboratory

Objectives:
The Instructor will:

1. Provide knowledge on water quality standard.
2. Exposure to wastewater disposal guideline and standard.
3. Provide hands on experience on water quality testing and wastewater characterizations

Learning Outcomes:
The students are expected to be able to:

1. Learn to test water quality parameters
2. Learn to characterize wastewater
3. Assess the quality with respect to standard guideline for suiatble decision making.

Laboratory Components:

1. Introduction to environmental engineering laboratory
2. Determination of alkalinity in water
3. Determination of hardness in water
4. Determination of TDS and TSS : Gravimetric method
5. Determination of pH, TDS or EC and turbidity
6. Determination of optimal coagulant dosing: Jar test
7. Determination of COD in wastewater
8. Determination of BOD in wastewater
9. Coliform test: demonstration and analysis
10. Air quality laboratory: Measurement of PM2.5 and PM10
11. Demonstration of weather monitoring station: Sampling and analysis

Text Books:

1. Metcalf & Eddy, Wastewater Engineering- Treatment and Reuse (Revised by G. Tchobanoglous, F. L. Burton and H.D. Stensel), Tata McGraw Hill, 4th Edn., 2004.H.
2. Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
3. M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Reference Books:

1. Nathanson and Jerry, Basic Environmental Technology: Water Supply, Waste Management and Pollution Control, Prentice Hall India Learning Pvt. Ltd., India 2009
2. Garg, Environmental Engineering: Water Supply Engineering, Khanna

Publishers, 18th Edn, India 2004
3. Manual on Sewerage and Sewage Treatment Systems, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 2013.
4. Manual on Water Supply and Treatment, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 1999.

Online Course Material:
1. Guidelines for drinking-water quality, World Health Organization 2022, ISBN 978-92-4-004506-4, https://www.who.int/publications/i/item/9789240045064
2. Drinking Water – Specification, Bureau of Indian Standard , IS 10500 : 2012, New Delhi, India, 110002

Reinforced Concrete Design Laboratory

Objectives:
This course aims at:
1. Providing an in-depth knowledge on the behaviour of RCC structures.
2. Highlighting the significance of working stress and limit state design methods.
3. Comprehensive development of the design and reinforcement detailing of beams, columns, slabs, staircases, footings.
4. Develop an understanding with respect to the importance of stiffness, strength, ductility and configuration which is useful for earthquake resistant design.
5. Practicing reinforcement detailing procedures of a RCC building considering wind and seismic forces as per Indian Standard codes and developing design skills.

Learning Outcomes:
Students will have the ability to:
1. Assess the behaviour of RCC structures and apply the appropriate design methodology and reinforcement detailing for different types of RCC structural elements i.e. beams, columns, slabs, staircases, foundations etc.
2. Design and provide reinforcement detailing of multistoried RCC buildings considering dead, wind and seismic forces as per Indian Standard codes.

Laboratory Contents:
1. Introduction of computer applications in the structural design.
2. Design of beam (under-reinforced, balanced, over-reinforced, doubly reinforced) using computer applications.
3. Design and detailing of slabs (one-way and two-way), columns, walls,

footing etc.

4. Modelling, analysis and design of a multi-storeyed framed structure.
5. Determination of the compressive strength of concrete.
6. Strength and deflection testing of a simply supported RCC beam.
7. Testing of RCC columns for strength subjected to concentric and eccentric loading conditions.
8. Non-destructive testing – rebound hammer test for concrete, ultrasonic pulse velocity test for concrete, rebar detector test for reinforcement etc.

Text Books:

1. U.S. Pillai, D. Menon (2016) Reinforced Concrete Design (3rd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 9780070141100.
2. S.N. Sinha (2014) Reinforced Concrete Design (3rd Ed) McGraw Hill Education Pvt. Ltd. ISBN 9789351342472.
3. A.K. Jain (2012) Reinforced Concrete - Limit State Design (7th Ed), NemChand Brothers. ISBN 9788185240664 50
4. P.C. Varghese (2010) Limit State Design of Reinforced Concrete (2nd Ed), Prentice Hall of India. ISBN 8120320395.
5. S.K. Mallick, A.K. Gupta (2002) Reinforced Concrete (2nd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 0070473323.

References Books:

1. Design of Reinforced Concrete Structures (2009) by Dr. Nirjhar Dhang on NPTEL. https://nptel.ac.in/courses/105/105/105105105/

Title	**Civil Engineering Laboratory – IV**	Number	CIP3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite / Requisite	Water Resources Engineering; Advanced Structural Analysis; AI/ML in Civil Engineering		

Water Resources Engineering Laboratory

Objectives:
In this course the students will:

1. Learn to describe the fundamental concepts of hydrology
2. Understand various methods of hydrologic analysis
3. Apply various hydrologic and hydraulic analysis methods to design stormwater drains, dams, irrigation and diversion structures

Learning Outcomes :
At the end of the course, the student should be able to:

1. Conceptualize different hydrologic process based on physical laws
2. Apply the concept of flood and reservoir routing in real data and solve the field issues
3. Find solutions for water resources issues such as urban flooding, water management and groundwater depletion.

Irrigation Engineering Design Laboratory:

1. Design of Irrigation Canal [lined and unlined]
2. Design of canal outlets
3. Design of Alluvial channels carrying clear and sediment laden water
4. Design the section of hydraulic jump
5. Design of cross drainage structures with typical plan and section
6. Design of profile of ogee spillway
7. Design of Trapezoidal Notch Fall, SARDA fall and Glacis Fall
8. Design of Cisterns
9. Analysis and Design of gravity dam
10. Analysis and Design of earth embankment dam
11. Problems related to energy dissipaters

Text Books:

1. Mays, L.W. *Water Resources Engineering*, John Wiley and sons, Inc., Kundli
2. Todd, D. K.and Mays, L. W. *Ground Water Hydrology,* John Wiley India Edition

Reference Books:

1. Chow, V.T., Maidment, D.R., and Mays L.W. Applied Hydrology. McGraw-Hill Inc., New York.
2. James, L.D., and Lee, R.R. Economics of water resources planning. McGraw-Hill Book Inc., New York.
3. Garg, S.K. Irrigation Engineering and Hydraulic Structures. Khanna Publishers, Delhi.
4. Linsley, R.K., Franzini, J.B., Freyberg, D.L., and Tchobanoglous, G. Water Resources Engineering, McGraw Hill Inc., Singapore.
5. Singh, V.P. Elementary Hydrology. Prentice Hall, Englewood Cliffs, New Jersey.

Advanced Structural Analysis Laboratory

Objectives:

This course aims at:

1. Introduction to advanced concepts of structural analysis and review the concepts of solid mechanics.
2. Further development of analytical skills to compute quantities like displacements, stresses etc. in different structures subjected to static forces.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
4. Gaining the basic understanding of structural stability.
5. Problem solving ability for static loads in structural engineering.

Learning Outcomes:

Students will have the ability to:

1. Analyze statically determinate and indeterminate trusses, beams and frames using matrix formulation.
2. Understand the fundamentals of the matrix formulation of structural analysis through finite element method.
3. Analyze cables, suspension bridges with three hinged and two hinged stiffening girders and statically determinate and indeterminate arches.
4. Determine bending moment, shear forces and axial force in the frames subjected to lateral and vertical loads using approximate method.
5. Perform comprehensive methods of structural analysis on real-world problems of practical interest.
6. Model real life structures for solving them in standard software and examine both local and global stability of the structural system.

Laboratory Contents:

1. Introduction of computer applications in structural analysis.
2. Create and analyze model to determine the shear force, bending moment,

deflection diagrams for statically indeterminate beams.

3. Create and analyze model to determine the shear force, bending moment, deflection diagrams for portal frames.
4. Create and analyze model to determine the shear force, bending moment, deflection diagrams for multi storeyed building.
5. Determine shear force, bending moment, deflection diagrams for multi-storey frames using approximate methods.
6. Modelling and analysis of 3D structures.

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
3. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
4. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.
5. J.B. Wilbur, C.H. Norris, S. Utku (2006) Elementary Structural Analysis, McGraw Hill, New York. ISBN 9781258242909.

Reference Books:

1. Advanced Structural Analysis (2012) by Prof. Devdas Menon on NPTEL. https://nptel.ac.in/courses/105/106/105106050/

AI/ML in Civil Engineering Laboratory

Objectives:

1. Comprehensive understanding of the problem settings and application of AI/ML in civil and infrastructure engineering.
2. Development of programming skills to execute AI/ML algorithms in civil and infrastructure engineering.
3. The ability to bridge the gap between traditional engineering physics and data-driven discovery.

Learning outcomes:

By the end of this course, students will have the ability to:

1. Analyze engineering problems within both deterministic and random (uncertain) environments.
2. Apply modern architectures (CNNs, Transformers, GNNs) to address challenges such as structural damage assessment, traffic forecasting, and uncertainty quantification.

3. Implement end-to-end ML pipelines, from raw sensor data to deployable infrastructure models.

Course Contents:

Tentative list of laboratory exercises:

1. Exploratory Data Analysis and Data Preprocessing Pipelines
2. Linear Regression and Logistic Regression using a single Perceptron
3. Regularization and performance metrics for Regression and Classification
4. Training Autoencoders on sensor data for unsupervised anomaly detection.
5. Automated Visual Inspection and classification (or image segmentation) using CNNs and Transfer Learning
6. Time Series Forecasting using LSTM & Transformers
7. Segmentation of multispectral remote sensing images
8. Grid Resilience Analysis using Graph Neural Networks
9. Interpreting and Auditing Deep Learning Models using xAI for Structural Safety or other critical systems, such as drinking water supply.
10. Building a fast surrogate model for flood routing, soil settlement under dynamic loading, or beam deflection, and making it Physics-Informed.

Text Books:

1. J.-A. Goulet (2020) Probabilistic Machine Learning for Civil Engineers, The MIT Press.
2. Goodfellow, I., Bengio, Y., and Courville, A. (2016) Deep Learning, MIT Press.
3. S. Shalev-Shwartz, and S. Ben-David (2014) Understanding Machine Learning From Theory to Algorithms, Cambridge University Press. ISBN 9781107057135.

Reference Books:

1. R. S. Sutton and A. G. Barto (2018) Reinforcement Learning: An Introduction (2nd Edition), MIT Press.
2. A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola (2023) Dive into Deep Learning, Cambridge University Press.
3. Y. Singh et al. (2023). *Machine Learning Applications in Civil Engineering*, Elsevier.
4. E. Stevens, L. Antiga, and T. Viehmann (2020) Deep Learning with PyTorch, Manning Publications.
5. C M Bishop and H Bishop, 2024, Deep Learning Foundations and Concepts, Springer Nature, Switzerland, AG 2024.

Online Course Material:

1. Raissi, M., Perdikaris, P. and Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal*

of Computational physics, *378*, pp. 686-707. https://doi.org/10.1016/j.jcp.2018.10.045

2. Torralba, A., Isola, P., & Freeman, W., Foundations of Computer Vision, The MIT Press, https://visionbook.mit.edu/

Title	**Steel Structure Design**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Structural Analysis		

Objectives:

This course aims at:

1. Providing an in-depth knowledge on the behaviour of steel structures.
2. Comprehensive development of the design and associated analysis of critical components such as joints, fasteners etc.
3. Practicing limit state design of structural steel members subjected to compressive, tensile and bending loads, including connections as per Indian Standard codes.

Learning Outcomes:

Students will have the ability to:

1. Design bolted and welded connections for steel structures.
2. Design tension members, understand the effect of shear lag and understand the design concept of axially loaded columns and column base connections.
3. Understand specific problems related to the design of laterally restrained and unrestrained steel beams.
4. Design structural systems such as roof trusses, gantry girders as per provisions of Indian Standard code for working stress and limit state methods.

Course Contents:

Mechanical properties and types of structural steel; Indianstructural steel products; brief on the steps involved in the design process ofsteel structural systems and their elements; type of loads; standards andspecifications; concept of allowable stress method and limit state designmethods for steel structures; relative advantages and limitations; strengths andserviceability. [6 Lectures]

Type of fasteners – rivets, bolts and welds; types of connections;relative advantages and limitations; modes of failure; concept of shear lag; prying forces and hanger connection; design of slip critical connections; designof joints; eccentrically loaded bolted bracket connections; welds; effective areaof welds; fillet and but welded connections; axially loaded connections; eccentrically loaded bracket connections. [9 Lectures]

Types of tension members and sections; behaviour oftension members modes of failure; slenderness ratio; net area; net effectivesections for plates, angles etc. in tension; design of plate and angle tensionmembers; design of built up tension

members; connections in tensionmembers; use of lug angles; design of tension splice. [8 Lectures]

Types of compression members and sections;behaviour and types of failures; short and slender columns; effective length;design of single section and compound angles - axially loaded solid section columns subjected to biaxial bending; design of built up laced and battenedtype columns; design of column bases; plate and gusseted bases for axiallyloaded columns; splices for columns. [7 Lectures]

Types of steel beam sections; behaviour of beams in flexure; classification ofcross sections; flexural strength and lateral stability of beams; shear strength; web buckling, crippling and defection of beams; design of laterally supportedbeams, solid rolled section beams, plated beams with cover plates, purlin in rooftrusses etc. [9 Lectures]

Text Books:

1. N. Subramanian (2015) Design of Steel Structures, Oxford University Press. ISBN 9780199460915.
2. M.L. Gambhir (2013) Fundamentals of Structural Steel Design (1st Ed), McGraw Hill Publication. ISBN 9781259006760.
3. S.K. Duggal (2019) Limit State Design of Steel Structures (3rd Ed), McGraw Hill Publication. ISBN 9789353164874.
4. K.S. Sai Ram (2015) Design of Steel Structures (2nd Ed), Pearson Education. ISBN 9789332542105.
5. M.R. Shiyekar (2017) Limit State Design in Structural Steel (3rd Ed), Prentice Hall of India Pvt. Ltd. ISBN 9788120353503.

References Books:

1. Design of Steel Structures I (2009) by Prof. S.R. Satish Kumar and Prof. A.R. Santha Kumar on NPTEL. https://nptel.ac.in/courses/105/106/105106112/

Title	**Advacned Structural Design**	Number	CIL7750
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	Reinforced Concrete Design; Steel Structure Design		

Objectives:

This course aims at:

1. Providing in-depth advance knowledge on the behaviour of RCC and steel structures.
2. Develop an understanding with respect to design of different structural components such as deep beam, shear wall etc.
3. Practicing detailing procedures and developing design skills of RCC and steel structures using Indian and foreign standard codes.

Learning Outcomes:

Students will have the ability to:

1. Assess the behaviour of RCC and steel structures and apply the appropriate design methodology and reinforcement detailing for different types of RCC and steel structural elements.
2. Design and provide detailing of multistoried RCC and steel buildings considering dead, wind and seismic forces as per Indian and international standard codes.

Course Contents:

Review of basics concepts: Design philosophy, modelling of loads, material characteristics. Durability and mix design of concrete. [5 Lectures]

Advanced reinforced concrete design: P-M, M-phi relationships, strut-and-tie method, design of deep beam and corbel; design using Indian, America and European standards. [9 Lectures]

Design of shear walls, compression field theory for shear design, design against torsion. [9 Lectures]

Advanced design ofsteel structures: Stability design; torsional buckling (pure, flexural and lateral); design using Indian and AISC Standards; Eurocode. [9 Lectures]

Design of beam-columns; Fatigue resistant design. [7 Lectures]

Text Books:

1. P.C. Varghese (2005) Advanced Reinforced Concrete Design (2nd Ed), Prentice Hall of India
2. C.-K. Wang, C.H. Solomon, J. A. Pincheira (2007) Reinforced Concrete Design (7th Ed), John Wiley and Sons.
3. J.G. MacGregor, J.K. Wight (2008) Reinforced Concrete: Mechanics and Design (5th Ed), Pearson Education.
4. U.S. Pillai, D. Menon (2016) Reinforced Concrete Design (3rd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 9780070141100.
5. N. Subramanian (2015) Design of Steel Structures, Oxford University Press. ISBN 9780199460915.

Online Course Material:

1. Design of Reinforced Concrete Structures (2009) by Dr. Nirjhar Dhang on NPTEL. https://nptel.ac.in/courses/105/105/105105105/
2. Design of Steel Structures I (2009) by Prof. S.R. Satish Kumar and Prof. A.R. Santha Kumar on NPTEL. https://nptel.ac.in/courses/105/106/105106112/

Title	**Dynamics of Structure**	Number	CIL4350
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	Mechanics of Solids		

Objectives:

This course aims at:

1. Introducing the behaviour of structures (especially framed building) to various dynamic loads, *viz.* earthquake, machine vibration etc.
2. Fundamental understanding to analyze structures subjected to dynamic excitation and computing quantities like displacements, forces, stresses etc.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
4. Problem solving ability for dynamic response in Civil Engineering design.

Learning Outcomes:

Students will have the ability to:

1. Establish dynamic equilibrium and equation of motion for structural systems.
2. Analyze the dynamic response and stress generation in single and multi degree of freedom systems for different types of loading.
3. Implement the basis for the classical energy formulations in dynamical systems.
4. Model structural damping and solve problems in frequency as well as time domain.
5. Solve the eigenvalue problems and gain knowledge on its useful properties.
6. Device methods to protect new and existing structures against seismic collapse.
7. Apply the structural dynamics theory to real-world problems like seismic analysis and design of structures.

Course Contents:

Introduction to dynamics of structure: Basic concepts of dynamic loading and response; degrees of freedom; types of mass-spring-damper system; equation of motion using direct equilibration, principle of virtual displacement and Hamilton's principle. Single-degree of freedom system: Development of equation of motion; statement of problem and solution; damped and undamped systems; free vibration response. [8 Lectures]

Solution of different loading cases: Responses to harmonic, periodic, arbitrary, pulse etc. excitations; numerical evaluation; generalized system. [8 Lectures]

Two-degree of freedom system: Development of equation of motion; statement of

problem and solution methods; free vibration response; damping; natural vibration frequencies and modes; computation of vibration properties. [9 Lectures]

Multiple degree of freedom system: Solving problems with more than two degree of freedom systems; static condensation.
Damping in structures: Estimating modal damping ratios; construction of damping matrix. [9 Lectures]

Introduction to Earthquake response: Response of linear elastic building subjected to seismic excitation; basic concepts of vibration control. [5 Lectures]

Text Books:

1. J.M. Biggs (1964) Introduction to Structural Dynamics, McGraw Hill, NY. ISBN 07-005255-7.
2. R.W. Clough and J. Penzien (1993) Dynamics of Structures (2nd Ed), McGraw Hill Inc Editor. ISBN 0-07-011392-0.
3. A.K. Chopra (2017) Dynamics of Structures Theory and Application to Earthquake Engineering (5th Ed), Prentice Hall, NJ. ISBN 978-0-134-55512-6.
4. A.K. Jain (2017) Dynamics of Structures with MATLAB® Applications, Pearson Education, Chennai, India. ISBN 978-9-332-55855-7
5. M. Paz and Y.H. Kim (2019) Structural Dynamics Theory and Computation (6th Ed), Springer Nature, Switzerland. ISBN 978-3-319-94742-6.

Online Course Material:

1. Structural Dynamics (2012) by Dr. Pradipta Banerji on NPTEL. https://nptel.ac.in/courses/105/101/105101006/

Title	**Finite Element Method**	Number	CIL7630
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

This course aims at:

1. Introduction to fundamental concepts of finite element analysis.
2. Development of analytical skills to compute quantities like displacements, stresses etc. in structures subjected to static and dynamic forces.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.

Learning Outcomes:

Students will have the ability to:

1. Establish equilibrium equation of structural systems.
2. Analyze the responses and stresses generated in structures subjected to different types of static loading.
3. Apply the FEM to solve real-world problems of the building, bridges etc.

Course Contents:

Fundamental concepts: Introduction to FEM; stresses and equilibrium; boundary conditions; strain-displacement and stress-strain relations; governing equation and its solution approximations (e.g. Collocation, Least Squares, Galerkin's method, the Ritz method). [6 Lectures]

Introduction to calculus of variations: Functionals; extremization; variational form from differential equation; principle of virtual work and minimum potential energy.
Concept of discretization of structures and shape functions; Lagrangian and serendipity elements; isoparametric formulation. [9 Lectures]

Analysis of framed structures: plane stress and plane strain problems; axisymmetric problems; 3D stress analysis. [9 Lectures]

Analysis of plate and shell.
Numerical integration: Numerical integration formulae; order of integration; error analysis and convergence. [8 Lectures]

Programming applications: Computer implementations of algorithms.
Application of FEM in dynamics: Eigenvalues and orthogonality. [7 Lectures]

Text Books:

1. J.N. Reddy (2003) An Introduction to the Finite Element Method (2nd Ed), Tata McGraw Hill.
2. C.S. Krishnamoorthy (1994) Finite Elements Analysis: Theory and Programming (2nd Ed), Tata McGraw Hill.
3. R.D. Cook, D.S. Malkus, M.E. Plesha (2002) Concepts and Applications of Finite Element Analysis (4th Ed), John Wiley & Sons.
4. O.C. Zienkiewicz, R.L. Taylor, J.Z. Zhu (2005) Finite Element Method Its Basis and Fundamentals (6th Ed), Elsevier

Reference Books:

1. S.S. Rao (1999) Finite Element Method in Engineering (3rd Ed), Butterworth Heinemann.
2. K.J. Bathe (2002) Finite Element Procedures, Prentice Hall of India Pvt. Ltd.

Title	**Prestressed Concrete Structures**	Number	CIL7670
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
This course aims at:

1. Fundamental understanding of the behaviour of prestressed concrete structures.
2. Developing skills for analysis and design of prestressed sections in beams, columns, bridges etc.

Learning Outcomes:
Students will have the ability to:

1. Assess the effect of losses and forces such as axial, flexure, shear and torsion in prestressing.
2. Design the beams, columns, bridge decks etc. based on prestressed concrete..

Course Contents:

Introduction to prestressing system: Basics of prestressed concrete structures and advantages; types of prestressing; pre- and post tensioning systems and devices; review of concrete-properties; prestressing steel– forms, types and properties. [8 Lectures]

Losses in prestress: Elastic shortening; friction; anchorage slip; force variation; creep; shrinkage.
Analysis of members: Analysis under axial load- permissible prestress based on transfer, stresses under service loads, ultimate strength of member; Analysis of members under flexure– at transfer and service, calculation of cracking moment, location of kern points and pressure lines, ultimate strength. [8 Lectures]

Design of member sections: For axial tension; flexure– type 1 and type 2; choice of section; limiting zone; Magnel's graphical method; tendon profile and reinforcement detailing.
Analysis and design for shear and torsion: Cracks and modes of failure due to shear and torsion; effect of prestressing force; limit state of collapse for shear and torsion; design of longitudinal and transverse reinforcement. (13 Lectures)

Design of member sections: For axial tension; flexure – type 1 and type 2; choice of section; limiting zone; Magnel's graphical method; tendon profile and reinforcement detailing.
Analysis and design for shear and torsion: Cracks and modes of failure due to shear and torsion; effect of prestressing force; limit state of collapse for shear and torsion; design of longitudinal and transverse reinforcement. [10 Lectures]

Text Books:

1. C.W. Dolan, H.R. Hamilton (2019) Prestressed Concrete Building, Design, and Construction, Springer. ISBN9783319978819.
2. N. Krishna Raju (2006) Prestressed Concrete (3rd Ed), TataMcGraw-Hill Publishing Company Ltd. ISBN0070634440.
3. E.G. Nawy (2005) Prestressed Concrete – A Fundamental Approach (5th Ed), Pearson. ISBN9780131497597.

Online Course Material:

1. Prestressed Concrete Structures (2009) by Prof. Amlan K. Sengupta and Prof. Devdas Menon on NPTEL. https://nptel.ac.in/courses/105/106/105106117/

Title	**Highway Geometric Design**	Number	CIL4390
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

The course will help students to understand the concepts involved in geometric design of intersections, horizontal & vertical alignment of roads & pedestrian facilities**.**

Learning Outcomes:

At the end of the course, the student should be able to:

1. Describe the urban street hierarchy and functional classification system.
2. Demonstrate the relationship between mobility and access and formulate a functional design process that accommodate the needs of all users and allows for street designs that are compatible with the surrounding area.
3. Identify and define the elements of a roadway cross-section.
4. Design alignment and grade elements including sight distance; horizontal and vertical curves; and terrain and acceptance grades for urban local and collector streets.
5. Define the functional area of an intersection and identify key design elements for intersections.
6. Identify pedestrian street crossing issues.

Course Contents:

Highway Cross Section Elements and Geometric Design: Functional Classification of Highway System; Design Control characteristics, Objectives and elements of Geometric Design, Pavement Surface characteristics, Camber, Objectives of Camber, design standards. [8 Lectures]

Horizontal and Vertical Alignment: Objectives of horizontal curves, Super elevation, Minimum Radius of Curve, Extra widening on Curves, Transition Curves, Types of Gradients, Vertical Curves – Types and Design, Grade Compensation, Sight Distances, Importance of Sight Distances for Horizontal and Vertical Curves. [12 Lectures]

Intersection Design: Types of Intersections; Design Principles for Intersections; Design of At-grade Intersections – Channelization, Objective; Traffic Islands and Design standards; Rotary Intersection – Concept and Design, Advantages and Disadvantages; Grade separated Interchanges – Types, warrants and Design standards. [7 Lectures]

Traffic signs and road markings: Types of Road Signs; Guidelines for the provision of Road Signs; Cautionary Signs, Regulatory Signs, Information Signs – Design standards; Road markings – Objective of Road Markings; Types of Road Markings; Role of Road markings in Road Safety and Traffic Regulation; Specification for Road Markings. Highway Appurtenances – Delineators, Traffic Impact Attenuators, Safety Barriers. [5 Lectures]

Miscellaneous Elements: Requirements of Pedestrians; Pedestrian facilities on Urban Roads; Cycle Tracks – Guidelines and Design standards; Bus bays – Types and Guide lines; Design of On-street and Off-street Parking facilities – Guidelines for lay out Design. [7 Lectures]

Text Books:

1. L.R. Kadiyali and N.B. Lal, Principles and Practice of Highway Engineering, Khanna Publications
2. C.E.G. Justo and S.K. Khanna, Highway Engineering, Nem Chand and Brothers.

Online Course Material:

1. IRC Codes for Signs, Markings and Mixed Traffic Control in Urban Areas.
2. Traffic Engineering and Transportation Planning, L.R.Kadiyali, Khanna Publications.

Online Course Material:

1. Geometric Design of Highways by Prof. Rajat Rastogi. NPTEL. https://nptel.ac.in/courses/105107220

Title	**Traffic System and ITS**	Number	CIL4360
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
The course will help students to understand the working and control of traffic engineering elements and its integration with advanced technologies such as ITS.

Learning Outcomes:
At the end of the course, the student should be able to:

1. Analyse the road traffic characteristics and can undertake various types of road trafficstudies
2. Suggest and design appropriate control measures for traffic congestion management
3. Understand the role of ANN and fuzzy logic in traffic control measures
4. Understand the fundamentals and taxonomy of Intelligent Transport System (ITS) andITS user services
5. Understand the integration of ITS to transportation planning
6. Have knowledge on various public transportation systems.

Course Contents:
Traffic system characteristics: Road, User and Vehicle characteristic Traffic studies: Volume studies, Origin-Destination Studies; Speed studies, Parking studies, Accident studies. [9 Lectures]

Traffic control devices: Definition, necessity, types; Road signs and markings definition, classification. Traffic signals - location, types, signal design by IRC method, Websters methods and HCM method, ANN and Fuzzy logic concepts in traffic control. [11 Lectures]

Intelligent Transport System [ITS]: Introduction, ITS user services - Travel and traffic management, Public Transport operations, Electronic payment, Commercial vehicle operation, Advanced vehicle control and safety system, Emergency management, Information management, Maintenance and construction management; ITS architecture and planning; Integration of ITS into transportation planning [14 Lectures]

Public Transportation System: Evolution of public transport system, Need and benefits; Classifications; Rapid Transit systems- BRTS, MRTS, Intermediate Public Transport; Public Transport system in India [5 Lectures]

Text Books:

1. Nicholas J Garber and Lester A Hoel. Traffic and Highway Engineering. Copyright: Brooks/Cole-Wadsworth Group\Thomson learning
2. Fred. L. Mannering, Walter P. Kilareski, and Scott S. Washburn. Principles of Highway Engineeringand Traffic Analysis. Copyright: John Wiley and Sons, Inc.
3. M A Chowdhary and A Sadek. Fundamentals of Intelligent Transportation systems planning. ArtechHouse Inc., US, 2003.

Reference Books:

1. Bob Williams. Intelligent transportation systems standards. Artech House, London, 2008
2. Sussman, J. Perspectives on Intelligent Transportation Systems (ITS). New York, NY: Springer, 2005
3. Daganzo, C. F. and Quyang, Y. Public Transportation Systems: Principles of system design, operations planning and real-time control. World Scientific Publishers.

Online Course Material:

1. Intelligent Transportation System – II, Lecture Notes in Transportation Systems Engineering by Prof. Tom V. Mathew
   https://www.civil.iitb.ac.in/tvm/nptel/592_ITS_2/web/web.html

Title	**Urban Transportation Planning**	Number	CIL4410
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
The course will provide students in-depth understanding on designing, conducting, and administering surveys to provide the data required for transportation planning and will make them understand travel demand modelling procedures.

Learning Outcomes:.
At the end of the course, the student should be able to:

1. Design, conduct, and administer surveys to provide the data required for transportation planning
2. Develop and calibrate trip generation rates for specific types of land use developments.
3. Estimate the traffic impact of new developments using the four-stage sequential models.
4. Assess the impact of the transportation project on the land use

Course Contents:
Urban Transportation Planning Process & Concepts:
Role of Transportation and Changing Concerns of Society in Transportation Planning; Transportation Problems and Problem Domain; Objectives and Constraints; Flow Chart for Transportation Planning Process- Inventory, Model Building, Forecasting and Evaluation Stages, Planning in System Engineering Framework; Concept of Travel Demand and its Modelling based on Consumer Behaviour of Travel Choices- Independent Variables, Travel Attributes. [8 Lectures]

Travel demand estimation: Assumptions in Demand Estimation- Sequential, Recursive and Simultaneous Process - Introduction to Transportation Planning Practices; Definition of Study Area, Zoning. Trip Generation Analysis: Trip Generation Models- Zonal Models, Category analysis, Household Models, Trip Attractions of Work Centres & Commercial Trips Trip Distribution Analysis: Trip End and Trip Interchange Models; Trip Distribution Models - Growth Factor Models, Gravity Models, Opportunity Models and their calibration; Estimation of Travel Demand based on link volume philosophy. [12 Lectures]

Mode Split and Route Split analysis:Mode Split Analysis- Mode Choice Behaviour, Competing Modes, Mode Split Curves, Probabilistic Models and Two Stage Mode Split Analysis; Route Split Analysis- Elements of Transportation Networks, Coding, Minimum Path Tress, Diversion Curves, All-or-Nothing Assignment, Capacity Restrained Assignment, Multipath Assignment. [12 Lectures]

Landuse-Transportation Models: Location models - Opportunity Models, Lowry based Landuse-Transportation Models Allocation Function, Constraints, Travel Demand Estimation – Iterative Solutions, Matrix Formulation, Dynamic and Disaggregated extensions. [7 Lectures]

Text Books:

1. Hutchinson, B.G., Principles of Urban Transportation SystemPlanning, Mc-Graw Hill
2. Khisty, C J., Transportation Engineering – An Introduction, Prentice-Hall, NJ
3. Kanafani, A., Transportation Demand Analysis, McGraw-Hill.

Reference Books:

1. Wilson, A.G, Urban and Regional Models in Geography and Planning, John Wiley and Sons.
2. Dickey, J.W., Metropolitan Transportation Planning, Tata Mc-Graw Hill 1980
3. ITE (1982), 'Transportation and Traffic Engineering Hand Book', Chapters 10,12, and 17, Prentice Hall, New Jersey
4. Oppenheim, N., Applied Models in Urban and Regional Analysis, Prentice-Hall, NJ.
5. Dickey J.W., et. al., Metropolitan Transportation Planning, Tata McGraw-Hill.
6. Gallion A.B and Eisner S., The Urban Pattern, Affiliated East-West Press, New Delhi.

Online Course Material:

1. Urban Transportation Planning by Prof. V. Thamizh Arasan. NPTEL. https://nptel.ac.in/courses/105106058

Title	**Ground Improvement Techniques**	Number	CIL4380
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
This course aims at :

1. Introducing the need of ground improvement and available techniques.
2. Introducing the aspects of soil shear strength modification for effective design of foundations.
3. Introducing recent soil modification techniques.

Learning Outcomes:
Students will have the ability to

1. Apply the ground modification techniques for efficient design of foundations
2. Utilize the concepts of preloading and dewatering in soils of cohesive nature
3. Stabilize the soils using different mechanical and chemical approaches

Course Contents:
Classical examples of soil failure, need of ground improvement, In-situ densification methods, In-situ densification of granular and cohesive soils; vibration at ground surface and at depth. [9 Lectures]

Preloading and dewatering, prefabricated vertical drains (PVDs), sand drains, geo-drains, filter requirements, spacing effects, stone columns, sumps and interceptor ditches, single and multi-stage well points, vacuum well points, horizontal weIIs, eIectro-osmosis. [10 Lectures].

Stabilization of soils, methods of soil stabilization- mechanical, cement, Iime, bitumen and polymer stabilization, use of industrial wastes like fly ash and granulated blast furnace slag, stabiIization of expansive soils; stabilization with cement, lime and chemicals, Grouting techniques, types of grouts, grouting equipment and machinery, injection methods, grout monitoring. [10 Lectures]

Geosynthetics, type of geosynthetics, geotextiles, geomembranes and gabions, Reinforced Earth Technology, concept of soil reinforcement; reinforcing materials; backfill criteria; design of reinforcement for internal stability; applications of reinforced earth structures; miscellaneous methods-soil nailing, thermal methods. [10 Lectures]

Text Books:

1. Hausmann, M.R. (1990). Engineering Principles of Ground Modification, McGraw-Hill International Editions. ISBN: 9780070272798.

2. Dr. Purushottama Raj (2005). Ground Improvement Techniques. Laxmi Publications. ISBN: 9788170088370.
3. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.

Online Course Material:

1. Ground Improvement Techniques by Prof. G.L. Sivakumar Babu on NTPEL. https://nptel.ac.in/courses/105/108/105108075/
2. Ground Improvement Techniques by Prof. Dilip Kumar Baidya on NTPEL. https://nptel.ac.in/courses/105105210

Title	**Introduction to Soil Dynamics**	Number	CIL4430
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

This course aims at :

1. Introducing basic concepts of soil dynamics.
2. Understanding the challenges involved in design of foundations subjected to dynamic loads.

Learning Outcomes:

Students will have the ability to:

1. Evaluate the stresses in soils due to dynamic loads
2. Analyze the dynamic behavior of soils through laboratory and field tests
3. Apply the principles of earthquake geotechnology in foundation design

Course Contents:

Nature of dynamic loads, vibration theory, single and two-degree freedom systems, free and forced vibrations, wave propagation in elastic rods, shear and compressional velocities. [6 Lectures]

Behavior of cyclically loaded soils, small and large strain dynamic soil properties, laboratory element tests and field tests, design principles for machine foundations. Elastic homogeneous half-space and lumped parameter solutions, codal recommendations. [10 Lectures]

Earthquake geotechnology, causes and effects of earthquakes, strong ground motion, measurement, characterization and estimation, theory of amplification, ground response analysis, seismic liquefaction and mitigation techniques, seismic hazard assessment, microzonation, earthquake early warning systems. [12 Lectures]

Seismic slope stability, seismic earth pressures on retaining structures, bearing capacity of foundations against seismic loads, introduction to soil-structure and soil-pipe interaction, seismic vibration isolation and damping systems for foundations and structures. [11 Lectures]

Text Books:

1. Kramer S.L. (1996). Geotechnical Earthquake Engineering. Prentice Hall, ISBN-10: 8131707180
2. Ishihara, K. (1996). Soil Behavior in Earthquake Geotechnics, Oxford Science Publications, ISBN-10: 0198562241
3. Prakash S. (1981). Soil dynamics. McGraw-Hill, ISBN-10: 0070506582

Online Course Material:

1. Geotechnical Earthquake Enginering by Prof. Deepankar Choudhury on NPTEL. https://nptel.ac.in/courses/105/101/105101134/

Title	**Advanced Geotechnical Engineering**	Number	CIL7660
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

This course aims at :

1. Introducing basic principles of soil mechanics and foundation design.
2. Understanding the complex response of soils to isotropic and anisotropic loading conditions.

Learning Outcomes:

Students will have the ability to

1. Assess the stress-strain response of soils to practical loading scenario
2. Draw stress paths and utilize them for applications in geotechnical engineering
3. Design shallow and pile foundations based on stress transfer mechanism
4. Assess the stability of retaining walls and slope under different loading conditions.

Course Contents:

Geotechnical lessons learnt from failures-case studies, brief review of soil behaviorunder different loading conditions, normal stresses and strains, volumetric stressesshear stresses and strains, elastic state and anisotropy, total and effective stresses. [6 Lectures]

Stress path, stress and strain invariants, mean and deviatoric stress, axisymmetriccondition, plain strain condition, plotting stress paths, Shear strength of soils, drained and undrained shear tests, stress anisotropy, pore water pressure parameters. Critical state soil mechanics, basic concepts, failure surface, normally consolidatedand over consolidated soils behavior, critical state boundary, elements of criticalstate model, failure stresses from critical state. [10 Lectures]

Shallow foundations, bearing capacity of soils, allowable loads and settlements,bearing capacity formulations, isolated footing design, bearing capacity from fieldload tests, settlement calculations, Pile foundations, types and installation methods,load capacity, pile group capacity, load transfer mechanism, lateral load capacity of pile foundations. Earth-retaining structures, basics of earth pressure theory, modes of failure ofearth retaining structures, stability of rigid and flexible retaining walls, mechanicallystabilized earth walls, Slope stability, slope failure types, two-dimensional slope stability analysis. [11 Lectures]

Structural health monitoring of geotechnical structures – shallow and deep

foundations, retaining walls, tunnels, underground stations, dams and embankments, interpretation of results, mitigation and requalification strategies. [6 Lectures]

Introduction to numerical and analytical modeling in Geomechanics: development of macros in excel, basic MATLAB programs and demo of geotechnical software-PLAXIS 2D and 3D, FLAC 2D and 3D, OASys, GeoSlope, L-Pile, P-disp/X-disp. [6 Lectures]

Text Books:

1. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.
2. Das B.M. (2013). Advanced Soil Mechanics (4th Ed). CRC Press. ISBN-10: 9780415506656
3. Knappett J.A. and Craig R.F. (2012). Craig's Soil Mechanics (8th Ed). CRC Press. ISBN-10: 9780415561266.

Online Course Material:

1. Soil Mechanics course in NPTEL. https://nptel.ac.in/courses/105/103/105103097/

Title	**Mathematical Modelling in Environmental Systems**	Number	CIL7650
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
To provide comprehensive understanding on scope and application of mathematical modelling to understand environmental processes and solve related problems.

Learning Outcomes:
Understanding basic principles of mathematical modelling used in solving complex environmental issues.

Course Content:
Environmental process and systems: Natural and Engineered Systems, Mass Balance, Reaction Kinetics, Phase equilibrium, Steady State, Response time, Micro and Macro Transport Processes. [8 Lectures]

Modelling of reactors: Homogeneous and heterogeneous reactors; Mixed batch reactors; sequencing batch reactor, mixed flow reactors, plug flow reactors. [10 Lectures]

Surface environment systems and Water Quality Modelling: Fundamental of modelling aquatic systems; Lakes, Rivers river and estuaries and lake hydrodynamics; Stratification and eutrophication; Dissolved oxygen model for streams; Temperature models. [14 Lectures]

Air quality modeling: Introduction to air quality models; Meteorology; Atmospheric stability and turbulence; Gaussian plume model and modifications, Dispersion models, Receptor models, Acid Rain Modelling, Introduction to Dispersion and receptor modelling software. [10 Lectures]

Text Books:

1. Weber, W.J. Jr., Process Dynamics in Environmental Systems, John Wiley & Sons Inc. (1996).
2. Thomann, R.V., and Muller, J.A., Principles of Surface Water Quality Modelling and Control, Harper International Edition, N.D. (1987).
3. Steven C.Chapra, Surface Water Quality Modelling, The McGraw-Hill Companies,Inc., New York (2009).
4. Deaton and Wine Brake, “Dynamic Modelling of Environmental Systems”, Wiley & sons (2002).

Title	**Green Building and Sustainable Materials**	Number	CIL7440
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

Objectives of this course are:

1. To introduce elements of green building construction and use of sustainable materials.
2. To familiarize students with different national and international green performance ratings.
3. To teach students methods of carbon footprint reduction and increase in energy efficiency of a building.

Learning Outcomes:

A graduate of this course will have a thorough idea of:

1. National and international green building assessment system and requirements.
2. Green building design elements and use of sustainable materials.
3. Techniques to reduce the carbon footprint of a green building.
4. Management of hydrological cycle, treatment of waste, and maintainance of indoor air quality.
5. Green building economics, business plan, and management.

Course Content:

Introduction to green building: Rationale for high-performance green building; green building progress and obstacles; trends in high-performance green building; the green building movement. [2 Lectures]

Ecological design: design vs ecological design; contemporary ecological design; future ecological design; thermodynamics: limits on recycling and the dissipation of materials. [2 Lectures]

Green building assessment: US green building assessment system; international green building assessment system; performance assessment. [2 Lectures]

LEED building rating system: history of LEED; structure of the LEED suite of building assessment systems; LEED credentials; the LEED process; LEED categories. [2 Lectures]

The Green Globes building assessment system: the Green Globes process; Green Globes verification and certification; structure of the ANSI/GBI 01-2010 standard; the Green Globes professional and Green Globes Assessor. [2 Lectures]

GRIHA/IGBC: Overview of IGBC; IGBC Certification Process; Overview of GRIHA; GRIHA Certification Process; Variants and Professionals. [2 Lectures]

Green building design process: conventional vs green building delivery systems; executing the green building project; the integrated design progress; role of Charrette in the design process; green building documentation requirements. [2 Lectures]

Sustainable site and landscape: land and landscape approaches for green buildings; sustainable landscapes; enhancing ecosystems; stormwater management; low-impact development; heat island mitigation; light trespass and pollution reduction; sustainable site initiative, Case study – IIT Jodhpur. [5 Lectures]

Life cycle assessment (LCA) for assessing the energy and carbon footprint reduction in buildings: LCA as a tool of building materials and products; environmental product declaration;materials and product certification systems; emerging construction materials and products; distinguishing between green building products and green building materials; Closing materials loops: design for deconstruction. [4 Lectures]

Energy and carbon footprint reduction – Embodied energy: Wall components and alternative systems for construction; Flooring systems and alternatives; Roofing systems and alternatives. [4 Lectures]

Energy and carbon footprint reduction: building energy issues; high-performance building energy design strategy; passive design strategy; building envelope; internal load reduction; active mechanical system; water-heating system; electrical power systems; innovative energy optimization strategies; renewable energy systems; fuel cells; smart buildings and energy management systems; ozone-depleting chemicals in HVAC&R systems; reducing carbon footprint of the built environment. [4 Lectures]

Building Information Modelling (BIM): Fundamentals of BIM, Various tools and their applications, BIM in building design, BIM Ecosystem, BIM aided green building design. [2 Lectures]

Built environment hydrological cycle: global water resource depletion; hydrological cycle terminology; high-performance building hydrological cycle strategy; designing the high-performance building hydrological cycle; water budget rules of thumb; sustainable stormwater management; landscaping water efficiency. [2 Lectures]

Indoor environmental quality: IEQ issues; design; main components of integrated IEQ design; emission from building materials; economic benefits of good indoor environmental quality. [2 Lectures]

Green building economics: the business case for high-performance green building; the economics of green building; quantifying green building benefits. [2 Lectures]

Text Books:

1. Charles J. Kibert (2013). Sustainable Construction. John Wiley & Sons. ISBN 978-1-118-33013-5
2. J. Newman and Ban Sang Choo (2003) Advanced Concrete Technology-Constituent Materials (1st Ed), Elsevier. ISBN 9780080489988.
3. S. Kubba (2010) LEED Practices, Certification, and Accreditation Hand book (1st Ed) Elsevier. ISBN 9780080958590.

Reference Books:

1. Ministry of Power, Energy Conservation Building Code 2018, Revised Version, Bureau of Energy Efficiency, 2018,
2. Indian Building Congress, Practical Handbook on Energy Conservation in Buildings, 1 st ed. Nabhi Publication, 2008.
3. TERI-Griha's Green Design practices (www.teriin.org/bcsd/griha/griha.htm)
4. Sartori, I., and Hestnes, A. G. "Energy use in the life cycle of conventional and low-energy buildings: A review article", Energy and Buildings., 20, 249-257.2007
5. Green Building Basics, California Integrated Waste Management Board (www.ciwmb.ca.gov/GREENBUILDING/Basics.htm#What)
6. Catarina Thormark. "A low energy building in a life cycle—its embodied energy, energy need for operation and recycling potential", Building and Environment., 37, 429-435.2001.

Online Course Material:

1. "NOC: Energy Efficiency, Acoustics and Daylighting in Building" by Prof. B. Bhattacharjee on NTPEL. https://nptel.ac.in/courses/105/102/105102175/

Title	**Railway and Airport Engineering**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	Transportation Engineering - I		

Objectives:
The course will help students to understand the fundamentals of railway and airport engineering, including the design and operation of their facilities.

Learning Outcomes:
At the end of the course, the student should be able to:

1. Identify and explain the functions of railway track components such as rails, sleepers, ballast, subgrade and fittings.
2. Explain the principles of railway traction, train resistance, hauling capacity, tractive effort and track stresses.
3. Design railway alignment and geometric elements, including gradients, cant, horizontal curves, transition curves and vertical curves.
4. Understand the requirements of airport planning including aircraft characteristics, site selection criteria and airport layout planning.
5. Design airport airside facilities such as runways, taxiways, aprons and circulation areas based on operational and geometric requirements.

Course Content:
Introduction to railways: History of railways - Indian and international perspective, gauge: factors affecting gauge choice, wheel and axle arrangement, Introduction to Bullet Train and Metro Trains. [4 Lectures]

Components and structural design of railway tracks: Rails, sleepers, ballast, subgrade, track fittings, railway cross-section, Traction and Resistances, hauling capacity, tractive effort, track stresses. [7 Lectures]

Alignment and geometric design: Track alignment – need and requirements, selection factors; speed on track – safe, permissible, equilibrium, cant excess, cant deficiency; various types of gradients, design of horizontal curves, transition curves and vertical curves, widening at curve. [9 Lectures]

Module IV: Points, Crossings, Signals: Turnouts – Types and Design; Crossings - Types and Design; Signals – Classification and functions, Types of signals, Signal placements; Interlocking on tracks: types. [7 Lectures]

Air transport: Overview, Aircraft characteristics, Airport planning, site selection, layout plan. [5 Lectures]

Design of airside and landside facilities: Runway orientation; geometric design of runways, taxiways, and aprons; circulation areas; hangars; terminal building functional areas and facilities; visual aids; drainage. [7 Lectures]

Text Books:

1. Satish Chandra and M. Agrawal, Railway Engineering, Oxford University Press, New Delhi
2. S. K. Khanna, M. G. Arora and S.S. Jain, 'Airport Planning & Design', Nem Chand & Bros., Roorkee

Reference Books:

1. S.C. Saxena and S. P. Arora, 'A Text Book of Railway Engineering', Dhanpat Rai & Sons, New Delhi
2. Robert Horonief and Francis X. McKelvey 'Planning & Design of Airports', McGraw Hill Publication
3. Alexander T. Wells and Seth Young, 'Airport Planning & Management', McGraw Hill Professionals

Online Course Material:

1. Transportation Engineering II by Prof. Rajat Rastogi. NPTEL. https://nptel.ac.in/courses/105107123

Title	**Hydraulic Structures and Machines**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

1. To understand the subsurface flow physics under diversion structures and master the theories of seepage and uplift pressure.
2. To perform rigorous stability analysis for gravity and embankment dams under various loading conditions.
3. To design complex energy dissipation systems and spillway profiles to ensure structural safety.
4. To analyze the performance and selection criteria for hydraulic turbines and pumps in water resource projects.

Learning outcomes:

1. Apply Khosla's Theory to design safe foundations against piping and uplift.
2. Evaluate the safety of dams against overturning, sliding, and internal erosion (piping).
3. Design river training works like Guide Banks and Groynes based on river morphology.
4. Select and size Impulse and Reaction turbines based on specific speed and head requirements.
5. Predict and mitigate cavitation in hydraulic machines.

Course Contents:

Weirs and Barrages: Creep Theories (Bligh's, Lane's, Khosla's), Design of Weir/Barrages on Permeable Foundations, Critical Exit Gradient and Safe Exit Gradient. [8 Lectures]

Dams, Spillways and Energy Dissipators: Dams: types, forces, failure types and causes, stability analysis of Gravity and Homogenous Embankment dams, dam outlet structures, dam terminal structures, Estimation of Reservoir Capacity, sediment flushing from reservoir; Spillway: Types, Design Considerations, Controlled and Uncontrolled Spillway Characteristics, Design of Uncontrolled Ogee Spillway, Design of Chute/Sid-Channel/Syphon/Piano-key/Morning-glory Spillway (any one is sufficient), Energy Dissipators, Stilling Basin, Design of Indian Standard Hydraulic Jump Type Stilling Basins, Spillway Crest Gates and their design considerations. [16 Lectures]

River Training Works: River Classification, Behavior of rivers (straight reaches, bends, meanders), Control and Training of Rivers: Objective, Classification, Methods (Leeves, Guide Banks, Spurs/Groynes, and Marginal Bunds), Stability Analysis of Leeves, Design of Guide Banks. [8 Lectures]

Hydraulic Machines: Hydraulic Turbines: Impulse and Reactive Turbines, Runaway Speed, Simplitude in turbines, draft tube, cavitation, Selection and Design of a Hydro-Turbine for a Specific Site Profile; Pumps: Centrifugal and Reciprocating. [7 Lectures]

Text Books:

1. Garg, S.K (2023). Irrigation Engineering and Hydraulic Structures. Khanna Publishers.
2. Novak, P., Nalluri, C. & Narayanan, R. (2007). Hydraulic Structures (4th ed.). Taylor & Francis.
3. Subramanya. K. (2007). Hydraulic Machines. McGraw-Hill Education

Reference Books:

1. Peterka, A. J. (1984). Hydraulic design of stilling basins and energy dissipators (USBR Engineering Monograph No. 25). U.S. Bureau of Reclamation.
2. Julien, P. Y. (1987). River mechanics. Cambridge University Press.
3. Mays, L.W. (2010), Water Resources Engineering, John Wiley & Sons, Inc.
4. U.S. Bureau of Reclamation. (1987). Design of Small Dams
5. Central Water Commission (CWC). Manual on River Training, Control and Management.
6. Different IS Codes: IS 6512, IS 6966, IS 4997

Title	**Quantity Estimation, Costing and Construction Management**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

1. To provide a foundational understanding of estimation, costing, and rate analysis in construction projects.
2. To introduce principles of engineering economics for effective financial decision-making.
3. To equip students with essential techniques of project planning, scheduling, and control using modern tools.
4. To familiarize students with bidding, tendering, and construction resource management practices.

Learning Outcomes:

1. Prepare quantity estimates for basic civil engineering works and buildings.
2. Perform rate analysis considering materials, labour, equipment, and overheads.
3. Understand and participate in bidding and tendering processes and interpret basic contract documents.
4. Apply engineering economics principles to evaluate project alternatives and financial decisions.
5. Develop project schedules using CPM and PERT techniques.
6. Monitor and control projects using tools like Earned Value Management and basic time–cost optimization.
7. Identify and manage construction resources including manpower and equipment effectively.

Course Contents:

Estimation, Costing and Engineering Economics (21 Hours)

Introduction to Estimation: Necessity of estimation, Units and measurements, Types and methods of estimate, Load bearing vs framed structures, Overview of Standard Methods of Measurement (SMM) . [2 Lectures]

Quantity Estimation and BOQ: Application of Standard Methods of Measurement (SMM) in quantity take-off, Quantities of: Brickwork, RCC, PCC Plastering, painting, finishing works, Basic residential buildings (flat & pitched roofs), Arches (brickwork & RCC basics), Joineries (doors, windows, ventilators), Steel estimation for RCC (basics), Introduction to spreadsheet/software based estimation, Preparation of Bill of Quantities (BOQ) . [8 Lectures]

Rate Analysis and Costing: Rate analysis (materials, labour, equipment), Labour productivity,

Equipment costing basics, Overheads, contingencies, profit. [4 Lectures]

Bidding, Tendering and Contracts Tendering process, Types of tenders, Basics of contracts and essentials, Procurement overview. [3 Lectures]

Construction Economics: Time value of money, Discounted cash flow basics, Present worth and comparison of alternatives, Basic cost-benefit analysis. [4 Lectures]

Construction and Project Management (18 Hours)
Construction Resources, Safety and Quality: Resource requirements, Manpower planning basics, Equipment selection factors, Introduction to construction safety, Introduction to construction quality. [4 Lectures]

Project Planning and Scheduling: Project scope and planning stages, Work Breakdown Structure (WBS), Bar charts, CPM and PERT, Time computation, Introduction to software-based scheduling. [8 Lectures]

Project Control and Optimization: Schedule monitoring and control, Resource levelling and allocation, Crashing of networks Earned Value Management (EVM), Time-cost trade-off. [4 Lectures]

Project Management Practices: Risk management basics, Leadership and teamwork, Stakeholder management, Project handover basics. [2 Lectures]

Text Books:

1. Jha, K. N. (2015). Construction Project Management: Theory and Practice. Pearson Education India.
2. Dutta, B. N. (2021). Estimating and Costing in Civil Engineering. UBS Publishers & Distributors Pvt. Ltd.

Reference Books:

1. Oberlender, G. D. (2014). Project Management for Engineering and Construction (2nd ed.). McGraw-Hill Education.
2. Harris, F., McCaffer, R., Baldwin, A., & Edum-Fotwe, F. (2021). Modern Construction Management (8th ed.). John Wiley & Sons.
3. Sawhney, A., Riley, M., & Irizarry, J. (Eds.). (2020). Construction 4.0: An Innovation Platform for the Built Environment. Routledge.
4. Sears, S. K., Sears, G. A., & Clough, R. H. (2010). Construction Project Management: A Practical Guide to Field Construction Management (5th ed.). John Wiley & Sons.
5. Walker, A. (2015). Project Management in Construction (6th ed.). John Wiley & Sons.
6. Kohli, D. D., & Kohli, R. C. (2013). A Textbook of Estimating, Costing and Accounts (Civil). S. Chand & Company Ltd.

Online Course Material:

1. Principles of Construction Management – Prof. Sudhir Misra, IIT Kanpur (https://nptel.ac.in/courses/105104161)

Title	**Non Destructive Evaluation of Structures**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

1. To understand the underlying principles of the different non-destructive tests (NDTs) for assessing concrete quality.
2. To understand the capabilities and limitations of the various NDTs for assessing concrete quality.
3. To familiarise with the relevant provisions in the standards and codes of practice relating to NDT of concrete.

Learning Outcomes:

Following this course, the student will have the ability to

1. Develop a testing plan for non-destructive evaluation of structures
2. Identify and perform the relevant NDT suitable for different scenarios
3. Analyse and interpret the NDT results obtained

Course Contents:

Philosophy, Importance and need of non-destructive testing, defects in concrete structures, deteriorations in concrete and steel, testing and quality control of concrete, Planning non-destructive evaluation of structures, Evaluation and Interpretation of NDT Results. [10 Lectures]

Visual Inspection – Introduction, general procedures, and applications, Fundamental principles, test procedures, applications, interpretation and limitations of Surface hardness test, penetration resistance test, pullout test, break-off test, core tests, and maturity methods. Stress wave methods for flaw detection: Ultrasonic pulse velocity, Impact echo, Impulse response, shear wave tomography. [10 Lectures]

NDT for evaluating corrosion - Surface resistivity, Half-cell potential, Polarisation resistance, Electromagnetic cover measurement, Direct measurement of corrosion rates. Test methods for measuring mass transport, carbonation, and moisture content. Chemical testing and allied techniques for concrete, Concrete Petrography. [10 Lectures]

Advanced Non-destructive techniques for concrete evaluation – Nuclear magnetic resonance, infrared thermography, electromagnetic tests for concrete, radioisotope gauge, Computer tomography, Magnetic-flux leakage. [9 Lectures]

Text Books:

1. Carino, N.J., and Malhotra, V.M. “Handbook on Non-destructive Testing of Concrete

(2nd Ed)" Taylor and Francis, 2003
2. Maierhofer, C., Reinhardt, H., and Dobmann, G. "Non-destructive evaluation of reinforced concrete structures Volume 2: Non- Non-destructive testing methods" Woodhead Publishing Series in Civil and Structural Engineering, 2010

Reference Books:

1. Bungey, J.H, Millard, S.G., and Grantham, M. "Testing of concrete in structures (4th Ed)", Taylor and Francis, 2006
2. Breysse D. "Non-Destructive Assessment of Concrete Structures: Reliability and Limits of Single and Combined Techniques" Springer 2012
3. International Atomic Energy Agency, "Guidebook on Non-destructive Evaluation of Concrete Structures", 2002
4. American Concrete Institute "Report on Nondestructive Test Methods for Evaluation of Concrete in Structures" Report by ACI Committee 228

Title	**Wastewater Engineering**	Number	CIL7090
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

The course aims to provide students with a solid foundation in the principles of wastewater treatment and the key parameters involved in the treatment processes. The course is designed to equip students with the technical skills needed to design various components of a wastewater treatment system, develop students' abilities to perform mass transfer, mass balance, and kinetic analyses and optimize the overall process.

Learning outcomes:

1. The students will develop fundamental principles and mechanisms of wastewater treatment.
2. The students will learn to design and evaluate critical components of treatment systems.
3. The students will learn to optimize and troubleshoot treatment processes for performance and compliance.
4. The students will learn to implement sustainable treatment technologies focusing on resource recovery and environmental stewardship.

Course Contents:

Introduction to biological wastewater treatment, wastewater quality parameters and standards, biological system: fundamentals of microbiology, bio metabolism, biological reaction kinetics. [4 lectures]

Reactor tanks: completely stirred tank reactor, plug flow reactor, enzyme reaction rate and kinetics, and effect of temperature, mass transfer and mass balance. [6 lectures]

Population equivalent, wastewater treatment scheme, centralized and de-centralized treatment plants, design of pump systems, pump head curve, design of equalization tank, parshall flume, design of screens, grit chamber, principles of sedimentation, design of primary sedimentation tank. [7 lectures]

Activated sludge process and its modifications for nutrient removal, process operation: (F/M), mean cell residence time, oxygen requirement, biofilm process, biological and chemical phosphorus removal. [8 lectures]

Process design and operation: A2O, SBR, MBR, MBBR, UASB, advancements in biological treatment, nature-based solutions, wastewater disinfection: Chlorination and UV System. [7 lectures]

Sludge production, characteristics, Sludge Handling: stabilization, thickening, dewatering, nutrient recovery, sludge pre-treatment, treatment. [7 lectures]

Text Books:

1. Metcalf & Eddy (2003) Wastewater Engineering: Treatment, Disposal and Reuse, 4 Th Edition, Tata McGraw Hill.
2. Peavy, H.S., Rowe, D.R., and Tchobanoglous, G. (1985) Environmental Engineering, McGraw Hill Book Company.

Reference Books:

1. Qasim, S.R. (1999) Wastewater Treatment Plants – Planning, Design and Operation, 2nd Edition, CRC Press.
2. Droste, R.L., Gehr, R.L. (2018) Theory and Practice of Water and Wastewater Treatment, 2nd Edition, J. Wiley.

Online course Material:

1. https://ocw.mit.edu/courses/1-85-water-and-wastewater-treatment-engineering-spring-2006/pages/lecture-notes/
2. Biological process design for wastewater treatment, NPTEL course material, Department of Civil Engineering, IIT Roorkee. https://archive.nptel.ac.in/courses/103/107/103107217/
3. Biological Wastewater Treatment by The Global Sanitation Graduate School (GSGS), IHE Delft Netherlands. https://studybwwt.online

Title	**Stability of Slopes**	Number	CIL7410
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
This course aims at:

1. To understand the influence of geology, rockmass properties, and hydrology on the stability of slopes.
2. To present basic types of analytical and numerical methods of slope stability analysis.
3. To introduce methods and instruments of slope monitoring and stabilization.
4. To enable planning and design of preventive measures, and redesign of failed slopes

Learning Outcomes:

1. Knowledge of data collection from field-work, aerial photography, geophysics, and geological map.
2. Knowledge of kinematic analysis of slopes using geological data and stereographic projection.
3. Knowledge of kinetic analysis of slopes and influence of critical geometrical and hydrological parameters.
4. Knowledge of available methods of slope stabilization by reinforcements and rock removal.
5. Knowledge of instrumental facilities to monitor slope movement and subsequent classification.

Course Contents:
Structural geology for slope analysis: Geological structures; Orientation of discontinuities; Introduction to geological maps; Identification of modes of slope instability. [3 Lectures]

Site investigation and geological data collection: Site reconnaissance; Geological mapping; Spacing, persistence, and roughness of discontinuities; Probabilistic analysis of structural geology; Drilling and logging. [4 Lectures]

Rock strength properties: Scale effect and rock strength; Shear strength of discontinuities; Laboratory testing of discontinuities; Hoek-Brown strength criterion for fractured rock mass; Rock durability and compressive strength. [4 Lectures]

Plane failure: Conditions of plane failure; Influence of groundwater on stability; Critical tension crack: identification and influence; Influence of critical slide plane inclination; Analysis of failure on a rough plane; Reinforcement of a slope; Seismic analysis of slopes. [5 Lectures]

Wedge failure: Conditions of wedge failure; Wedge analysis including cohesion, friction, and water pressure; Wedge stability chart for friction; Compression wedg analysis. [5 Lectures]

Circular failure: Conditions of circular failure; Methods of analysis; Derivation and use of circular failure charts; Location of critical slide surface and tension crack; Bishop's and Janbu's method of failure analysis. [5 Lectures]

Toppling failure: Types of toppling failure; Kinematic analysis of block toppling failure; Limit equilibrium analysis of toppling. [5 Lectures]

Stabilization of slopes: Stablization programs: Rock slope Inventory, Hazard rating analysis; Stabilization by rock reinforcements; Stabilization by rock removal; Measures against rock fall. [4 Lectures]

Movement monitoring: Types of slope movement: Initial response, Regressive and progressive movements, Long-term creep; Surface monitoring methods; Data interpretation. [4 Lectures]

Text Books:

1. Duncan C Wyllie, Christopher W. Mah (2005) Rock Slope Engineering, 4th Ed, Spon Press (Taylor & Francis Group), ISBN 0–415–28001–X (pbk).
2. Robin Chowdhury, Phil Flentje, Gautam Bhattacharya (2010) Geotechnical SlopeAnalysis. CRC Press (Taylor & Francis Group), ISBN: 978-0-415-46974-6 (Hbk).

Title	**Flow in Open Channels**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
The course discusses the fluid mechanics of steady and unsteady free surface flow.

Learning outcomes:

1. Understand channel types, flow classifications, and basic equations governing flow and velocity distributions.
2. Learn about specific energy, critical flow conditions, methods for computing critical flow, and concepts like specific force.
3. Study shear stress distribution, apply Chezy's and Manning's equations, and understand uniform flow computation.
4. Learn the gradually varied flow (GVF), analyze flow profiles, and compute GVF for different conditions.
5. Understand flow dynamics over sharp-crested weirs, spillways, sluice gates, and hydraulic jumps.
6. Explore unsteady flow principles, Saint-Venant equations, surge propagation, and wave behavior in unsteady conditions.

Course contents:
Basic flow concepts: Types of channels, classification of flows, basic equations, velocity distribution, velocity coefficients, pressure distribution. [3 Lectures]

Energy and momentum principles: Specific energy, critical flow, section factor for critical flow computation, first hydraulic exponent, computation of critical flow, specific force, channel transitions, and chokes. [6 Lectures]

Uniform flows: Shear stress distribution, Chezy's equation, Manning's equation, conveyance of a channel, section factor for uniform flow computation, second hydraulic exponent, computation of uniform flow, Hydraulically Efficient Channel Section, Critical Slope. [6 Lectures]

Gradually varied flows: Differential equation of gradually varied flow (GVF), classification and analysis of flow profiles, Control Section, computation of GVF, Advanced Numerical Methods for solving GVF. [6 Lectures]

Rapidly varied flows: Flow over sharp crested weir, spillways, flow under sluice gate, types of hydraulic jump, general equation for hydraulic jump in prismatic channels . [6 Lectures]

Unsteady flows: Unsteady flow in open channels, derivation of Saint-Venant equations, surge propagation, negative and positive surges, different types of waves in River and Ocean systems. [6 Lectures]

Hydraulics of Mobile Bed Channel: Sediment Properties, Initiation of motion, Bed Load, Sediment Load. [6 Lectures]

Text Books:

1. M. Hanif Chaudhry (2008) Open-channel flow. Springer
2. V. T. Chow (1959) Open channel hydraulics. Mc-Graw Hill
3. S. Dey (2024) Fluvial hydrodynamics. Springer

Reference Books:

1. H. Chanson (2004) The hydraulics of open channel flow: an introduction. Elsevier
2. S. Montes (1998) The hydraulics of open channel flow. ASCE Press
3. K. Subramanya (2019) Flow in open channels. Mc-Graw Hill

Title	**Hydropower Engineering**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

1. Inform students about the role of hydropower in the country's energy security and different aspects of hydropower site selection.
2. Introduce students to geotechnical, hydrological, and other related engineering an design aspects of conventional large dams.

Learning outcomes:

1. Develop an understanding of regional hydropower potential in the country and different technologies related to hydropower development.
2. Assessment of site suitability for hydropower project using hydrological analysis and geotechnical investigation
3. Learning design aspects of embankment and concrete gravity dam and relate structures such as inlet, spillway and others.

Course Contents:

Introduction: Hydropower resource, potential and market, Hydropower sites, Types of hydropower plants – Large, Small and Pumped Storage Hydropower. [5 lectures]

Tidal power: Tidal barrage power plant, operation and design, Tidal lagoons, Barrage construction techniques, Turbine speed regulation. [3 lectures]

Hydropower turbines: Classification and working principles, Impulse and reaction turbines; Propeller and Kaplan Turbines, Deriaz turbine. [5 lectures]

Hydrological Analysis for site selection: Methods for stream gauging, rainfall, runoff and its estimation by different methods, peak flood estimation, demonstration of discharge measuring instruments Hydrological analysis, flow duration studies, assessment of power potential. [4 lectures]

Geo-technical Investigation of dam site: Geological environments: rocks and soils; Geological, geotechnical, and geophysical investigation; Trenching and drilling. [4 lectures]

Design of Embankment Dam: freeboard; Slope protection; Embankment crest details; Dimensioning and tolerances; Conduits through embankments; Flood control structures; Seismic stability analysis of embankments; Liquefaction of dam embankments and foundations. [4 lectures]

Concrete face rockfill dams: Selection of dam types; Rockfill zones; Concrete face; Construction aspects; Settlements, and displacements of the face slab, and joints; Leakage; Strength and compressibility of rock foundations; Strength of concrete in the dam; Strength of the concrete-rock contact. [3 lectures]

Stability of Concrete gravity dams freeboard, different load on dam: hydrostatic and hydrodynamic load, uplift, earthquake forces, Silt and ice load, load combinations[4 lectures]:

Spillway and their design: design of diversion works and intake structures, Reservoir Planning and Management. [3 lectures]

Foundation preparation: General foundation preparation for embankmentdams; Cutoff foundation for embankment dams; Width and batter slopes; Slopemodification; Foundation preparation for concrete gravity dams. [4 lectures]

Text Books:

1. Chow, V.T. (2017), Applied Hydrology. McGraw Hills
2. Garg, S.K (2023), Irrigation Engineering and Hydraulic Structures, Khanna Publishers
3. Robin Fell, Patrick MacGregor and David Stapledon, Graeme Bell (2005), Geotechnical Engineering of Dams. A. A. Balkema publishers. ISBN 04 1536 440 x.

Reference Books:

1. ASCE Hydropower Task Committee (2007), Civil Works for hydroelectric Facilities, ASCE
2. Varshney, R. S. (2001).Hydro Power Structures. Nem Chand & Bros.
3. Hossein Samadi-Boroujeni (Editor) (2012), Hydropower – Practice and Application, INTECH

Title	**Statistics and Probability in Civil Engineering**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		
Antirequisite	Any department / academic unit specific statistics and/or probability course		

Objectives:
The course aims to provide students with a broad understanding of statistical methods and techniques utilized in different fields of Civil Engineering.

Learning outcomes:

1. Understand fundamental concepts such as joint, conditional probabilities, and their applications in assessing risks and inherent uncertainty in different fields of Civil Engineering
2. Gain knowledge about different probability distributions and statistical procedures (such as descriptive statistics or visualization) for data interpretation.
3. Designing and conducting experiments to support data related decisions in different fields of Civil Engineering.
4. Develop critical thinking and problem-solving abilities to interpret data accurately and communicate findings effectively.

Course Contents:
Statistics and Probability: Descriptive Statistics; Probability concepts; Sample Space and Random Variable; Joint Probability; Conditional and Marginal Probability; Bayes theorem, Chebyshev's Theorem [5 Lectures]

Probability Distribution and Their Applications: Discrete and Continuous Probability Distribution; Civil Engineering Applications; Sampling Distribution; Central limit Theorem [5 Lectures].

Concepts of sampling and monitoring: Simple random, systematic, ranked, stratified, composite and cluster sampling; sampling from normal distribution, Ranked set sampling, sampling in wild (quadrant, Recapture, transect) Completely randomized design, randomized block design; Control charts [3 Lectures].

Frequency Analysis; Risk and Reliability in Civil Engineering Design; Concept of uncertainty [2 Lectures]

Hypothesis Testing and Non-parametric Test: Null and alternative hypothesis; Type-I and II errors; one-tail and two-tail tests for mean and proportion (one- and two-

population), inferences about population variance, goodness of fit (Kolmogorov-smirnov test, Anderson Darling test), Man Whitney test, Wald statistic [6 Lectures]

Uni- and Multi-variate Data Analysis: Simple and Multiple Linear Regression, Pearson Correlation and other measures of Association; Principal Component Analysis; ANOVA test (one- and two-way); Monte Carlo methods [7 Lectures]

Time Series Analysis: Stationary time series; Realization and Ensemble; Trend and Periodicity Analysis; Data Transformation; Time Series Modeling (auto-regressive, moving average, auto-regressive moving average model with Exogenous Inputs) [7 Lectures]

Copula: Sklar's Theorem; Properties of Copula Function; Survival Copula; Common Copula Function; Prediction using Copula [4 Lectures]

Programming assignments may be given to support the understanding of theory.

Text Books:

1. Maity, R. (2022). Statistical methods in hydrology and hydroclimatology. 2nd Edition, Springer.
2. Manly, B.F.J. (2008), Statistics for Environmental science and Management, Chapman and Hall/CRC Press.
3. Johnson, R.A. (2020), Miller and Freund's Probability and Statistics for Engineering, Pearson

Reference Books:

1. Ang, A. H-S. and W. H. Tang, Probability Concepts in Engineering: Emphasis on Applications to Civil and Environmental Engineering, John Wiley & Sons, Inc., USA, Second Edition, 2006.
2. Benjamin, J.R. and Cornell, C.A., Probability, Statistics and Decision for Civil Engineers, McGraw-Hill, New York, 2014.
3. Wilks, D S. (2011), Statistical Methods in Atmospheric Sciences, Academic Press.

New Courses or Major Modifications

Title	**Hydraulic Engineering**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

The instructor will

1. Provide a rigorous and comprehensive foundation in the fundamental laws governing fluid flow
2. Help the students to understand the importance and application of fluid mechanics
3. Exemplify concepts of similarity and model testing

Learning Outcomes :

The students are expected to have the ability to:

1. Understand and apply the fundamental principles of fluid statics, kinematics, and dynamics to analyse forces, pressures, and flow conditions in civil engineering systems
2. Derive and apply the governing equations of fluid motion to solve engineering problems
3. Design physical hydraulic models, interpret experimental results, and extend findings to prototype-scale engineering predictions.

Course Contents:

Introduction: Definition and properties of fluid, Newtonian and Non-Newtonian Fluids, Dimensions and units, Continuum Hypothesis, Knudsen Number, Capillary and surface tension. [3 Lectures]

Pressure and Fluid Statics: Pascal's Law, Manometry, hydrostatic force on submerged planar and curved surfaces, buoyancy and stability, rigid body motion. [5 Lectures]

Fluid Kinematics: Graphical Description of Fluid Flow, Vorticity, Circulation, Stream and Potential Functions, rotational-irrotational flows, Bernoulli's equation and its applications. [5 Lectures]

Fluid Dynamics: Reynold Transport Theorem, Acceleration of fluid, Deformations in fluid particles, derivation of Euler's equation and Navier-Stokes equations. [9 Lectures]

Internal flow: Reynolds number regimes and Reynold experiments, Laminar and Turbulent flow in pipes, Head Losses in pipes, Flow measurement in pipes, Pipe network analysis, Introduction of water hammer effect and cavitation, Moody's chart. [6 Lectures]

External flow: Boundary layer over flat plate, boundary layer thickness, Prandtl-boundary layer equations, momentum integral equation, pressure gradient & separation, drag coefficients. [6 Lecture]

Dimensional analysis and similarity: Dimensionless Numbers, Scale and Similitude, Buckingham Pi theorem and examples. [2 Lectures]

Flow in Open Channel: Discharge in open channel, Concept of Specific Energy and Alternate Depths, Hydraulic Jump. [3 Lecture]

Text Books:

1. Hibbler, R.C., (2022), Fluid Mechanics, 2nd Edition, Pearson.
2. Modi, P.N. and Seth, S.M., Hydraulics and Fluid Mechanics Including Hydraulic Machines. Standard Book House, New Delhi.
3. White, F. M. (2003), Fluid Mechanics, 5th Edition, McGraw-Hill.

Reference Books:

1. Munson, B. R., Rothmayer, A. P., Okiishi, T'H., and Huebsch, W. W. Fundamentals of Fluid Mechanics, 7th Edition, John Wiley and Sons.
2. Fox, R. W., Pitchard, P. J., and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
3. Kundu, P. K, Cohen, I. M and Dowling, D. R, Fluid Mechanics, 6th Edition, Associated Press.
4. Cimbala, J.M. and Cengel, Y.A., (2010), Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

Online Course Material:

1. Prof. S. Dutta, Fluid Mechanics, NPTEL Course Material, Civil Engineering, IIT Guwahati, https://nptel.ac.in/courses/105103192/
2. Prof. M. S. Afzal, Hydraulic Engineering, NPTEL Course Material, Civil Engineering, IIT Kharagpur, https://nptel.ac.in/courses/105105203
3. Prof. T. I. Eldho, Fluid Mechanics, NPTEL Course Material, Civil Engineering, IIT Bombay, https://nptel.ac.in/courses/105101082

Title	**Building Materials and Construction Technology**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

1. Understand the fundamental behavior of different building materials
2. To introduce the various building materials available for construction
3. Understand the different components of building and their functions

Learning Outcome:

Following this course, the student will have the ability to:

1. Identify and select the right building materials for various applications
2. Identify the correct construction method for different constructions

Course Contents:

Introduction to materials – atomic structure, energy and entropy, structure of solids, Mechanical properties – stress-strain relationships, elastic behavior and elastic constants, failure under different conditions, fracture and toughness, viscoelasticity, surfaces – energy, wetting, electrical and thermal properties, variability in materials and its implications. [9 Lectures]

Classification, physical, mechanical properties and testing of commonly used building materials – Stones, bricks, cement, aggregates, chemical admixtures, supplementary cementitious materials, lime, timber, paints, glass, rebars, polymers etc.

Concrete – Fresh and hardened properties of concrete, test methods, grades of concrete, batching, mixing, placing and curing, Introduction of concrete mix design. [15 Lectures]

Building Construction: Components of building, stone and brick masonry, framed and loading bearing structures, shallow and deep foundations, floors and roofs, lintels, arches and staircases, doors, windows and ventilators, finishing materials, flooring, joints, waterproofing, formworks, scaffolding, shoring and underpinning. [15 Lectures]

Text Books:

1. P.C. Varghese (2008) Building Materials, Prentice-Hall of India, New Delhi.
2. B. C. Punmia (1993) Building Construction, Laxmi Publications, New Delhi.

Reference Books:

1. V. Raghavan (1990) Materials Science and Engineering, Prentice Hall.
2. J.F. Young, S. Mindess, R.J. Gray & A. Bentur (1998) The Science and

Technology of Civil Engineering Materials, Prentice Hall.
3. P.K. Mehta and P.J.M. Monteiro (2006) Concrete: Microstructure, properties and materials, McGraw Hill.
4. Edward Allen and Joseph Iano (2008) Fundamentals of Building Construction, 5th Edition, John Wiley & Sons Inc., New Jersey.
5. M. Soutsos and P. Domone (2018) Construction Materials, Their Nature and Behavior, Fifth Edition, CRC Press.

Online Course Material:

1. NPTEL course - NOC: Modern Construction Materials, Prof. Ravindra Gettu Prof. Aslam Kunhi Mohamed, IIT Madras.
   http://www.nptel.ac.in/courses/105106053/

Title	**Geoinformatics**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

This course aims at:

1. To introduce the basic principles of remote sensing, GPS, Surveying and GIS.
2. To teach application of geo-informatics in geoscience, ecology and environment study, agriculture, and disaster management.

Learning Outcomes:

A graduate of this course will have a thorough idea of:

1. Multispectral, thermal infrared, and microwave remote sensing.
2. Surveying Principles and handling modern surveying equipment
3. Different map projections and use of GPS.
4. Component GIS and applications.
5. Different cartographic models, implementation, validation, and visualization.

Course Contents:

Surveying and Global Positioning System: Surveying Fundamentals: Introduction to modern surveying; Coordinate systems; Measurement of distances and angles; Introduction to Total Station. Basic Geodesy, Map and Projections: Introduction, Basic Geodesy, Shape of Earth, Map and its types, Map Scale and Symbol, Map Projection and its Classification; Georeferencing. Global Positioning System (GPS): Components and operational principles; GPS receivers; Differential GPS (DGPS) concepts and types; Real-time kinematic positioning; limitations, and areas of application. [13 Lectures]

Remote Sensing and Photogrammetry: Physics of RS: Interaction of EMR with atmosphere and Earth's surface; Atmospheric windows. Photogrammetry: Types of aerial photographs and their geometry; Aerial cameras and Use of Drones; Image displacement. Stereoscopic Vision: Binocular observations; Stereoscopic pairs; Types of stereoscope; Measurement of height from photographs; Stereo model. Multispectral and Hyperspectral imaging: Spectral signatures of vegetation, soil, and water, Sensor Utilized; Image correction and interpretation Advanced Remote Sensing: Thermal Infrared (emissivity, characteristics, and applications); Microwave region (SAR characteristics and interpretation); LIDAR principles, LIDAR point cloud and its processing. [15 Lectures]

Geographical Information System and Applications: GIS Concepts Components and spatial data representation; Relationships of spatial objects; GIS functions;

Integration of Remote sensing and GIS. Cartographic Modelling: Inductive and deductive modelling; Model flowcharting; Conflict resolution, verification, and implementation, GIS Model Output. Visualization and Applications: Web mapping and visualization; Virtual and Immersive GIS, Mapping temporal data. Applications in Environment, Agriculture, Public Health, Water Resources, and Disaster Management. [11 Lectures]

Text Books:

1. Satheesh Gopi, R. Sathikumar, and N. Madhu (2025) Advanced Surveying: Total Station, GPS, GIS & Remote Sensing (2nd Ed). Pearson India.
2. T. M. Lillesand, R. W. Kiefer, J. W. Chipman (2004) Remote Sensing and Image Interpretation (5th Ed). Wiley. ISBN: 0-471-15227-7
3. G. S. Srivastava (2014) An Introduction to Geoinformatics. McGraw Hill Education (India) Private Limited. New Delhi. India. ISBN: 9781259058462.
4. Michael N. DeMers (2008) Fundamentals of Geographic Information Systems (4th Ed). Wiley. ISBN: 978-0-470-12906-7.

Reference Books:

1. Basudeb Bhatta (2021) Remote Sensing and GIS (3rd Ed). Oxford University Press.
2. L. E. Johnson (2009) Geographic Information Systems in Water Resources Engineering. CRC Press. Taylor & Francis Group. ISBN 978-1-4200-6913-6.
3. B. Tomaszewski (2020) Geographic Information Systems (GIS) for Disaster Management (2nd Ed). CRC Press.

Online Course Material:

1. Prof. Arun K. Saraf (2016) Introduction to Geographic Information System. NPTEL. https://nptel.ac.in/courses/105/107/105107155/
2. Prof. Arun K. Saraf (2017) Introduction to Remote Sensing. NPTEL. https://nptel.ac.in/courses/121/107/121107009/

Title	**Transportation Engineering - I**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
The course will help students to understand the principles of transportation engineering, including geometric design, pavement materials and design, pavement construction, evaluation, and maintenance.

Learning Outcomes:
At the end of the course, the student should be able to:

1. Design cross section elements, horizontal and vertical alignment.
2. Determine the characteristics of pavement materials.
3. Design flexible and rigid pavements.
4. Explain the construction procedures for different pavement layers and the equipment used in highway construction.
5. Evaluate the functional and structural condition of pavements using appropriate methods and recommend maintenance or rehabilitation measures.

Course Content:
Introduction to Transportation Engineering:Transportation system, Introduction to Transportation Engineering, Functional classification of roads, Terrain classification, Factors affecting transportation, Highway development in India**.** [4 Lectures]

Geometric design: Factors affecting geometric design; Alignment – types, requirements, factors influencing and stages; Highway design process - Highway cross sectional elements, Sight distance; Highway alignment: Horizontal alignment - horizontal curve, Super elevation, extra widening, Transition curve, Setback distance, Visibility, Vertical alignment - gradient, grade compensation, summit curve, valley curve; alignment for hill roads. [10 Lectures]

Pavement materials and design: Pavement characteristics, materials, Types of pavements - Flexible and Rigid pavements, Wheel load calculations, Flexible Pavement materials- soil, aggregate, bitumen; Tests on pavement materials; Bituminous mix design, Marshall stability; Rigid pavement- joints, analysis of stresses; IRC method for design of flexible and rigid pavement; Pavement failures, IIT Pave Software. [11 Lectures]

Pavement Construction: Construction of various layers, equipment used for highway construction and quality control aspects. [6 Lectures]

Pavement Evaluation and Maintenance: Failures of flexible and rigid pavements, condition indicators, various types of functional and structural evaluation techniques, maintenance and rehabilitation alternatives, Design of overlays. [8 Lectures]

Text Books:

1. N. J. Garber, L.A. Hoel. Traffic and Highway Engineering.
2. Saxena S. C. Textbook of Highway and Traffic Engineering, CBS Publishers.
3. Wright Dixon, Highway Engineering, Wiley India.

Reference Books:

1. S. K. Khanna, C. E. G. Justo, and A. Veeraragavan. Highway Engineering, Nem Chand and Bros. Publishers.
2. I.S. specifications on concrete, aggregates and bituminous materials.
3. David Croney, Design and performance of road pavements, McGraw Hill.
4. James H. Banks, Introduction to Transportation Engineering, McGraw Hill.
5. R. Srinivasa Kumar, Textbook of Highway Engineering, University Press.
6. Chakroborty, P., Animesh Das. Principles of Transportation Engineering, PHI Learning Private Limited.

Online Course Material:

1. Introduction to Transportation Engineering by Prof. K.S. Reddy and Prof. Bhargab Maitra. NPTEL. https://nptel.ac.in/courses/105105107

Title	**Engineering Hydrology**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
To provide an understanding of hydrologic processes, analysis methods, and design applications with emphasis on mathematical and statistical analysis in water resources management.

Learning outcomes:

1. Become familiar with engineering aspects of surface and sub-surface hydrology
2. Utilize knowledge of hydrological processes in solving engineering problems within hydrology and water resources
3. Develop the understanding of the major impact of changing climate on the water cycle.

Course Contents:
Introduction: Engineering hydrology and its applications, Hydrological cycle, Concept of Catchment, Water balance and fluxes, World Water Balance, Residence time, Climate Change Impacts. [4 Lectures]

Precipitation: Introduction, Forms of precipitation, Measurement of Precipitation, Mean Precipitation over an Area, Analysis of precipitation data, Precipitation variability and monsoon characteristics in India, Depth–Area–Duration Relationships, Maximum Intensity/Depth–Duration–Frequency Relationship, Probable Maximum Precipitation (PMP). [6 Lectures]

Abstractions from precipitation: Introduction, Evaporation and Evapotranspiration: factors affecting evaporation, measurement and estimation of evaporation, Reservoir Evaporation and Methods for Its Reduction, Transpiration, Measurement of Evapotranspiration, Potential Evapotranspiration and its estimation, Actual Evapotranspiration; Interception; Depression Storage; Infiltration: Infiltration Capacity and factors affecting Infiltration capacity, Measuring and modeling infiltration, Infiltration Indices. [7 Lectures]

Runoff and streamflow: Types of runoff, Runoff characteristics of stream, Catchment factors affecting runoff, Flow–Duration and Flow–Mass Curve, Estimation of Reservoir Volume, Hydrograph, Factors affecting Storm Hydrograph, Components of a Hydrograph, Base Flow Separation, Effective Rainfall, Unit Hydrograph and its derivation, Unit Hydrographs of Different Durations, Synthetic Unit Hydrograph. [8

Lectures]

Droughts, Floods and Flood Routing: Introduction to Hydrologic Extremes, Drought, Drought Indices, Flood Estimation: Rational Formula, Runoff Coefficient and Unit hydrograph method, Flood Frequency Analysis, Design Flood and Design Storm, Risk, reliability and Safety Factor in hydrologic design, Introduction to Flood routing, Level Pool Routing, Hydrological Channel Routing, Hydraulic Methods of Flood Routing, Flood Control, Flood in changing climate. [9 Lectures]

Groundwater: Sources and form of Groundwater, Saturated Formations, Water Table, Aquifer Properties, Porosity and Permeability, Groundwater Movement, Hydraulics of wells, Recharge and its techniques, Groundwater Resources of India. [5 Lectures]

Text Books:

1. Subramanya, K. (2024). Engineering hydrology (6th ed.). McGraw Hill Education (India) Pvt Ltd
2. Ojha, C. S. P., Berndtsson, R., & Bhunya, P. (2008). Engineering hydrology. Oxford University Press.
3. Viessman Jr., W., Lewis, G. (2002). Introduction to Hydrology. 5th Edition, Pearson

Reference Books:

1. Chow, V.T., Applied Hydrology, McGraw-Hill
2. Todd, D.K., & Mays, W.L. (2004) Groundwater Hydrology, 3rd Edition, Wiley

Online Course Material:

1. Margulis, S. (2016). Introduction to hydrology. https://margulis-group.github.io/textbook/
2. Prof. Rajib Maity (2012). Advanced Hydrology. NPTEL. https://nptel.ac.in/courses/105105214

Title	**Air Pollution and Solid Waste Management**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

1. To explain and classify sources, types, impacts of air pollutants, along with relevant standards and legislation.
2. To analyze the role of meteorological parameters in air pollutant dispersion and evaluate control strategies.
3. To explain and characterize solid waste sources, composition, and generation, along with regulatory and management frameworks.
4. To apply and evaluate integrated solid waste management principles, including waste minimization, treatment, and disposal technologies.

Learning Outcomes:

Upon successful completion of the course, students will be able to:

1. Classify and assess air pollutants and interpret their impacts on health, environment, and regulatory compliance.
2. Analyze and predict pollutant dispersion using meteorological concepts such as lapse rate, stability, and plume behavior.
3. Evaluate and justify the selection of air pollution monitoring and control technologies for specific applications.
4. Design and propose integrated solid waste management strategies including collection, processing, treatment, and disposal.

Course Contents:

Air pollution

Sources, Types and classification of air pollutants, air pollution episodes, effect of air pollutants on human health, vegetation and materials, air quality and emission standards and legislation. [7 Lectures]

Effect of meteorology, Lapse rate, Atmospheric Stability, Plume behaviour and dispersion, Stack height. [4 Lectures]

Engineering emission control concepts, Selection and basic design approach for air pollution control equipments, source control technologies, Indoor air pollution sources and control. [9 Lectures]

Municipal Solid Waste Management

Fundamentals of Integrated Solid Waste Management, legislation and regulations. [4 Lectures]

Waste generation, collection, and transportation; Characteristics of solid waste.[6 Lectures]

Transfer stations, material recovery facility; biological treatment methods (such as composting and anaerobic digestion), incineration, pyrolysis and gasification, and sanitary landfilling. [9 Lectures]

Text Books:

1. Noel De Nevers, Air Pollution Control Engineering, 3rd Ed., Waveland Press, Inc., 2016
2. Tchobanoglous, G., Theisen, H., and Vigil, S.A. (2014) Integrated Solid Waste Management: Engineering Principles and Management Issues, McGraw Hill Book Company, Singapore.
3. Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.

Reference Books:

1. Air Pollution Control: A Design Approach, C. David Cooper & F. C. Alley, 4th Ed., Waveland Press, Inc., 2011
2. CPHEEO Manual of Solid Waste Management, GOI Publication, 2016
3. M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Title	**Water and Wastewater Engineering**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
The Instructor will:

1. Provide an overview on water demand, water quality and treatment unit operation
2. Provide an overview on wastewater characteristics and treatment unit operation
3. Provide knowledge on water distribution system

Learning Outcomes:
The students are expected to be able to:

1. Understand water treatment unit operation and would be able to design it
2. Understand wastewater treatment unit operation and would be able to design it
3. Analyse water distribution system for further design

Course Contents:
Water demand and Water quality assessment: Water demand; Water quality parameters: Physical, chemical and biological parameters; Water quality standards. [4 Lectures]

Water treatment theory and operation:Theory and design parameters for various water treatment unit processes such as screening; sedimentation tank; coagulation-flocculation; filtration; disinfection; hardness; Exposure to advanced water treatment processes. [12 Lectures]

Water distribution system:Design of water supply pipeline; Pumping; Head loss and water distribution network analysis. [4 Lectures]

Wastewater characterization: Wastewater characterization : physical, chemical and biological parameters. [5 Lectures]

Wastewater treatment and sludge disposal: Introduction to Unit processes in wastewater treatment plant and design criteria; Focus on secondary treatment: suspended growth system (e.g., activated sludge process) and attached growth systems (e.g., trickling filter); nutrient removal; Introduction to natural

system; Exposure to advanced treatment process and design software. [14 Lectures]

Text Books:

1. Metcalf & Eddy, Wastewater Engineering- Treatment and Reuse (Revised by G. Tchobanoglous, F. L. Burton and H.D. Stensel), Tata McGraw Hill, 4th Edn., 2004.H.
2. Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
3. M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Reference Books:

1. Nathanson and Jerry, Basic Environmental Technology: Water Supply, Waste Management and Pollution Control, Prentice Hall India Learning Pvt. Ltd., India 2009
2. Garg, Environmental Engineering: Water Supply Engineering, Khanna Publishers, 18th Edn, India 2004
3. Manual on Sewerage and Sewage Treatment Systems, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 2013.
4. Manual on Water Supply and Treatment, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 1999.

Online Course Material:

1. Guidelines for drinking-water quality, World Health Organization 2022, ISBN 978-92-4-004506-4, https://www.who.int/publications/i/item/9789240045064
2. Drinking Water – Specification, Bureau of Indian Standard , IS 10500 : 2012, New Delhi, India, 110002

Title	**Transportation Engineering - II**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

The course will introduce students to the principles of traffic engineering, traffic operations, and transportation planning. It aims to develop an understanding of traffic flow characteristics, traffic control systems, travel demand estimation, and urban transportation planning for sustainable mobility systems.

Learning Outcomes:

At the end of the course, the student should be able to:

1. Analyze traffic flow characteristics and interpret speed–flow–density relationships
2. Conduct traffic engineering studies and evaluate traffic operations at different road facilities
3. Understand transportation planning principles and carry out surveys involved in transport planning
4. Apply four-step travel demand forecasting models

Course Content:

Introduction to Transportation systems: Overview of transportation systems and their role in economic and social development, Elements of traffic engineering and transport planning and Current transportation challenges. [2 Lectures]

Traffic Engineering: Transportation system, Types of facilities, Factors affecting traffic performance, Categories of traffic flow, Traffic stream components and characteristics, Analysis of speed, Flow and density relationships, Fundamental diagram of traffic flow, Trajectory diagrams. [9 Lectures]

Traffic Operations: Theories of traffic flow; Traffic studies (volume, speed, delay, parking, accident studies); Design of control strategies for simple systems like intersections, roundabouts, freeways, etc., Introduction to traffic signs and signals, Capacity and level of services of various transportation facilities, Introduction of Intelligent Transportation System (ITS), software tools, AI/ML applications. [11 Lectures]

Transportation Planning: Characteristics and objectives of planning, Urban Transportation Planning- Comprehensive planning process, multimodal mobility, Transit-Oriented Development (TOD), NMT planning, Transportation data needs, and sources, types of trips, Study area and analysis zone, Transportation surveys/data collection methods. [8 Lectures]

Travel demand estimation: Four-step transportation planning process, Trip generation, Trip distribution, Modal split, Route assignment, Traffic impact assessment studies, Activity based models. [9 Lectures]

Text Books:

1. N. J. Garber, L.A. Hoel. Traffic and Highway Engineering, 4th Edition
2. R. P. Roess, E.S. Prassas, W.R. McShane. Traffic Engineering, 4th edition, Pearson.
3. C. S. Papacostas, P. D. Prevedouros. Transportation Engineering and Planning, Third Edition, Pearson

Reference Books:

1. Coleman A. O'Flaherty. Transport Planning and Traffic Engineering, CRC Press
2. K. Khanna, C. E. G. Justo, A. Veeraragavan. Highway Engineering, Tenth Edition, NemChand & Bros.
3. Dr. L. R. Kadiyali, Traffic Engineering and Transport Planning, Khanna Publishers.
4. James H. Banks, Introduction to Transportation Engineering, McGraw Hill
5. Saxena S. C. Textbook of Highway and Traffic Engineering, CBS Publishers.

Online Course Material:

1. 1. Introduction to Transportation Engineering by Prof K.S. Reddy and Prof Bhargab Maitra.NPTEL. https://nptel.ac.in/courses/105105107
2. Urban Transportation Planning by Prof. V. Thamizh Arasan. NPTEL. https://nptel.ac.in/courses/105106058

Title	**Geotechnical Engineering**	**Number**	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P	3-1-0 [4]
Offered For	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objective:
To introduce geotechnical investigation methods and the design principles of slopes, retaining structures, and shallow and deep foundations.

Learning outcomes:
Students will be able to:

1. Plan and interpret geotechnical site investigations methods and data
2. Assess slope stability problems
3. Evaluate earth pressure on retaining structures
4. Evaluate bearing capacity and settlement of shallow foundations
5. Understand design principles of deep foundations

Course Contents:
Site Investigation and Data Collection: Objectives of site investigation, Methods of subsurface exploration, Boring, drilling, and logging methods, Sampling techniques (disturbed and undisturbed), Rock Mass Characterization, Rock Mass Classification – RMR, Q-System, and GSI.; Standard Penetration Test (SPT), Cone Penetration Test (CPT), Geophysical exploration methods, Preparation of soil/engineering geology investigation reports. [10 Lectures]

Shallow Foundations: Types of shallow foundations, Bearing capacity concepts, Modes of shear failure, Terzaghi bearing capacity theory, Bearing capacity factors, Effect of water table, Plate load test, Settlement of foundations, Safe bearing capacity. [8 Lectures]

Deep Foundations: Types of piles, Methods of pile installation, Load carrying capacity of piles, shaft resistance and base resistance, Pile load tests, Group action of piles, Settlement of pile groups, Introduction to drilled piers and well foundations. [7 Lectures]

Lateral Earth Pressure and Retaining Structures: Types of earth pressure, Earth pressure at rest, Rankine earth pressure theory, Coulomb earth pressure theory, Retaining wall types, Stability of retaining walls. [7 Lectures]

Stability of Slopes: Types of slopes and slope failures, Factor of safety concept,

Stability of infinite slopes, Stability of finite slopes, Plane failure and circular failure, Method of slices, Bishop simplified method. [7 Lectures]

Text Books:

1. Das, Braja M. (2015). Principles of Foundation Engineering, SI (7th Edition). Cengage Learning.
2. Ranjan, Gopal, and A. S. R. Rao. 2023. *Basic and Applied Soil Mechanics*. 5th ed. New Delhi: New Age International Private Limited.
3. Budhu, Muni. (2011). Soil Mechanics and Foundations (3rd Edition). John Wiley & Sons, Hoboken, New Jersey.
4. Murthy, V. N. S. (2002). Geotechnical Engineering: Principles and Practices of Soil Mechanics and Foundation Engineering. New York: Marcel Dekker Inc.
5. Deb, D., Verma AK (2016) Fundamentals and Applications of Rock Mechanics. PHI Learning Pvt. Ltd.

Title	**AI/ML in Civil Engineering**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Mathematics I, Mathematics II		
Antirequisite	Any department / academic unit specific AI/ML course		

Objectives:

1. Comprehensive understanding of the problem settings and application of AI/ML in civil and infrastructure engineering.
2. Development of programming skills to execute AI/ML algorithms in civil and infrastructure engineering.
3. The ability to bridge the gap between traditional engineering physics and data-driven discovery.

Learning outcomes:

By the end of this course, students will have the ability to:

4. Analyze engineering problems within both deterministic and random (uncertain) environments.
5. Apply modern architectures (CNNs, Transformers, GNNs) to address challenges such as structural damage assessment, traffic forecasting, and uncertainty quantification.
6. Implement end-to-end ML pipelines, from raw sensor data to deployable infrastructure models.

Course Contents:

Foundations and Information Theory: Review of AI/ML: Fundamentals of data-driven infrastructure engineering; Machine Learning Paradigms: Supervised, unsupervised, and reinforcement, Bias-Variance Tradeoff, probability and information theory basics, including Shannon entropy and KL divergence; modern optimization algorithms, including stochastic gradient descent and Adam; support vector machines and feature embedding spaces; Universal Approximation Theorem; autograd and backpropagation. [12 Lectures]

Vision, Representation, and Sequential Intelligence: Autoencoder; Stacked Autoencoder; Computer vision in infrastructure: CNN architectures, Pooling Layer, Transfer Learning; image classification and semantic segmentation for automated damage assessment; sequential sensor data and time series: Recurrent Neural Networks and LSTMs; attention mechanisms; Input Embedding and Transformer architecture; Transformers for real-time structural health monitoring and traffic flow forecasting, Introduction to Graph Neural Network. [15 Lectures]

Surrogate Modeling and Active Learning: Introduction to surrogate modeling; dimensionality reduction of complex physical fields; Gaussian processes and neural

network surrogates. Physics-aware ML: Brief conceptual introduction to Physics-Informed Neural Networks (PINNs) and embedding governing differential equations into AI loss functions. Active learning: Bayesian optimization, uncertainty quantification, and optimal data sampling strategies for data-scarce engineering environments, Recent advances in the field of AI relevant to Infrastructure Engineering (Sensor Network and Edge Computing, Agentic AI in Infrastructure Engineering) [12 Lectures]

Text Books:

1. J.-A. Goulet (2020) Probabilistic Machine Learning for Civil Engineers, The MIT Press.
2. Goodfellow, I., Bengio, Y., and Courville, A. (2016) Deep Learning, MIT Press.
3. S. Shalev-Shwartz, and S. Ben-David (2014) Understanding Machine Learning From Theory to Algorithms, Cambridge University Press. ISBN 9781107057135.

Reference Books:

1. R. S. Sutton and A. G. Barto (2018) Reinforcement Learning: An Introduction (2nd Edition), MIT Press.
2. A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola (2023) Dive into Deep Learning, Cambridge University Press.
3. Y. Singh et al. (2023). *Machine Learning Applications in Civil Engineering*, Elsevier.
4. E. Stevens, L. Antiga, and T. Viehmann (2020) Deep Learning with PyTorch, Manning Publications.
5. Brunton, S. L., & Kutz, J. N. (2019). *Data-driven science and engineering: Machine learning, dynamical systems, and control*. Cambridge University Press.

Online Course Material:

1. Raissi, M., Perdikaris, P. and Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, pp. 686-707. https://doi.org/10.1016/j.jcp.2018.10.045
2. Torralba, A., Isola, P., & Freeman, W., Foundations of Computer Vision, The MIT Press, https://visionbook.mit.edu/

Title	**Water Resources Engineering**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:

1. To quantify water demand for varied sectors and estimate available surface and groundwater resources.
2. To master the hydraulic design of storage, diversion, and conveyance structures.
3. To apply computational tools (like EPANET) for the analysis of complex water distribution networks.
4. To integrate watershed management principles for sustainable water resource development.

Learning Outcomes :

A graduate of this course will have a thorough idea of::

1. Predict future water demands and estimate reservoir capacity.
2. Design stable alluvial and non-alluvial canals using Lacey's and Kennedy's theories.
3. Analyze and optimize pipe networks using the Hardy-Cross method and software.
4. Design hydraulic profiles for spillways, energy dissipators, and cross-drainage works.
5. Develop a River Basin Master Plan incorporating soil conservation and rainwater harvesting.

Course Contents:

Water Demand: Introduction, Role of Water in Modern Lifestyle, Estimating Urban Water Demand, Irrigation requirement, Industrial Water Demand, Projecting future demand, Concept of Water Equivalent.. [4 Lectures]

Available Water Resources: Introduction; Overview of Available Water Resources in India; Surface Water Resources: Stream flow Measurement Techniques, Hydrograph, Flow-Duration and Flow-mass Curve, Reservoir Capacity Estimation using Sequent Peak Algorithm; Groundwater Resources: Occurrence, Aquifer Properties, Hydraulics of Wells, Yield, Artificial Recharge. [8 Lectures]

Water Retention: Dam: types and their role in water retention, Suitable Site; Spillways: Design Considerations, Location of Spillway, Spillway Types; Reservoir: Different Storage in Reservoir, Reservoir Losses, Reservoir Sedimentation, Trap Efficiency, Methods to Control Silting, Safe Yield, Multi-Purpose Reservoir

Operation. [10 Lectures]

Water Conveyance: Intake Structures: Introduction; design considerations; Pipe Network: Design of Pipe Network, Turbulent Flow in Pipes, Minor Loss, Hardy-Cross Analysis, Software Analysis (for instance EPANET) of pipe network; Canal System: Alignment, Command Area, Time and Capacity Factors, Water losses in canal, Canal Falls, Canal Regulation, Alluvial vs Non-Alluvial Canal, Regime Theories, Design and economics of Canal Lining, Cross Drainage Works. [8 Lectures]

Integrated Watershed Management (IWM): Watershed Concepts: Definition, types, and characterization, River Basin Master Plan; Soil Erosion and Conservation: Mechanics and estimation of water erosion, Structural and vegetative measures (contouring, terracing, gully plugs); Rainwater Harvesting: Rooftop harvesting, recharge pits, and check dams; Land Use Management: Impact of deforestation and urbanization on runoff, Concept of Low Impact Development; Participatory Management. [9 Lectures]

Text Books:

1. Mays, L.W. *Water Resources Engineering*, John Wiley and sons, Inc., Kundli
2. Todd, D. K.and Mays, L. W. *Ground Water Hydrology,* John Wiley India Edition
3. Garg, S.K. Irrigation Engineering and Hydraulic Structures. Khanna Publishers, Delhi.

References Books:

1. Chow, V.T., Maidment, D.R., and Mays L.W. Applied Hydrology. McGraw-Hill Inc., New York.
2. James, L.D., and Lee, R.R. Economics of water resources planning. McGraw-Hill Book Inc., New York.
3. Todd, D. K.and Mays, L. W. *Ground Water Hydrology,* John Wiley India Edition
4. Linsley, R.K., Franzini, J.B., Freyberg, D.L., and Tchobanoglous, G. Water Resources Engineering, McGraw Hill Inc., Singapore.
5. Rohitashw Kumar, Vijay P. Singh, and Sabah Parvaze. Watershed Planning and Management, CRC Press
6. Subramanya, K. (2024). Engineering hydrology (6th ed.). McGraw-Hill Education (India) Pvt Ltd
7. Rossman, L. et al., (2000) EPANET 2.2 User Manual. U.S. Environmental Protection Agency

Title	**Railway and Airport Engineering**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	Transportation Engineering - I		

Objectives:
The course will help students to understand the fundamentals of railway and airport engineering, including the design and operation of their facilities.

Learning Outcomes:
At the end of the course, the student should be able to:

1. Identify and explain the functions of railway track components such as rails, sleepers, ballast, subgrade and fittings.
2. Explain the principles of railway traction, train resistance, hauling capacity, tractive effort and track stresses.
3. Design railway alignment and geometric elements, including gradients, cant, horizontal curves, transition curves and vertical curves.
4. Understand the requirements of airport planning including aircraft characteristics, site selection criteria and airport layout planning.
5. Design airport airside facilities such as runways, taxiways, aprons and circulation areas based on operational and geometric requirements.

Course Content:
Introduction to railways: History of railways - Indian and international perspective, gauge: factors affecting gauge choice, wheel and axle arrangement, Introduction to Bullet Train and Metro Trains. [4 Lectures]

Components and structural design of railway tracks: Rails, sleepers, ballast, subgrade, track fittings, railway cross-section, Traction and Resistances, hauling capacity, tractive effort, track stresses. [7 Lectures]

Alignment and geometric design: Track alignment – need and requirements, selection factors; speed on track – safe, permissible, equilibrium, cant excess, cant deficiency; various types of gradients, design of horizontal curves, transition curves and vertical curves, widening at curve. [9 Lectures]

Module IV: Points, Crossings, Signals: Turnouts – Types and Design; Crossings - Types and Design; Signals – Classification and functions, Types of signals, Signal placements; Interlocking on tracks: types. [7 Lectures]

Air transport: Overview, Aircraft characteristics, Airport planning, site selection, layout plan. [5 Lectures]

Design of airside and landside facilities: Runway orientation; geometric design of runways, taxiways, and aprons; circulation areas; hangars; terminal building functional areas and facilities; visual aids; drainage. [7 Lectures]

Text Books:

1. Satish Chandra and M. Agrawal, Railway Engineering, Oxford University Press, New Delhi
2. S. K. Khanna, M. G. Arora and S.S. Jain, 'Airport Planning & Design', Nem Chand & Bros., Roorkee

Reference Books:

1. S.C. Saxena and S. P. Arora, 'A Text Book of Railway Engineering', Dhanpat Rai & Sons, New Delhi
2. Robert Horonief and Francis X. McKelvey 'Planning & Design of Airports', McGraw Hill Publication
3. Alexander T. Wells and Seth Young, 'Airport Planning & Management', McGraw Hill Professionals

Online Course Material:

1. Transportation Engineering II by Prof. Rajat Rastogi. NPTEL. https://nptel.ac.in/courses/105107123

Title	**Statistics and Probability in Civil Engineering**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
The course aims to provide students with a broad understanding of statistical methods and techniques utilized in different fields of Civil Engineering.

Learning outcomes:

1. Understand fundamental concepts such as joint, conditional probabilities, and their applications in assessing risks and inherent uncertainty in different fields of Civil Engineering
2. Gain knowledge about different probability distributions and statistical procedures (such as descriptive statistics or visualization) for data interpretation.
3. Designing and conducting experiments to support data related decisions in different fields of Civil Engineering.
4. Develop critical thinking and problem-solving abilities to interpret data accurately and communicate findings effectively.

Course Contents:
Statistics and Probability: Descriptive Statistics; Probability concepts; Sample Space and Random Variable; Joint Probability; Conditional and Marginal Probability; Bayes theorem, Chebyshev's Theorem [5 Lectures]

Probability Distribution and Their Applications: Discrete and Continuous Probability Distribution; Civil Engineering Applications; Sampling Distribution; Central limit Theorem [5 Lectures].

Concepts of sampling and monitoring: Simple random, systematic, ranked, stratified, composite and cluster sampling; sampling from normal distribution, Ranked set sampling, sampling in wild (quadrant, Recapture, transect) Completely randomized design, randomized block design; Control charts [3 Lectures].

Frequency Analysis; Risk and Reliability in Civil Engineering Design; Concept of uncertainty [2 Lectures]

Hypothesis Testing and Non-parametric Test: Null and alternative hypothesis; Type-I and II errors; one-tail and two-tail tests for mean and proportion (one- and two-population), inferences about population variance, goodness of fit (Kolmogorov-smirnov test, Anderson Darling test), Man Whitney test, Wald statistic

[6 Lectures]

Uni- and Multi-variate Data Analysis: Simple and Multiple Linear Regression, Pearson Correlation and other measures of Association; Principal Component Analysis; ANOVA test (one- and two-way); Monte Carlo methods [7 Lectures]

Time Series Analysis: Stationary time series; Realization and Ensemble; Trend and Periodicity Analysis; Data Transformation; Time Series Modeling (auto-regressive, moving average, auto-regressive moving average model with Exogenous Inputs) [7 Lectures]

Copula: Sklar's Theorem; Properties of Copula Function; Survival Copula; Common Copula Function; Prediction using Copula [4 Lectures]

Programming assignments may be given to support the understanding of theory.

Text Books:

1. Maity, R. (2022). Statistical methods in hydrology and hydroclimatology. 2nd Edition, Springer.
2. Manly, B.F.J. (2008), Statistics for Environmental science and Management, Chapman and Hall/CRC Press.
3. Johnson, R.A. (2020), Miller and Freund's Probability and Statistics for Engineering, Pearson

Reference Books:

1. Ang, A. H-S. and W. H. Tang, Probability Concepts in Engineering: Emphasis on Applications to Civil and Environmental Engineering, John Wiley & Sons, Inc., USA, Second Edition, 2006.
2. Benjamin, J.R. and Cornell, C.A., Probability, Statistics and Decision for Civil Engineers, McGraw-Hill, New York, 2014.
3. Wilks, D S. (2011), Statistical Methods in Atmospheric Sciences, Academic Press.

Title	**Hydraulic Structures and Machines**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

1. To understand the subsurface flow physics under diversion structures and master the theories of seepage and uplift pressure.
2. To perform rigorous stability analysis for gravity and embankment dams under various loading conditions.
3. To design complex energy dissipation systems and spillway profiles to ensure structural safety.
4. To analyze the performance and selection criteria for hydraulic turbines and pumps in water resource projects.

Learning outcomes:

1. Apply Khosla's Theory to design safe foundations against piping and uplift.
2. Evaluate the safety of dams against overturning, sliding, and internal erosion (piping).
3. Design river training works like Guide Banks and Groynes based on river morphology.
4. Select and size Impulse and Reaction turbines based on specific speed and head requirements.
5. Predict and mitigate cavitation in hydraulic machines.

Course Contents:

Weirs and Barrages: Creep Theories (Bligh's, Lane's, Khosla's), Design of Weir/Barrages on Permeable Foundations, Critical Exit Gradient and Safe Exit Gradient. [8 Lectures]

Dams, Spillways and Energy Dissipators: Dams: types, forces, failure types and causes, stability analysis of Gravity and Homogenous Embankment dams, dam outlet structures, dam terminal structures, Estimation of Reservoir Capacity, sediment flushing from reservoir; Spillway: Types, Design Considerations, Controlled and Uncontrolled Spillway Characteristics, Design of Uncontrolled Ogee Spillway, Design of Chute/Sid-Channel/Syphon/Piano-key/Morning-glory Spillway (any one is sufficient), Energy Dissipators, Stilling Basin, Design of Indian Standard Hydraulic Jump Type Stilling Basins, Spillway Crest Gates and their design considerations. [16 Lectures]

River Training Works: River Classification, Behavior of rivers (straight reaches, bends, meanders), Control and Training of Rivers: Objective, Classification, Methods (Leeves, Guide Banks, Spurs/Groynes, and Marginal Bunds), Stability Analysis of Leeves, Design of Guide Banks. [8 Lectures]

Hydraulic Machines: Hydraulic Turbines: Impulse and Reactive Turbines, Runaway Speed, Simplitude in turbines, draft tube, cavitation, Selection and Design of a Hydro-Turbine for a Specific Site Profile; Pumps: Centrifugal and Reciprocating. [7 Lectures]

Text Books:

1. Garg, S.K (2023). Irrigation Engineering and Hydraulic Structures. Khanna Publishers.
2. Novak, P., Nalluri, C. & Narayanan, R. (2007). Hydraulic Structures (4th ed.). Taylor & Francis.
3. Subramanya. K. (2007). Hydraulic Machines. McGraw-Hill Education

Reference Books:

1. Peterka, A. J. (1984). Hydraulic design of stilling basins and energy dissipators (USBR Engineering Monograph No. 25). U.S. Bureau of Reclamation.
2. Julien, P. Y. (1987). River mechanics. Cambridge University Press.
3. Mays, L.W. (2010), Water Resources Engineering, John Wiley & Sons, Inc.
4. U.S. Bureau of Reclamation. (1987). Design of Small Dams
5. Central Water Commission (CWC). Manual on River Training, Control and Management.
6. Different IS Codes: IS 6512, IS 6966, IS 4997

Title	**Quantity Estimation, Costing and Construction Management**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite			

Objectives:

1. To provide a foundational understanding of estimation, costing, and rate analysis in construction projects.
2. To introduce principles of engineering economics for effective financial decision-making.
3. To equip students with essential techniques of project planning, scheduling, and control using modern tools.
4. To familiarize students with bidding, tendering, and construction resource management practices.

Learning Outcomes:

1. Prepare quantity estimates for basic civil engineering works and buildings.
2. Perform rate analysis considering materials, labour, equipment, and overheads.
3. Understand and participate in bidding and tendering processes and interpret basic contract documents.
4. Apply engineering economics principles to evaluate project alternatives and financial decisions.
5. Develop project schedules using CPM and PERT techniques.
6. Monitor and control projects using tools like Earned Value Management and basic time–cost optimization.
7. Identify and manage construction resources including manpower and equipment effectively.

Course Contents:

Estimation, Costing and Engineering Economics (21 Hours)

Introduction to Estimation: Necessity of estimation, Units and measurements, Types and methods of estimate, Load bearing vs framed structures, Overview of Standard Methods of Measurement (SMM) . [2 Lectures]

Quantity Estimation and BOQ: Application of Standard Methods of Measurement (SMM) in quantity take-off, Quantities of: Brickwork, RCC, PCC Plastering, painting, finishing works, Basic residential buildings (flat & pitched roofs), Arches (brickwork & RCC basics), Joineries (doors, windows, ventilators), Steel estimation for RCC (basics), Introduction to spreadsheet/software based estimation, Preparation of Bill of Quantities (BOQ) . [8 Lectures]

Rate Analysis and Costing: Rate analysis (materials, labour, equipment), Labour productivity,

Equipment costing basics, Overheads, contingencies, profit. [4 Lectures]

Bidding, Tendering and Contracts Tendering process, Types of tenders, Basics of contracts and essentials, Procurement overview. [3 Lectures]

Construction Economics: Time value of money, Discounted cash flow basics, Present worth and comparison of alternatives, Basic cost-benefit analysis. [4 Lectures]

Construction and Project Management (18 Hours)
Construction Resources, Safety and Quality: Resource requirements, Manpower planning basics, Equipment selection factors, Introduction to construction safety, Introduction to construction quality. [4 Lectures]

Project Planning and Scheduling: Project scope and planning stages, Work Breakdown Structure (WBS), Bar charts, CPM and PERT, Time computation, Introduction to software-based scheduling. [8 Lectures]

Project Control and Optimization: Schedule monitoring and control, Resource levelling and allocation, Crashing of networks Earned Value Management (EVM), Time-cost trade-off. [4 Lectures]

Project Management Practices: Risk management basics, Leadership and teamwork, Stakeholder management, Project handover basics. [2 Lectures]

Text Books:

1. Jha, K. N. (2015). Construction Project Management: Theory and Practice. Pearson Education India.
2. Dutta, B. N. (2021). Estimating and Costing in Civil Engineering. UBS Publishers & Distributors Pvt. Ltd.

Reference Books:

1. Oberlender, G. D. (2014). Project Management for Engineering and Construction (2nd ed.). McGraw-Hill Education.
2. Harris, F., McCaffer, R., Baldwin, A., & Edum-Fotwe, F. (2021). Modern Construction Management (8th ed.). John Wiley & Sons.
3. Sawhney, A., Riley, M., & Irizarry, J. (Eds.). (2020). Construction 4.0: An Innovation Platform for the Built Environment. Routledge.
4. Sears, S. K., Sears, G. A., & Clough, R. H. (2010). Construction Project Management: A Practical Guide to Field Construction Management (5th ed.). John Wiley & Sons.
5. Walker, A. (2015). Project Management in Construction (6th ed.). John Wiley & Sons.
6. Kohli, D. D., & Kohli, R. C. (2013). A Textbook of Estimating, Costing and Accounts (Civil). S. Chand & Company Ltd.

Online Course Material:

1. Principles of Construction Management – Prof. Sudhir Misra, IIT Kanpur (https://nptel.ac.in/courses/105104161)

Title	**Non Destructive Evaluation of Structures**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

1. To understand the underlying principles of the different non-destructive tests (NDTs) for assessing concrete quality.
2. To understand the capabilities and limitations of the various NDTs for assessing concrete quality.
3. To familiarise with the relevant provisions in the standards and codes of practice relating to NDT of concrete.

Learning Outcomes:

Following this course, the student will have the ability to

1. Develop a testing plan for non-destructive evaluation of structures
2. Identify and perform the relevant NDT suitable for different scenarios
3. Analyse and interpret the NDT results obtained

Course Contents:

Philosophy, Importance and need of non-destructive testing, defects in concrete structures, deteriorations in concrete and steel, testing and quality control of concrete, Planning non-destructive evaluation of structures, Evaluation and Interpretation of NDT Results. [10 Lectures]

Visual Inspection – Introduction, general procedures, and applications
Fundamental principles, test procedures, applications, interpretation and limitations of
Surface hardness test, penetration resistance test, pullout test, break-off test, core tests, and maturity methods.
Stress wave methods for flaw detection: Ultrasonic pulse velocity, Impact echo, Impulse response, shear wave tomography. [10 Lectures]

NDT for evaluating corrosion - Surface resistivity, Half-cell potential, Polarisation resistance, Electromagnetic cover measurement, Direct measurement of corrosion rates.
Test methods for measuring mass transport, carbonation, and moisture content. Chemical testing and allied techniques for concrete, Concrete Petrography. [10 Lectures]

Advanced Non-destructive techniques for concrete evaluation – Nuclear magnetic resonance, infrared thermography, electromagnetic tests for concrete, radioisotope gauge, Computer tomography, Magnetic-flux leakage. [9 Lectures]

Text Books:

1. Carino, N.J., and Malhotra, V.M. “Handbook on Non-destructive Testing of Concrete (2nd Ed)” Taylor and Francis, 2003
2. Maierhofer, C., Reinhardt, H., and Dobmann, G. “Non-destructive evaluation of reinforced concrete structures Volume 2: Non- Non-destructive testing methods” Woodhead Publishing Series in Civil and Structural Engineering, 2010

Reference Books:

1. Bungey, J.H, Millard, S.G., and Grantham, M. “Testing of concrete in structures (4th Ed)”, Taylor and Francis, 2006
2. Breysse D. “Non-Destructive Assessment of Concrete Structures: Reliability and Limits of Single and Combined Techniques” Springer 2012
3. International Atomic Energy Agency, “Guidebook on Non-destructive Evaluation of Concrete Structures”, 2002
4. American Concrete Institute “Report on Nondestructive Test Methods for Evaluation of Concrete in Structures” Report by ACI Committee 228

Title	**Flow in Open Channels**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
The course discusses the fluid mechanics of steady and unsteady free surface flow.

Learning outcomes:

1. Understand channel types, flow classifications, and basic equations governing flow and velocity distributions.
2. Learn about specific energy, critical flow conditions, methods for computing critical flow, and concepts like specific force.
3. Study shear stress distribution, apply Chezy's and Manning's equations, and understand uniform flow computation.
4. Learn the gradually varied flow (GVF), analyze flow profiles, and compute GVF for different conditions.
5. Understand flow dynamics over sharp-crested weirs, spillways, sluice gates, and hydraulic jumps.
6. Explore unsteady flow principles, Saint-Venant equations, surge propagation, and wave behavior in unsteady conditions.

Course contents:
Basic flow concepts: Types of channels, classification of flows, basic equations, velocity distribution, velocity coefficients, pressure distribution. [3 Lectures]

Energy and momentum principles: Specific energy, critical flow, section factor for critical flow computation, first hydraulic exponent, computation of critical flow, specific force, channel transitions, and chokes. [6 Lectures]

Uniform flows: Shear stress distribution, Chezy's equation, Manning's equation, conveyance of a channel, section factor for uniform flow computation, second hydraulic exponent, computation of uniform flow, Hydraulically Efficient Channel Section, Critical Slope. [6 Lectures]

Gradually varied flows: Differential equation of gradually varied flow (GVF), classification and analysis of flow profiles, Control Section, computation of GVF, Advanced Numerical Methods for solving GVF. [6 Lectures]

Rapidly varied flows: Flow over sharp crested weir, spillways, flow under sluice gate, types of hydraulic jump, general equation for hydraulic jump in prismatic channels . [6 Lectures]

Unsteady flows: Unsteady flow in open channels, derivation of Saint-Venant equations, surge propagation, negative and positive surges, different types of waves in River and Ocean systems. [6 Lectures]

Hydraulics of Mobile Bed Channel: Sediment Properties, Initiation of motion, Bed Load, Sediment Load. [6 Lectures]

Text Books:

1. M. Hanif Chaudhry (2008) Open-channel flow. Springer
2. V. T. Chow (1959) Open channel hydraulics. Mc-Graw Hill
3. S. Dey (2024) Fluvial hydrodynamics. Springer

Reference Books:

1. H. Chanson (2004) The hydraulics of open channel flow: an introduction. Elsevier
2. S. Montes (1998) The hydraulics of open channel flow. ASCE Press
3. K. Subramanya (2019) Flow in open channels. Mc-Graw Hill

Title	**Soil Mechanics**	Number	CIE2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered For	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	-		

Objectives:
To introduce the fundamental engineering principles of soil mechanics including their formation, classification, hydraulic behavior, stress distribution, consolidation and shear strength.

Learning outcomes:
Students will have the ability to:

1. Identify and classify soils based on index properties.
2. Assess soil phase relationships and engineering indices.
3. Analyze seepage and permeability problems.
4. Determine stress distribution in soil due to applied loads.
5. Evaluate consolidation and settlement behaviour.
6. Determine shear strength parameters of soils.

Course Contents:
Soil Formation: Phase Relationships, Index Properties and Compaction: Geological soil formation and classification, Types of soils, Soil particle size and shape, Clay minerals and clay-water interaction, Three phase system of soils, Phase relationships, Index properties of Soils, Grain size distribution, Atterberg limits, Soil classification systems (USCS), Relative density of sands, Mechanism of compaction, Standard and modified Proctor tests. [9 Lectures]

Permeability and Seepage: Darcy's law, Hydraulic gradient and seepage velocity, Coefficient of permeability, Constant head permeability test, Falling head permeability test, Permeability of stratified soils, Field permeability of soils, Seepage through soils, Flow net construction and applications, Quick sand condition and piping. [7 Lectures]

Effective Stress and Stress Distribution: Effective stress principle, Pore water pressure, Capillary rise in soils, Vertical stresses in soil mass, Stress distribution due to surface loads, Boussinesq's theory, Westergaard's theory, Influence charts. [7 Lectures]

Compressibility and Consolidation: Compressibility of soils, One-dimensional consolidation, Terzaghi consolidation theory, Consolidation test (oedometer), Determination of coefficient of consolidation, Primary and secondary consolidation, Consolidation settlement. [8 Lectures]

Shear Strength of Soils: Concept of shear strength, Mohr circle of stress, Mohr–Coulomb failure criterion, Stress conditions in soil mass, Direct shear test, Triaxial compression test, Unconfined compression test, Drained and undrained shear strength. [8 Lectures]

Text books:

1. Budhu, M. (2010). *Soil mechanics and foundations*. John Wiley and Sons.
2. Murthy, V. N. S. (2002). Principles and practices of soil mechanics and foundation engineering. *New York: Marcek Decker INC*.
3. Ranjan, G., & Rao, A. S. R. (2011). *Basic and applied soil mechanics*. New Age International.

Reference Books:

1. Terzaghi, K., Peck, R. B., & Mesri, G. (1996). *Soil mechanics in engineering practice*. John wiley & sons.
2. Lambe, T. W. (1951). *Soil testing for engineers* (Vol. 72, No. 5, p. 406). LWW.

Minor Modifications

Title	**Advanced Structural Analysis**	Number	CIL3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Structural Analysis		

Objectives:

This course aims at:

1. Introduction to advanced concepts of structural analysis and review the concepts of solid mechanics.
2. Further development of analytical skills to compute quantities like displacements, stresses etc. in different structures subjected to static forces.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
4. Gaining the basic understanding of structural stability.
5. Problem solving ability for static loads in structural engineering.

Learning Outcomes:

Students will have the ability to:

1. Analyze statically determinate and indeterminate trusses, beams and frames using matrix formulation.
2. Understand the fundamentals of the matrix formulation of structural analysis through finite element method.
3. Analyze cables, suspension bridges with three hinged and two hinged stiffening girders and statically determinate and indeterminate arches.
4. Determine bending moment, shear forces and axial force in the frames subjected to lateral and vertical loads using approximate method.
5. Perform comprehensive methods of structural analysis on real-world problems of practical interest.
6. Model real life structures for solving them in standard software and examine both local and global stability of the structural system.

Course Contents:

Indeterminate structure: Static and kinematic indeterminacy; Force method of analysis of indeterminate structures: Fixed and continuous beams, frames etc. - analysis by consistent deformation method; application of moment area and conjugate beam methods for fixed beams; theorem of three moments for continuous beams. [7 Lectures]

Displacement method of analysis of indeterminate structures: Slope deflection method: no chord rotation and with chord rotation (derivation); analysis of continuous beams; beams with overhang; analysis of rigid frames; frames with

sloping legs; settlement effects; moment distribution method. [5 Lectures]

Matrix Method of Analysis: Flexibility matrix method of analysis; stiffness matrix method of analysis; stiffness matrix for a beam-column element; direct stiffness approach; transformation; static condensation; rigid end offset. [6 Lectures]

Introduction to FEM: Energy methods and variational principles; shape functions; use of virtual work method for generating stiffness matrix of structure. [4 Lectures]

Stability of structure: Concept of structural stability; classical stability analysis of beams and frames; energy method of the stability analysis of beams, frames and plates. [6 Lectures]

Plastic analysis: Introduction to plastic theory; concept of plastic hinge; plastic modulus; shape factor; redistribution of moments; collapse mechanism; plastic analysis of beams and portal frames by equilibrium and mechanism methods; moment-curvature for arbitrary composite section. [5 Lectures]

Introduction to structural dynamics: Basic concepts of dynamic loading and response; types of mass-spring damper system; equation of motion using direct equilibration, Hamilton's principle. [2 Lectures]
Single-degree of freedom system: Development of equation of motion; statement of problem and solution; damped and undamped systems; free vibration response. [4 Lectures]

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
3. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
4. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.
5. J.B. Wilbur, C.H. Norris, S. Utku (2006) Elementary Structural Analysis, McGraw Hill, New York. ISBN 9781258242909.

Online Course Material:

1. Advanced Structural Analysis (2012) by Prof. Devdas Menon on NPTEL. https://nptel.ac.in/courses/105/106/105106050/

Title	**Green Building and Sustainable Materials**	Number	CIL7440
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:

Objectives of this course are:

1. To introduce elements of green building construction and use of sustainable materials.
2. To familiarize students with different national and international green performance ratings.
3. To teach students methods of carbon footprint reduction and increase in energy efficiency of a building.

Learning Outcomes:

A graduate of this course will have a thorough idea of:

1. National and international green building assessment system and requirements.
2. Green building design elements and use of sustainable materials.
3. Techniques to reduce the carbon footprint of a green building.
4. Management of hydrological cycle, treatment of waste, and maintainance of indoor air quality.
5. Green building economics, business plan, and management.

Course Content:

Introduction to green building: Rationale for high-performance green building; green building progress and obstacles; trends in high-performance green building; the green building movement. [2 Lectures]

Ecological design: design vs ecological design; contemporary ecological design; future ecological design; thermodynamics: limits on recycling and the dissipation of materials. [2 Lectures]

Green building assessment: US green building assessment system; international green building assessment system; performance assessment. [2 Lectures]

LEED building rating system: history of LEED; structure of the LEED suite of building assessment systems; LEED credentials; the LEED process; LEED categories. [2 Lectures]

The Green Globes building assessment system: the Green Globes process; Green Globes verification and certification; structure of the ANSI/GBI 01-2010 standard; the Green Globes professional and Green Globes Assessor. [2 Lectures]

GRIHA/IGBC: Overview of IGBC; IGBC Certification Process; Overview of GRIHA; GRIHA Certification Process; Variants and Professionals. [2 Lectures]

Green building design process: conventional vs green building delivery systems; executing the green building project; the integrated design progress; role of Charrette in the design process; green building documentation requirements. [2 Lectures]

Sustainable site and landscape: land and landscape approaches for green buildings; sustainable landscapes; enhancing ecosystems; stormwater management; low-impact development; heat island mitigation; light trespass and pollution reduction; sustainable site initiative, Case study – IIT Jodhpur. [5 Lectures]

Life cycle assessment (LCA) for assessing the energy and carbon footprint reduction in buildings: LCA as a tool of building materials and products; environmental product declaration;materials and product certification systems; emerging construction materials and products; distinguishing between green building products and green building materials; Closing materials loops: design for deconstruction. [4 Lectures]

Energy and carbon footprint reduction – Embodied energy: Wall components and alternative systems for construction; Flooring systems and alternatives; Roofing systems and alternatives. [4 Lectures]

Energy and carbon footprint reduction: building energy issues; high-performance building energy design strategy; passive design strategy; building envelope; internal load reduction; active mechanical system; water-heating system; electrical power systems; innovative energy optimization strategies; renewable energy systems; fuel cells; smart buildings and energy management systems; ozone-depleting chemicals in HVAC&R systems; reducing carbon footprint of the built environment. [4 Lectures]

Building Information Modelling (BIM): Fundamentals of BIM, Various tools and their applications, BIM in building design, BIM Ecosystem, BIM aided green building design. [2 Lectures]

Built environment hydrological cycle: global water resource depletion; hydrological cycle terminology; high-performance building hydrological cycle strategy; designing the high-performance building hydrological cycle; water budget rules of thumb; sustainable stormwater management; landscaping water efficiency. [2 Lectures]

Indoor environmental quality: IEQ issues; design; main components of integrated IEQ design; emission from building materials; economic benefits of good indoor environmental quality. [2 Lectures]

Green building economics: the business case for high-performance green building; the economics of green building; quantifying green building benefits. [2 Lectures]

Text Books:

1. Charles J. Kibert (2013). Sustainable Construction. John Wiley & Sons. ISBN 978-1-118-33013-5
2. J. Newman and Ban Sang Choo (2003) Advanced Concrete Technology-Constituent Materials (1st Ed), Elsevier. ISBN 9780080489988.
3. S. Kubba (2010) LEED Practices, Certification, and Accreditation Hand book (1st Ed) Elsevier. ISBN 9780080958590.

Reference Books:

1. Ministry of Power, Energy Conservation Building Code 2018, Revised Version, Bureau of Energy Efficiency, 2018,
2. Indian Building Congress, Practical Handbook on Energy Conservation in Buildings, 1 st ed. Nabhi Publication, 2008.
3. TERI-Griha's Green Design practices (www.teriin.org/bcsd/griha/griha.htm)
4. Sartori, I., and Hestnes, A. G. "Energy use in the life cycle of conventional and low-energy buildings: A review article", Energy and Buildings., 20, 249-257.2007
5. Green Building Basics, California Integrated Waste Management Board (www.ciwmb.ca.gov/GREENBUILDING/Basics.htm#What)
6. Catarina Thormark. "A low energy building in a life cycle—its embodied energy, energy need for operation and recycling potential", Building and Environment., 37, 429-435.2001.

Online Course Material:

1. "NOC:Energy Efficiency, Acoustics and daylighting in Building" by Prof. B. Bhattacharjee on NTPEL. https://nptel.ac.in/courses/105/102/105102175/

Title	**Ground Improvement Techniques**	Number	CIL4380
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
This course aims at :

1. Introducing the need of ground improvement and available techniques.
2. Introducing the aspects of soil shear strength modification for effective design of foundations.
3. Introducing recent soil modification techniques.

Learning Outcomes:
Students will have the ability to

1. Apply the ground modification techniques for efficient design of foundations
2. Utilize the concepts of preloading and dewatering in soils of cohesive nature
3. Stabilize the soils using different mechanical and chemical approaches

Course Contents:
Classical examples of soil failure, need of ground improvement, In-situ densification methods, In-situ densification of granular and cohesive soils; vibration at ground surface and at depth. [9 Lectures]

Preloading and dewatering, prefabricated vertical drains (PVDs), sand drains, geo-drains, filter requirements, spacing effects, stone columns, sumps and interceptor ditches, single and multi-stage well points, vacuum well points, horizontal weIIs, eIectro-osmosis. [10 Lectures].

Stabilization of soils, methods of soil stabilization- mechanical, cement, Iime, bitumen and polymer stabilization, use of industrial wastes like fly ash and granulated blast furnace slag, stabiIization of expansive soils; stabilization with cement, lime and chemicals, Grouting techniques, types of grouts, grouting equipment and machinery, injection methods, grout monitoring. [10 Lectures]

Geosynthetics, type of geosynthetics, geotextiles, geomembranes and gabions, Reinforced Earth Technology, concept of soil reinforcement; reinforcing materials; backfill criteria; design of reinforcement for internal stability; applications of reinforced earth structures; miscellaneous methods-soil nailing, thermal methods. [10 Lectures]

Text Books:

1. Hausmann, M.R. (1990). Engineering Principles of Ground Modification,

McGraw-Hill International Editions. ISBN: 9780070272798.
2. Dr. Purushottama Raj (2005). Ground Improvement Techniques. Laxmi Publications. ISBN: 9788170088370.
3. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.

Online Course Material:

1. Ground Improvement Techniques by Prof. G.L. Sivakumar Babu on NTPEL. https://nptel.ac.in/courses/105/108/105108075/
2. Ground Improvement Techniques by Prof. Dilip Kumar Baidya on NTPEL. https://nptel.ac.in/courses/105105210

Title	**Stability of Slopes**	Number	CIL7410
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (CIE)	Type	Program Elective (PE)
Prerequisite	-		

Objectives:
This course aims at:

1. To understand the influence of geology, rockmass properties, and hydrology on the stability of slopes.
2. To present basic types of analytical and numerical methods of slope stability analysis.
3. To introduce methods and instruments of slope monitoring and stabilization.
4. To enable planning and design of preventive measures, and redesign of failed slopes

Learning Outcomes:

1. Knowledge of data collection from field-work, aerial photography, geophysics, and geological map.
2. Knowledge of kinematic analysis of slopes using geological data and stereographic projection.
3. Knowledge of kinetic analysis of slopes and influence of critical geometrical and hydrological parameters.
4. Knowledge of available methods of slope stabilization by reinforcements and rock removal.
5. Knowledge of instrumental facilities to monitor slope movement and subsequent classification.

Course Contents:
Structural geology for slope analysis: Geological structures; Orientation of discontinuities; Introduction to geological maps; Identification of modes of slope instability. [3 Lectures]

Site investigation and geological data collection: Site reconnaissance; Geological mapping; Spacing, persistence, and roughness of discontinuities; Probabilistic analysis of structural geology; Drilling and logging. [4 Lectures]

Rock strength properties: Scale effect and rock strength; Shear strength of discontinuities; Laboratory testing of discontinuities; Hoek-Brown strength criterion for fractured rock mass; Rock durability and compressive strength. [4 Lectures]

Plane failure: Conditions of plane failure; Influence of groundwater on stability; Critical tension crack: identification and influence; Influence of critical slide plane inclination; Analysis of failure on a rough plane; Reinforcement of a slope; Seismic analysis of slopes. [5 Lectures]

Wedge failure: Conditions of wedge failure; Wedge analysis including cohesion, friction, and water pressure; Wedge stability chart for friction; Compression wedg analysis. [5 Lectures]

Circular failure: Conditions of circular failure; Methods of analysis; Derivation and use of circular failure charts; Location of critical slide surface and tension crack; Bishop's and Janbu's method of failure analysis. [5 Lectures]

Toppling failure: Types of toppling failure; Kinematic analysis of block toppling failure; Limit equilibrium analysis of toppling. [5 Lectures]

Stabilization of slopes: Stablization programs: Rock slope Inventory, Hazard rating analysis; Stabilization by rock reinforcements; Stabilization by rock removal; Measures against rock fall. [4 Lectures]

Movement monitoring: Types of slope movement: Initial response, Regressive and progressive movements, Long-term creep; Surface monitoring methods; Data interpretation. [4 Lectures]

Text Books:

1. Duncan C Wyllie, Christopher W. Mah (2005) Rock Slope Engineering, 4th Ed, Spon Press (Taylor & Francis Group), ISBN 0–415–28001–X (pbk).
2. Robin Chowdhury, Phil Flentje, Gautam Bhattacharya (2010) Geotechnical SlopeAnalysis. CRC Press (Taylor & Francis Group), ISBN: 978-0-415-46974-6 (Hbk).

Laboratory Courses

Title	**Civil Engineering Laboratory – I**	Number	CIP2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-4 [2]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite / Requisite	Building Materials and Construction Technology; Geoinformatics		

Building Materials and Construction Technology Laboratory

Objectives:
To familiarize the students with testing of building materials

Learning Outcome:
Following this course, the student will have the ability to:

1. Assess and interpret the standard materials testing data as per the relevant standards
2. Prepare reports and communicate the materials testing reports
3. Recommend the right building materials based on the material testing reports

List of Experiments:
Cements

1. Specific Gravity, Consistency, Initial and Final Setting Time
2. Blaine's air permeability test, and Sieve Analysis
3. Mortar Strength Test

Aggregates

4. Specific gravity, bulk density and water absorption of coarse and fine aggregates
5. Sieve analysis of fine and coarse aggregates, and Flakiness and elongation index tests
6. Aggregate crushing strength tests

Concrete

7. Workability Tests on Concrete – Slump, Compaction Factor, and Vebe Tests
8. Mechanical Properties of Concrete – 7 and 28 Days compressive/split tensile/flexural strengths, and Modulus of Elasticity of Concrete
9. Self Compacting Concrete - Slump flow, L-Box, V-funnel and J-Ring tests

Rebar

10. Tests on Rebars - tension, stress-strain, and pullout from concrete for the bond

properties

Miscellaneous

11. Bricks – Dimensions check, Compressive strength, water absorption, importance of frog

Text Books:

1. P.C. Varghese (2008) Building Materials, Prentice-Hall of India, New Delhi.
2. M. Soutsos and P. Domone (2018) Construction Materials, Their Nature and Behavior, Fifth Edition, CRC Press.

Reference Books:

1. Relevant BIS standards

Geoinformatics Laboratory

Objectives:

This course aims at:

1. To introduce the basic principles of remote sensing, GPS, Surveying and GIS.
2. To teach application of geo-informatics in geoscience, ecology and environment study, agriculture, and disaster management.

Learning Outcomes:

A graduate of this course will have a thorough idea of:

1. Multispectral, thermal infrared, and microwave remote sensing.
2. Surveying Principles and handling modern surveying equipment
3. Different map projections and use of GPS.
4. Component GIS and applications.
5. Different cartographic models, implementation, validation, and visualization.

List of laboratory exercises:

1. Establishment of Local Control Network and Vertical Profiling using modern surveying Instruments (Total Station or Theodolite, Digital level and EDM).
2. Field data collection using handheld and Differential GPS.
3. Georeferencing when coordinate values are given/not given in a GIS platform.
4. Creating maps using vector data models and cartographic guidelines.
5. Working with Digital Elevation Model (DEM) and terrain analysis.
6. Drone data acquisition using Multi spectral sensor and Lidar sensor. Image data processing and generation of ortho-rectified Images (ORI).
7. Site suitability analysis for service buildings using vector and raster data.
8. Procurement and preparation of land use and land cover map using satellite

images for a specified study area.

9. Developing DEM and CHM using LIDAR point cloud data.
10. Network analysis to locate suitable routes.
11. Creating a Web app serving a GIS model (created using model builder) in a GIS platform.

Note: Lab. will have hands-on use of Google Earth Engine, QGIS, ERDAS IMAGINE, ArcGIS, or other relevant software (any one or two tools will suffice).

A surveying excursion for 1 week (or less period) may be planned for nearby places to make students familiar with practical experience in mapping, and using surveying equipment in real-world environments.

Text Books:

1. Satheesh Gopi, R. Sathikumar, and N. Madhu (2025) Advanced Surveying: Total Station, GPS, GIS & Remote Sensing (2nd Ed). Pearson India.
2. T. M. Lillesand, R. W. Kiefer, J. W. Chipman (2004) Remote Sensing and Image Interpretation (5th Ed). Wiley. ISBN: 0-471-15227-7
3. G. S. Srivastava (2014) An Introduction to Geoinformatics. McGraw Hill Education (India) Private Limited. New Delhi. India. ISBN: 9781259058462.
4. Michael N. DeMers (2008) Fundamentals of Geographic Information Systems (4th Ed). Wiley. ISBN: 978-0-470-12906-7.

Reference Books:

1. Basudeb Bhatta (2021) Remote Sensing and GIS (3rd Ed). Oxford University Press.
2. L. E. Johnson (2009) Geographic Information Systems in Water Resources Engineering. CRC Press. Taylor & Francis Group. ISBN 978-1-4200-6913-6.
3. B. Tomaszewski (2020) Geographic Information Systems (GIS) for Disaster Management (2nd Ed). CRC Press.

Online Course Material:

1. Prof. Arun K. Saraf (2016) Introduction to Geographic Information System. NPTEL. https://nptel.ac.in/courses/105/107/105107155/
2. Prof. Arun K. Saraf (2017) Introduction to Remote Sensing. NPTEL. https://nptel.ac.in/courses/121/107/121107009/

Title	**Civil Engineering Laboratory – II**	Number	CIP2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite	Hydraulic Engineering		
Requisite	Structural Analysis; Soil Mechanics		

Structural Analysis Laboratory

Objectives:
This course aims at:

1. Introduction to fundamental concepts of structural analysis.
2. Development of analytical skills to compute quantities like displacements, stresses etc. in structures
3. subjected to static forces.
4. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
5. Problem solving ability for static loads in Civil Engineering design of structures.

Learning Outcomes:
Students will have the ability to:

1. Establish equilibrium equations of structural systems.
2. Analyze the responses and stresses generated in structures subjected to different types of static loading.
3. Apply the structural analysis methods to solve real-world problems of the building, bridges etc.

List of Experiments:

1. Verification of moment area theorem for the slope and deflection of beam.
2. Verification of Clerk-Maxwell's reciprocal theorem.
3. Experimentally determining the elastic properties of an elastic beam.
4. Experimental and analytical study of defections in a cantilever beam.
5. To study the behaviour of struts and columns with various support end conditions.
6. Determination of the deflection of a pin connected truss through analytical, graphically and experimental calculations.
7. Determine the horizontal thrust in a three hinged arch for a given system of loads experimentally and verify it with theoretical calculations.
8. Experimental and analytical study of sway in portal frame.

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata
3. McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
4. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
5. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.

Online Course Material:

1. Structural Analysis I (2009) by Prof. Siddhartha Ghosh and Prof. R.S. Jangid on NPTEL. https://nptel.ac.in/courses/105/101/105101085/

Hydraulic Engineering Laboratory

Objectives:
The instructor will

1. Provide hands-on experience with the fundamental theoretical principles of fluid mechanics
2. Introduce various forms of flow measuring devices and techniques

Learning outcomes:
Upon completion of this laboratory course, students will be able to:

1. verify fundamental theoretical principles of fluid mechanics through hands-on experimental measurement
2. calibrate and operate various instruments and assess their accuracy, key hydraulic parameters and suitability for different flow conditions
3. compare experimental results with theoretical principles, and draw evidence-based engineering conclusions

Contents:

1. Observation of streamline formation and flow behaviour in a transparent flow channel
2. Determination of metacentric height of floating bodies
3. Determination of the Reynolds number for fluid flow through a pipe and thereby identify the nature of the flow regime
4. Verification of Bernoulli's equation experimentally and calculation of the total energy at different points
5. Determination of the coefficient of discharge of a venturi meter
6. Determination of the coefficient of discharge of an orifice meter
7. Study the impact of a jet on different types of vanes:

8. Hemispherical Vane
9. Flat Plate Vane
10. Conical vane
11. Study and compare the water surface profile of a forced vortex
12. Determination of head loss in pipes having different diameters, different materials and different roughness
13. Determination of pressure drop at different types of bends and valves
14. Determination of the friction factor for laminar and turbulent flow through pipes
15. Determine the coefficient of discharge for weirs and notches (rectangular, triangular and trapezoidal)

Text Books:

1. Fox, R. W., Pitchard, P. J., and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
2. White, F. M. (2003), Fluid Mechanics, 5th Edition, McGraw-Hill.
3. Cimbala, J.M. and Cengel, Y.A., (2010), Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

Soil Mechanics Laboratory

Objective:

This course aims at:

1. Familiarizing students with the physical and engineering properties of soil.
2. Introducing the laboratory testing equipment and their functioning principles.
3. Introducing basic field soil testing techniques.

Learning outcomes: Students will be able to:

1. Classify the soils roughly based on visual identification
2. Determine the index and engineering properties of soils
3. Use simple and complex field and laboratory testing techniques to determine the shear strength of soils

Course Contents:

1. Visual classification of soils
2. Specific gravity and moisture content determination
3. Dry and wet sieve analysis of soils
4. Atterberg limits: liquid limit; plastic limit and shrinkage limit
5. Permeability of soils through constant head and falling head tests
6. Field unit weight through sand replacement and core cutter tests
7. Laboratory compaction: light weight and heavy weight
8. Consolidation of soil
9. Unconfined compression and CBR test

10. Triaxial and direct shear test

Text Books:

1. Das B.M. (2009). Soil mechanics laboratory manual (7th Ed). OUP USA International Edition. ISBN 10-0195367596.
2. Fratta D. Aguettant J. Smith L.R. (2007). Introduction to Soil Mechanics Laboratory Testing. CRC Press-Taylor & Francis Group (1st Ed). ISBN-10: 1138430242
3. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.

Online Course Materials:

1. NPTEL Soil Mechanics/Geotechnical Engineering-I, IIT Kharagpur, https://nptel.ac.in/courses/105105168
2. MIT Open Course Ware, Advanced Soil Mechanics, https://ocw.mit.edu/courses/1-361-advanced-soil-mechanics-fall-2004/pages/lecture-notes/

Title	**Civil Engineering Laboratory - III**	Number	CIP3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite / Requisite	Transportation Engineering – I; Water and Wastewater Engineering; Reinforced Concrete Design		

Transportation Engineering Laboratory

Objectives:
The instructor will:

1. Introduce students to laboratory testing procedures related to highway construction materials and pavement engineering.
2. Provide hands-on experience in evaluating the engineering properties of aggregates and bituminous materials used in transportation infrastructure.

Learning Outcomes:
The students will have the ability to:

1. Perform standard laboratory tests on aggregates and bituminous materials as per relevant IS/IRC specifications.
2. Determine engineering properties of pavement materials and assess their suitability for highway construction.

Contents:

1. 10% fine value test
2. Aggregate Impact Test
3. Los Angles Abrasion Test
4. Specific Gravity and Water Absorption Test on Aggregates
5. Shape Tests
6. Penetration Test and Softening Point Test
7. Viscosity Test
8. Ductility Test
9. Flash & Fire Test
10. Specific Gravity Tests on Bitumen
11. Marshall Stability Test

Text Books:

1. Khanna, S.K., Justo, C.E.G. and A Veeraragavan, "Highway Material and Pavement Testing Manual", Nem Chand & Bros
2. IRC and IS Codes related to Highway Materials Testing

Environmental Engineering Laboratory

Objectives:
The Instructor will:

1. Provide knowledge on water quality standard.
2. Exposure to wastewater disposal guideline and standard.
3. Provide hands on experience on water quality testing and wastewater characterizations

Learning Outcomes:
The students are expected to be able to:

1. Learn to test water quality parameters
2. Learn to characterize wastewater
3. Assess the quality with respect to standard guideline for suiatble decision making.

Laboratory Components:

1. Introduction to environmental engineering laboratory
2. Determination of alkalinity in water
3. Determination of hardness in water
4. Determination of TDS and TSS : Gravimetric method
5. Determination of pH, TDS or EC and turbidity
6. Determination of optimal coagulant dosing: Jar test
7. Determination of COD in wastewater
8. Determination of BOD in wastewater
9. Coliform test: demonstration and analysis
10. Air quality laboratory: Measurement of PM2.5 and PM10
11. Demonstration of weather monitoring station: Sampling and analysis

Text Books:

- Metcalf & Eddy, Wastewater Engineering- Treatment and Reuse (Revised by G. Tchobanoglous, F. L. Burton and H.D. Stensel), Tata McGraw Hill, 4th Edn., 2004.H.
- Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
- M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Reference Books:

1. Nathanson and Jerry, Basic Environmental Technology: Water Supply, Waste Management and Pollution Control, Prentice Hall India Learning Pvt. Ltd., India 2009

2. Garg, Environmental Engineering: Water Supply Engineering, Khanna Publishers, 18th Edn, India 2004
3. Manual on Sewerage and Sewage Treatment Systems, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 2013.
4. Manual on Water Supply and Treatment, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 1999.

Online Course Material:

1. Guidelines for drinking-water quality, World Health Organization 2022, ISBN 978-92-4-004506-4, https://www.who.int/publications/i/item/9789240045064
2. Drinking Water – Specification, Bureau of Indian Standard , IS 10500 : 2012, New Delhi, India, 110002

Reinforced Concrete Design Laboratory

Objectives:

This course aims at:

1. Providing an in-depth knowledge on the behaviour of RCC structures.
2. Highlighting the significance of working stress and limit state design methods.
3. Comprehensive development of the design and reinforcement detailing of beams, columns, slabs, staircases, footings.
4. Develop an understanding with respect to the importance of stiffness, strength, ductility and configuration which is useful for earthquake resistant design.
5. Practicing reinforcement detailing procedures of a RCC building considering wind and seismic forces as per Indian Standard codes and developing design skills.

Learning Outcomes:

Students will have the ability to:

1. Assess the behaviour of RCC structures and apply the appropriate design methodology and reinforcement detailing for different types of RCC structural elements i.e. beams, columns, slabs, staircases, foundations etc.
2. Design and provide reinforcement detailing of multistoried RCC buildings considering dead, wind and seismic forces as per Indian Standard codes.

Laboratory Contents:

1. Introduction of computer applications in the structural design.
2. Design of beam (under-reinforced, balanced, over-reinforced, doubly

reinforced) using computer applications.

3. Design and detailing of slabs (one-way and two-way), columns, walls, footing etc.
4. Modelling, analysis and design of a multi-storeyed framed structure.
5. Determination of the compressive strength of concrete.
6. Strength and deflection testing of a simply supported RCC beam.
7. Testing of RCC columns for strength subjected to concentric and eccentric loading conditions.
8. Non-destructive testing – rebound hammer test for concrete, ultrasonic pulse velocity test for concrete, rebar detector test for reinforcement etc.

Text Books:

1. U.S. Pillai, D. Menon (2016) Reinforced Concrete Design (3rd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 9780070141100.
2. S.N. Sinha (2014) Reinforced Concrete Design (3rd Ed) McGraw Hill Education Pvt. Ltd. ISBN 9789351342472.
3. A.K. Jain (2012) Reinforced Concrete - Limit State Design (7th Ed), NemChand Brothers. ISBN 9788185240664 50
4. P.C. Varghese (2010) Limit State Design of Reinforced Concrete (2nd Ed), Prentice Hall of India. ISBN 8120320395.
5. S.K. Mallick, A.K. Gupta (2002) Reinforced Concrete (2nd Ed), Tata McGraw-Hill Publishing Co. Ltd. ISBN 0070473323.

References Books:

1. Design of Reinforced Concrete Structures (2009) by Dr. Nirjhar Dhang on NPTEL. https://nptel.ac.in/courses/105/105/105105105/

Title	**Civil Engineering Laboratory – IV**	Number	CIP3XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech. (CIE)	Type	Program Compulsory (PC)
Prerequisite / Requisite	Water Resources Engineering; Advanced Structural Analysis; AI/ML in Civil Engineering		

Water Resources Engineering Laboratory

Objectives:
In this course the students will:

1. Learn to describe the fundamental concepts of hydrology
2. Understand various methods of hydrologic analysis
3. Apply various hydrologic and hydraulic analysis methods to design stormwater drains, dams, irrigation and diversion structures

Learning Outcomes :
At the end of the course, the student should be able to:

1. Conceptualize different hydrologic process based on physical laws
2. Apply the concept of flood and reservoir routing in real data and solve the field issues
3. Find solutions for water resources issues such as urban flooding, water management and groundwater depletion.

Irrigation Engineering Design Laboratory:

1. Design of Irrigation Canal [lined and unlined]
2. Design of canal outlets
3. Design of Alluvial channels carrying clear and sediment laden water
4. Design the section of hydraulic jump
5. Design of cross drainage structures with typical plan and section
6. Design of profile of ogee spillway
7. Design of Trapezoidal Notch Fall, SARDA fall and Glacis Fall
8. Design of Cisterns
9. Analysis and Design of gravity dam
10. Analysis and Design of earth embankment dam
11. Problems related to energy dissipaters

Text Books:

1. Mays, L.W. *Water Resources Engineering*, John Wiley and sons, Inc., Kundli
2. Todd, D. K.and Mays, L. W. *Ground Water Hydrology,* John Wiley India Edition

Reference Books:

1. Chow, V.T., Maidment, D.R., and Mays L.W. Applied Hydrology. McGraw-Hill Inc., New York.
2. James, L.D., and Lee, R.R. Economics of water resources planning. McGraw-Hill Book Inc., New York.
3. Garg, S.K. Irrigation Engineering and Hydraulic Structures. Khanna Publishers, Delhi.
4. Linsley, R.K., Franzini, J.B., Freyberg, D.L., and Tchobanoglous, G. Water Resources Engineering, McGraw Hill Inc., Singapore.
5. Singh, V.P. Elementary Hydrology. Prentice Hall, Englewood Cliffs, New Jersey.

Advanced Structural Analysis Laboratory

Objectives:

This course aims at:

1. Introduction to advanced concepts of structural analysis and review the concepts of solid mechanics.
2. Further development of analytical skills to compute quantities like displacements, stresses etc. in different structures subjected to static forces.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
4. Gaining the basic understanding of structural stability.
5. Problem solving ability for static loads in structural engineering.

Learning Outcomes:

Students will have the ability to:

1. Analyze statically determinate and indeterminate trusses, beams and frames using matrix formulation.
2. Understand the fundamentals of the matrix formulation of structural analysis through finite element method.
3. Analyze cables, suspension bridges with three hinged and two hinged stiffening girders and statically determinate and indeterminate arches.
4. Determine bending moment, shear forces and axial force in the frames subjected to lateral and vertical loads using approximate method.
5. Perform comprehensive methods of structural analysis on real-world problems of practical interest.
6. Model real life structures for solving them in standard software and examine both local and global stability of the structural system.

Laboratory Contents:

1. Introduction of computer applications in structural analysis.
2. Create and analyze model to determine the shear force, bending moment, deflection diagrams for statically indeterminate beams.
3. Create and analyze model to determine the shear force, bending moment, deflection diagrams for portal frames.
4. Create and analyze model to determine the shear force, bending moment, deflection diagrams for multi storeyed building.
5. Determine shear force, bending moment, deflection diagrams for multi-storey frames using approximate methods.
6. Modelling and analysis of 3D structures.

Text Books:

1. C. K Wang (2010) Intermediate Structural Analysis (1st Ed), Tata McGraw Hill. ISBN 9780070702493.
2. S. Timoshenko, D.H. Young, J.V. Rao, S. Patil (2013) Engineering Mechanics (In SI Units) (5th Ed), Tata McGraw-Hill Education Pvt. Ltd. ISBN 9781259062667.
3. R.C. Hibbeler (2018) Structural Analysis (10th Ed), Pearson Education. ISBN 9780134610672.
4. D. Menon (2018) Structural Analysis (2nd Ed), Narosa Publications. ISBN 9788184875973.
5. J.B. Wilbur, C.H. Norris, S. Utku (2006) Elementary Structural Analysis, McGraw Hill, New York. ISBN 9781258242909.

Reference Books:

1. Advanced Structural Analysis (2012) by Prof. Devdas Menon on NPTEL. https://nptel.ac.in/courses/105/106/105106050/

AI/ML in Civil Engineering Laboratory

Objectives:

1. Comprehensive understanding of the problem settings and application of AI/ML in civil and infrastructure engineering.
2. Development of programming skills to execute AI/ML algorithms in civil and infrastructure engineering.
3. The ability to bridge the gap between traditional engineering physics and data-driven discovery.

Course Contents:

Tentative list of laboratory exercises:

1. Exploratory Data Analysis and Data Preprocessing Pipelines
2. Linear Regression and Logistic Regression using a single Perceptron

3. Regularization and performance metrics for Regression and Classification
4. Training Autoencoders on sensor data for unsupervised anomaly detection.
5. Automated Visual Inspection and classification (or image segmentation) using CNNs and Transfer Learning
6. Time Series Forecasting using LSTM & Transformers
7. Segmentation of multispectral remote sensing images
8. Grid Resilience Analysis using Graph Neural Networks
9. Interpreting and Auditing Deep Learning Models using xAI for Structural Safety or other critical systems, such as drinking water supply.
10. Building a fast surrogate model for flood routing, soil settlement under dynamic loading, or beam deflection, and making it Physics-Informed.

Text Books:

1. J.-A. Goulet (2020) Probabilistic Machine Learning for Civil Engineers, The MIT Press.
2. Goodfellow, I., Bengio, Y., and Courville, A. (2016) Deep Learning, MIT Press.
3. S. Shalev-Shwartz, and S. Ben-David (2014) Understanding Machine Learning From Theory to Algorithms, Cambridge University Press. ISBN 9781107057135.

Reference Books:

1. R. S. Sutton and A. G. Barto (2018) Reinforcement Learning: An Introduction (2nd Edition), MIT Press.
2. A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola (2023) Dive into Deep Learning, Cambridge University Press.
3. Y. Singh et al. (2023). *Machine Learning Applications in Civil Engineering*, Elsevier.
4. E. Stevens, L. Antiga, and T. Viehmann (2020) Deep Learning with PyTorch, Manning Publications.
5. C M Bishop and H Bishop, 2024, Deep Learning Foundations and Concepts, Springer Nature, Switzerland, AG 2024.

Online Course Material:

1. Raissi, M., Perdikaris, P. and Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, pp. 686-707. https://doi.org/10.1016/j.jcp.2018.10.045
2. Torralba, A., Isola, P., & Freeman, W., Foundations of Computer Vision, The MIT Press, https://visionbook.mit.edu/

CHECKLIST FOR THE NEW COURSE APPROVAL

Hydraulic Engineering

1. Justification for the proposal in line with the philosophy of the program.
   This course provides a rigorous and comprehensive foundation in the fundamental laws governing fluid. The course also helps the students understand the importance and application of fluid mechanics. Hence, the course lays the basic foundation for more advanced water resources courses, which are required for any Civil Engineering Graduate.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Saran Aadhar, Dr. Deepak Swami

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   No lab component in this course. Lab exercises are separated as a different course.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources. **I therefore recommend that the revised B.Tech. course syllabi in Water Resources be approved.**	Thank you for the review. No change required.
2.	Prof. Pramod Kr. Shrma	Professor in Hydraulic Engineering,	No major comments. It looks fine.	Thank you for the feedback. No major change required.

		Department of Civil Engineering Indian Institute of Technology Roorkee		
3.	Prof. Atul Dhar	Professor, School of Engineering, Indian Institute of Technology Mandi	The proposed courses are well aligned with the standard fluid mechanics courses in the UG curriculum of Civil, Mechanical, Chemical, Aerospace engineering etc. Few minor points may be taken into consideration for improvement: 1.In internal flows: Introduction of water hammer effect and cavitation concepts may also be added. 2. Lectures in introduction may be reduced and open channel may be in created. 3. The lab should not specify the theory course as prerequisite. It is better to give students option to take both theory and lab courses simultaneously.	Thank you for the feedback. The introduction to water hammer effect and cavitation is now added to the internal flow section. Due to limited lecture length of this course, flow in the open channel is designed as a different course.
4.	Prof. C.S.P. Ojha	Professor, Department of Civil Engineering Indian Institute of Technology Roorkee	1. Knudsen Number is already used in defining continuum hypothesis, so can be dropped. 2. Hydrostatic force on submerged surface can be replaced “hydrostatic force on submerged planar and curved surfaces”. 3. “Acceleration field of fluid” can be written as “acceleration of fluid”. 4. Reynolds number regimes can be “Reynolds number regimes and Reynold experiments”.	Suggestions added as deemed fit.

			5. “Pipe network” can be “Pipe network analysis”. Kindly add Hardy-Cross method. 6. It will be good to add “displacement thickness, momentum thickness, and energy thickness.” 7. In place of discharge in open channel, please mention velocity variations in open channel. You can Area-velocity method to compute discharge.	

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	**Hydraulic Engineering**	Number	CIL2XX0
Department	Civil & Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B. Tech. in Civil Engineering	Type	Program Core
Prerequisite	None		

Objectives

The instructor will

- provide a rigorous and comprehensive foundation in the fundamental laws governing fluid flow.
- help the students understand the importance and application of fluid mechanics.
- exemplify concepts of similarity and model testing.

Learning Outcomes

The students are expected to have the ability to:

- understand and apply the fundamental principles of fluid statics, kinematics, and dynamics to analyse forces, pressures, and flow conditions in civil engineering systems
- derive and apply the governing equations of fluid motion to solve engineering problems
- design physical hydraulic models, interpret experimental results, and extend findings to prototype-scale engineering predictions.

Contents

Introduction [3 Lecture]: Definition and properties of fluid, Newtonian and Non-Newtonian Fluids, Dimensions and units, Continuum Hypothesis, Knudsen Number, Capillary and surface tension.

Pressure and Fluid Statics [5 Lectures]: Pascal's Law, Manometry, hydrostatic force on submerged planar and curved surfaces, buoyancy and stability, rigid body motion.

Fluid Kinematics [5 Lectures]: Graphical Description of Fluid Flow, Vorticity, Circulation, Stream and Potential Functions, rotational-irrotational flows, Bernoulli's equation and its applications.

Fluid Dynamics [9 Lectures]: Reynold Transport Theorem, Acceleration of fluid, Deformations in fluid particles, derivation of Euler's equation and Navier-Stokes equations.

Internal flow [6 Lectures]: Reynolds number regimes and Reynold experiments, Laminar and Turbulent flow in pipes, Head Losses in pipes, Flow measurement in pipes, Pipe network analysis, Introduction of water hammer effect and cavitation, Moody's chart.

External flow [6 Lecture]: Boundary layer over flat plate, boundary layer thickness, Prandtl-boundary layer equations, momentum integral equation, pressure gradient & separation, drag coefficients.

Dimensional analysis and similarity [2 Lectures]: Dimensionless Numbers, Scale and Similitude, Buckingham Pi theorem and examples.

Flow in Open Channel [3 Lecture]: Discharge in open channel, Concept of Specific Energy and Alternate Depths, Hydraulic Jump.

Textbooks

1. Hibbler, R.C., (2022), Fluid Mechanics, 2nd Edition, Pearson.
2. Modi, P.N. and Seth, S.M., Hydraulics and Fluid Mechanics Including Hydraulic Machines. Standard Book House, New Delhi.
3. White, F. M. (2003), Fluid Mechanics, 5th Edition, McGraw-Hill.

Reference Books

1. Munson, B. R., Rothmayer, A. P., Okiishi, T'H., and Huebsch, W. W. Fundamentals of Fluid Mechanics, 7th Edition, John Wiley and Sons.
2. Fox, R. W., Pitchard, P. J., and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
3. Kundu, P. K, Cohen, I. M and Dowling, D. R, Fluid Mechanics, 6th Edition, Associated Press.
4. Cimbala, J.M. and Cengel, Y.A., (2010), Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

Self-Learning Material

1. Prof. S. Dutta, Fluid Mechanics, NPTEL Course Material, Civil Engineering, IIT Guwahati, https://nptel.ac.in/courses/105103192/
2. Prof. M. S. Afzal, Hydraulic Engineering, NPTEL Course Material, Civil Engineering, IIT Kharagpur, https://nptel.ac.in/courses/105105203
3. Prof. T. I. Eldho, Fluid Mechanics, NPTEL Course Material, Civil Engineering, IIT Bombay, https://nptel.ac.in/courses/105101082

CHECKLIST FOR THE NEW COURSE APPROVAL

Hydraulic Engineering Lab

1. Justification for the proposal in line with the philosophy of the program.
   This course 'Hydraulic Engineering' provides a rigorous and comprehensive foundation in the fundamental laws governing fluid. This course 'Hydraulic Engineering Lab' is a laboratory component for 'Hydraulic Engineering' course. The lab experiments are designed to introduce students to engineering aspects of different hydraulic phenomena, which is required for effective application of hydraulic engineering principles in the field.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Saran Aadhar, Dr. Deepak Swami

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   Adequate infrastructure is available for running lab exercises .

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources. **I therefore recommend that the revised B.Tech. course syllabi in Water Resources be approved.**	Thank you for the review. No change required.
2.	Prof. Pramod Kr. Shrma	Professor in Hydraulic	No major comments. It looks fine.	Thank you for the feedback. No major change required.

		Engineering, Department of Civil Engineering Indian Institute of Technology Roorkee		
3.	Prof. Atul Dhar	Professor, School of Engineering, Indian Institute of Technology Mandi	The proposed courses are well aligned with the standard fluid mechanics courses in the UG curriculum of Civil, Mechanical, Chemical, Aerospace engineering etc. Few minor points may be taken into consideration for improvement: 1.In internal flows: Introduction of water hammer effect and cavitation concepts may also be added. 2. Lectures in introduction may be reduced and open channel may be in created. 3. The lab should not specify the theory course as prerequisite. It is better to give students option to take both theory and lab courses simultaneously.	Thank you for the feedback. The introduction to water hammer effect and cavitation is now added to the internal flow section. Due to limited lecture length of this course, flow in the open channel is designed as a different course. The prerequisite for lab course is removed, as per the feedback.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	Hydraulic Engineering Lab	Number	CIP2XXX
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	0-0-2 [1]
Offered for	B. Tech. in Civil Engineering	Type	Program Core
Prerequisite			

Objectives

The instructor will

- provide hands-on experience with the fundamental theoretical principles of fluid mechanics.
- introduce various forms of flow measuring devices and techniques.

Learning outcomes

Upon completion of this laboratory course, students will be able to:

- verify fundamental theoretical principles of fluid mechanics through hands-on experimental measurement.
- calibrate and operate various instruments and assess their accuracy, key hydraulic parameters and suitability for different flow conditions.
- compare experimental results with theoretical principles and draw evidence-based engineering conclusions.

Contents:

- Observation of streamline formation and flow behaviour in a transparent flow channel
- Determination of metacentric height of floating bodies
- Determination of the Reynolds number for fluid flow through a pipe and thereby identify the nature of the flow regime
- Verification of Bernoulli's equation experimentally and calculation of the total energy at different points
- Determination of the coefficient of discharge of a venturi meter
- Determination of the coefficient of discharge of an orifice meter
- Study the impact of a jet on different types of vanes:
  - ✓ Hemispherical Vane
  - ✓ Flat Plate Vane
  - ✓ Conical vane
- Study and compare the water surface profile of a forced vortex
- Determination of head loss in pipes having different diameters, different materials and different roughness
- Determination of pressure drop at different types of bends and valves
- Determination of the friction factor for laminar and turbulent flow through pipes
- Determine the coefficient of discharge for weirs and notches (rectangular, triangular and trapezoidal)

Textbook:

1. White, F. M. (2003), Fluid Mechanics, 5th Edition, McGraw-Hill.

Reference Books

1. Fox, R. W., Pitchard, P. J., and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
2. Cimbala, J.M. and Cengel, Y.A., (2010), Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

CHECKLIST FOR THE NEW COURSE APPROVAL

Building Materials and Construction Technology

1. Justification for the proposal in line with the philosophy of the program.

   The proposed course aligns with the philosophy of the Department of Civil and Infrastructure Engineering at Indian Institute of Technology Jodhpur by strengthening interdisciplinary, industry-relevant, and research-oriented learning in emerging areas of infrastructure engineering.

2. Total number of electives in the department/Center/School already approved by the Senate.

   Program core

3. Name of faculty members who will teach the proposed course.

   Dr. Sreejith Krishnan, Dr. Yogiraj Sargam, Dr. Anusha S. Basavaraj

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
	Prof. Ananth Ramaswamy	Professor, IISc	The distinction between ductile and brittle materials, quasi brittle materials, failure?, ductile response, criteria for accepting or rejecting materials.	Added
	Prof. Ravindra Gettu	Professor, IIT Madras	There should be an introduction to Building Materials and Construction Technology -Chemical admixtures should be included -Cover finishing materials, dry walls, flooring -Include construction equipment -Building Services should be introduced -Project could be for the students to draw the layout for a small single story home or office, decide the location, foundation, structural materials and finishes	Changed the name of the course Added chemical admixtures Added finishing materials, walls and flooring

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	**Building Materials and Construction Technology**	Number	CILXXXX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0
Offered for	B.Tech	Type	
Prerequisite	Not applicable		

Objectives:

- Understand the fundamental behavior of different building materials
- To introduce the various building materials available for construction
- Understand the different components of building and their functions

Learning Outcome:

Following this course, the student will have the ability to:

- Identify and select the right building materials for various applications
- Identify the correct construction method for different constructions

Course Contents:

- **Module 1 (9 Hours)**

Introduction to materials – atomic structure, energy and entropy, structure of solids, Mechanical properties – stress - strain relationships, elastic behavior and elastic constants, failure under different conditions, fracture and toughness, viscoelasticity, surfaces – energy, wetting, electrical and thermal properties, variability in materials and its implications.

- **Module 2 (15 Hours)**

Classification, physical, mechanical properties and testing of commonly used building materials – Stones, bricks, cement, aggregates, chemical admixtures, supplementary cementitious materials, lime, timber, paints, glass, rebars, polymers etc.

Concrete – Fresh and hardened properties of concrete, test methods, grades of concrete, batching, mixing, placing and curing, Introduction of concrete mix design

- **Module 3 (15 Hours)**

Building Construction – Components of building, stone and brick masonry, framed and loading bearing structures, shallow and deep foundations, floors and roofs, lintels, arches and staircases, doors, windows and ventilators, finishing materials, flooring, joints, waterproofing, formworks, scaffolding, shoring and underpinning

Textbooks:

- P.C. Varghese (2008) Building Materials, Prentice-Hall of India, New Delhi.
- B. C. Punmia (1993) Building Construction, Laxmi Publications, New Delhi.

Reference books:

- V. Raghavan (1990) Materials Science and Engineering, Prentice Hall.
- J.F. Young, S. Mindess, R.J. Gray & A. Bentur (1998) The Science and Technology of Civil Engineering Materials, Prentice Hall.
- P.K. Mehta and P.J.M. Monteiro (2006) Concrete: Microstructure, properties and materials, McGraw Hill.
- Edward Allen and Joseph Iano (2008) Fundamentals of Building Construction, 5th Edition, John Wiley & Sons Inc., New Jersey.
- M. Soutsos and P. Domone (2018) Construction Materials, Their Nature and Behavior, Fifth Edition, CRC Press.

Online Resources:

NPTEL course - NOC: Modern Construction Materials, Prof. Ravindra Gettu Prof. Aslam Kunhi Mohamed, IIT Madras. http://www.nptel.ac.in/courses/105106053/

CHECKLIST FOR THE NEW COURSE APPROVAL

Building Materials and Construction Technology Laboratory

1. Justification for the proposal in line with the philosophy of the program.

   The proposed course aligns with the philosophy of the Department of Civil and Infrastructure Engineering at Indian Institute of Technology Jodhpur by strengthening interdisciplinary, industry-relevant, and research-oriented learning in emerging areas of infrastructure engineering.

2. Total number of electives in the department/Center/School already approved by the Senate.

   Program core

3. Name of faculty members who will teach the proposed course.

   Dr. Sreejith Krishnan, Dr. Yogiraj Sargam, Dr. Anusha S. Basavaraj

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.

5. Experts' comments on the proposed courses:

S. No.	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Prof. Ananth Ramaswamy	Professor, Indian Institute of Science	While the theory course makes reference to masonry the lab course is silent . Tests on bricks, need to moist it prior to use, the role of the frog on the brick and perhaps some basic masonry unit tests on wallets-compression, and flexure? Tests on rebar - tension, stress-strain, smooth and tor steel, and pullout from concrete for the bond properties. All the above without overloading. Last but not the least criteria for acceptance and rejection of materials-concrete?	Added – tests on brick (need of moist brick and role of frog); tests on rebar tension, stress-strain, smooth and tor steel, and pullout from concrete for the bond properties.
2	Prof. Ravindra Gettu	Professor, IIT Madras	-Include demo of the effectiveness of superplasticizers -Drop VeBe and Compaction factor -Add fibre reinforced concrete	Dropped VeBe and compaction factor Added fibre reinforced concrete

			Project could be the mix design for different performance requirements	

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	**Building Materials and Construction Technology Laboratory**	Number	CIP2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-2
Offered for	B.Tech.	Type	PC
Prerequisite	Not applicable		

Objectives:

- To familiarize the students with testing of building materials

Learning Outcome:

Following this course, the student will have the ability to:

- Assess and interpret the standard materials testing data as per the relevant standards
- Prepare reports and communicate the materials testing reports
- Recommend the right building materials based on the material testing reports

List of Experiments:

Cements

- Specific Gravity, Consistency, Initial and Final Setting Time
- Blaine's air permeability test, and Sieve Analysis
- Mortar Strength Test

Aggregates

- Specific gravity, bulk density and water absorption of coarse and fine aggregates
- Sieve analysis of fine and coarse aggregates, and flakiness and elongation index tests
- Crushing strength tests

Concrete

- Workability Tests on Concrete – Slump cone
- Effectiveness of chemical admixtures – Marsh cone, flow table
- Mechanical Properties of Concrete – 7 and 28 Days compressive/split tensile/flexural strengths, and modulus of elasticity of Concrete

- Self Compacting Concrete - Slump flow, L-Box, V-funnel and J-Ring tests
- Fiber reinforced Concrete - mix design for different performance requirements

Rebar

- Tests on Rebars - tension, stress-strain, and pullout from concrete for the bond properties

Miscellaneous

- Bricks – Dimensions check, Compressive strength, water absorption, importance of frog,

Text Books:

- P.C. Varghese (2008) Building Materials, Prentice-Hall of India, New Delhi.
- M. Soutsos and P. Domone (2018) Construction Materials, Their Nature and Behavior, Fifth Edition, CRC Press.

Reference Books:

- Relevant BIS standards

CHECKLIST FOR THE NEW COURSE APPROVAL

Geoinformatics

1. Justification for the proposal in line with the philosophy of the program.
   This course introduces students to concepts of basic principles of remote sensing, GPS, Surveying and GIS. The knowledge of these technologies are essential for planning and implementation of any large scale Civil construction works.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Saran Aadhar

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   The computation lab exercises can be carried out in computer labs (under computer center). However, some equipment, such as Total station, level, Theodolite, compass, GPS receivers etc are required for conducting field experiments. These equipment need to be procured for the lab.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Rajib Maity	Professor, Department of Civil Engineering, IIT Kharagpur	In brief, all three courses are good and nicely developed. Being said that, three minor changes can be made as follows: Fundamentals of Hydrology: One or two lectures may be included (incomplete indications are marked in the attached document in track change). Geoinformatics: All good. Expand LIDAR somewhere. Secondly, I am bit doubtful whether 12 lectures for Module 1 will be sufficient or not. Statistics and Probability in Civil Engineering: All good. One minor edit shown in the attached file.	All changes have been implemented. The lectures schedule is updated as per comments.

2.	Shri Rohit Agrawal	Director, Rajasthan Geo-Spatial Directorate, Survey of India	Pl find inputs from this GD highlighted in red color for your further action	The majority of suggested lab exercises and changes in lecture component is done. However as the course is intended to be an introduction to Geoinformatics for Civil Engineering 2nd year B.Tech Students, more advanced topics (such digital image processing, programming, Open GIS implementation etc.) are not added in the lecture component. Additionally, these topics can not be added due to limited lecture hours (39 hours) in a course.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	Geoinformatics	Number	CIL2XXX
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-2 [4]
Offered for	B. Tech.		
Prerequisite			

Objectives

- To introduce the basic principles of remote sensing, GPS, Surveying and GIS.
- To teach application of geo-informatics in geoscience, ecology and environment study, agriculture, and disaster management.

Learning outcomes:
A graduate of this course will have a thorough idea of:

- Multispectral, thermal infrared, and microwave remote sensing.
- Surveying Principles and handling modern surveying equipment
- Different map projections and use of GPS.
- Component GIS and applications.
- Different cartographic models, implementation, validation, and visualization.

Contents:
Module I: Surveying and Global Positioning System (13 Lectures)

- Surveying Fundamentals: Introduction to modern surveying; Coordinate systems; Measurement of distances and angles; Introduction to Total Station.
- Basic Geodesy, Map and Projections: Introduction, Basic Geodesy, Shape of Earth, Map and its types, Map Scale and Symbol, Map Projection and its Classification; Georeferencing.
- Global Positioning System (GPS): Components and operational principles; GPS receivers; Differential GPS (DGPS) concepts and types; Real-time kinematic positioning; limitations, and areas of application.

Module II: Remote Sensing and Photogrammetry (15 Lectures)

- Physics of RS: Interaction of EMR with atmosphere and Earth's surface; Atmospheric windows.
- Photogrammetry: Types of aerial photographs and their geometry; Aerial cameras and Use of Drones; Image displacement. Stereoscopic Vision: Binocular observations; Stereoscopic pairs; Types of stereoscope; Measurement of height from photographs; Stereo model.
- Multispectral and Hyperspectral imaging: Spectral signatures of vegetation, soil, and water, Sensor Utilized; Image correction and interpretation
- Advanced Remote Sensing: Thermal Infrared (emissivity, characteristics, and applications); Microwave region (SAR characteristics and interpretation); LIDAR principles, LIDAR point cloud and its processing.

Module II: Geographical Information System and Applications (11 Lectures)

- GIS Concepts: Components and spatial data representation; Relationships of spatial objects; GIS functions; Integration of Remote sensing and GIS.
- Cartographic Modelling: Inductive and deductive modelling; Model flowcharting; Conflict resolution, verification, and implementation, GIS Model Output.

- Visualization and Applications: Web mapping and visualization; Virtual and Immersive GIS, Mapping temporal data.
- Applications in Environment, Agriculture, Public Health, Water Resources, and Disaster Management.

Tentative list of laboratory exercises:

- Establishment of Local Control Network and Vertical Profiling using modern surveying Instruments (Total Station or Theodolite, Digital level and EDM).
- Field data collection using handheld and Differential GPS.
- Georeferencing when coordinate values are given/not given in a GIS platform.
- Creating maps using vector data models and cartographic guidelines.
- Working with Digital Elevation Model (DEM) and terrain analysis.
- Drone data acquisition using Multi spectral sensor and Lidar sensor. Image data processing and generation of ortho-rectified Images (ORI).
- Site suitability analysis for service buildings using vector and raster data.
- Procurement and preparation of land use and land cover map using satellite images for a specified study area.
- Developing DEM and CHM using LIDAR point cloud data.
- Network analysis to locate suitable routes.
- Creating a Web app serving a GIS model (created using model builder) in a GIS platform.

Note: Labs will have hands-on use of Google Earth Engine, QGIS, ERDAS IMAGINE, ArcGIS, or other relevant software (any one or two tools will suffice).

A surveying excursion for 1 week (or less period) may be planned for nearby places to make students familiar with practical experience in mapping, and using surveying equipment in real-world environments.

Text Books

1. Satheesh Gopi, R. Sathikumar, and N. Madhu (2025) Advanced Surveying: Total Station, GPS, GIS & Remote Sensing (2nd Ed). Pearson India.
2. T. M. Lillesand, R. W. Kiefer, J. W. Chipman (2004) Remote Sensing and Image Interpretation (5th Ed). Wiley. ISBN: 0-471-15227-7
3. G. S. Srivastava (2014) An Introduction to Geoinformatics. McGraw Hill Education (India) Private Limited. New Delhi. India. ISBN: 9781259058462.
4. Michael N. DeMers (2008) Fundamentals of Geographic Information Systems (4th Ed). Wiley. ISBN: 978-0-470-12906-7.

Other Books:

- Basudeb Bhatta (2021) Remote Sensing and GIS (3rd Ed). Oxford University Press.
- L. E. Johnson (2009) Geographic Information Systems in Water Resources Engineering. CRC Press. Taylor & Francis Group. ISBN 978-1-4200-6913-6.
- B. Tomaszewski (2020) Geographic Information Systems (GIS) for Disaster Management (2nd Ed). CRC Press.

Online Resources

- Prof. Arun K. Saraf (2016) Introduction to Geographic Information System. NPTEL. https://nptel.ac.in/courses/105/107/105107155/
- Prof. Arun K. Saraf (2017) Introduction to Remote Sensing. NPTEL. https://nptel.ac.in/courses/121/107/121107009/
- Prof. Rishikesh Bharti (2019) Remote Sensing and GIS. NPTEL. https://nptel.ac.in/courses/105/103/105103193/

CHECKLIST FOR THE NEW COURSE APPROVAL

Transportation Engineering-I

1. Justification for the proposal in line with the philosophy of the program.
   Transportation engineering is a core component of civil engineering curriculum worldwide and is widely taught in all IITs and NITs. Transportation Engineering-I covers geometric design of highways and the material components, design, construction and maintenance of pavements. Transportation engineering is fundamental to developing civil engineers capable of addressing contemporary challenges in sustainability and social welfare. Through rigorous training in highway design, students gain the technical foundation necessary to create infrastructure that serves communities safely and efficiently. This aligns with the department's mission to nurture leaders who contribute meaningfully to society's development.

2. Total number of electives in the department/Center/School already approved by the Senate.
   35

3. Name of faculty members who will teach the proposed course.
   Dr. Ranju Mohan, Dr. Bhupendra Singh, Dr. Akshay Gupta, Dr. Subham Jain

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   Not applicable

5. Experts' comments on the proposed courses:

Name of Expert	Designation of Expert	Comments received	Action Taken
Prof Indrajit Ghosh	Professor, Department of Civil Engineering, IIT Roorkee	No practical?	There is a separate lab course called Transportation Engineering Lab course
		Module 1: "Road classification" can be termed as functional classification of roads	"Road classification" has now been modified to "Functional classification of roads".
		Module 1: Do not miss NHDTP, Bharatmala Poriyojana, PM Gati Shakti under highway development in India	"Highway development in India" has now been added to Module 1.

		Module 2: Remove "highway classification"	"Highway classification" has now been removed.
		Module 2: Don't miss these aspects: Requirements of Highway alignment, factors influencing highway alignment, stages of alignment selection, alignment for hill roads	"Alignment – types, requirements, factors influencing and stages"; "alignment for hill roads" have now been added to Module 2
		Module 5: Types of surface and sub-surface failures, and remedial measures	"Types of surface and sub-surface failures" will be covered in "Failures of flexible and rigid pavements". "Remedial measures" will be covered in "Maintenance and rehabilitation alternatives".
		Module 5: Design of overlays based on Benkelman Beam and Falling Weight Deflectometer (FWD)	"Design of overlays" has now been added to Module 5
Prof Dharamveer Singh	Professor, Department of Civil Engineering, IIT Bombay	You may consider some soft tool along with theory course	IIT Pave Software added to Module 3

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.

Title	**Transportation Engineering-I**	Number	**CI2XX0**
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0
Offered for	B.Tech (CIE)	Type	Program Compulsory (PC)
Prerequisite	Not applicable		

Objectives

The course will help students to understand the principles of transportation engineering, including geometric design, pavement materials and design, pavement construction, evaluation, and maintenance.

Learning Outcomes

At the end of the course, the student should be able to:

- Design cross section elements, sight distance, horizontal and vertical alignment
- Determine the characteristics of pavement materials
- Design flexible and rigid pavements
- Explain the construction procedures for different pavement layers and the equipment used in highway construction.
- Evaluate the functional and structural condition of pavements using appropriate methods and recommend maintenance or rehabilitation measures.

Course Content

Module I: Introduction to Transportation Engineering [Lectures - 4]:
Transportation system, Introduction to Transportation Engineering, Functional classification of roads, Terrain classification, Factors affecting transportation, Highway development in India.

Module II: Geometric design [Lectures – 10]:
Factors affecting geometric design; Alignment – types, requirements, factors influencing and stages; Highway design process - Highway cross sectional elements, Sight distance; Highway alignment: Horizontal alignment - horizontal curve, Super elevation, extra widening, Transition curve, Setback distance, Visibility, Vertical alignment - gradient, grade compensation, summit curve, valley curve; alignment for hill roads.

Module III: Pavement materials and design [Lectures - 11]:
Pavement characteristics, materials, Types of pavements - Flexible and Rigid pavements, Wheel load calculations, Flexible Pavement materials- soil, aggregate, bitumen; Tests on pavement materials; Bituminous mix design, Marshall stability; Rigid pavement- joints, analysis of stresses; IRC method for design of flexible and rigid pavement; Pavement failures, IIT Pave Software.

Module IV: Pavement Construction [Lectures - 6]
Construction of various layers, equipment used for highway construction and quality control aspects.

Module V: Pavement Evaluation and Maintenance [Lectures - 8]
Failures of flexible and rigid pavements, condition indicators, various types of functional and structural evaluation techniques, maintenance and rehabilitation alternatives, Design of overlays.

Textbooks:

- N. J. Garber, L.A. Hoel. Traffic and Highway Engineering.
- Saxena S. C. Textbook of Highway and Traffic Engineering, CBS Publishers.
- Wright Dixon, Highway Engineering, Wiley India.

Additional Reference

- S. K. Khanna, C. E. G. Justo, and A. Veeraragavan. Highway Engineering, Nem Chand and Bros. Publishers.
- I.S. specifications on concrete, aggregates and bituminous materials.
- David Croney, Design and performance of road pavements, McGraw Hill
- James H. Banks, Introduction to Transportation Engineering, McGraw Hill
- R. Srinivasa Kumar, Textbook of Highway Engineering, University Press
- Chakroborty, P., Animesh Das. Principles of Transportation Engineering, PHI Learning Private Limited

CHECKLIST FOR THE NEW COURSE APPROVAL

Course Title: Transportation Engineering Lab

1. Justification for the proposal in line with the philosophy of the program.

   The proposed Transportation Engineering Lab is designed to provide hands-on exposure to pavement materials, highway engineering design practices, and testing methodologies. It aligns with the department's vision of developing sustainable and resilient infrastructure by enabling students to understand the performance of construction materials used in transportation systems.

2. Total number of electives in the department/Center/School already approved by the Senate.

   35

3. Name of faculty members who will teach the proposed course.
   Dr. Ranju Mohan, Dr. Bhupendra Singh, Dr. Akshay Gupta, Dr. Subham Jain

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   Yes.

5. Experts' comments on the proposed courses:

S. No.	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Dr. Surender Singh	Associate Professor, Civil Engineering Department, IIT Madras	Since bitumen is classified based on viscosity, it will be better to include viscosity related tests.	We have added viscosity test to lab course syllabus.
			Will it be possible to include sieve analysis and blending? You might be already doing this in Marshall stability test.	Yes, you are correct. It has already been part of Marshall stability test.
			Are OMC, MDD, and CBR covered in geotechnical engineering laboratory? If not, these could be included here.	Yes. OMC, MDD and CBR will be covered under geotechnical engineering (soil mechanics) laboratory.
			In the shape test, will you be including angularity number? If not, can it be included as it takes into	The shape test is restricted to elongation and flakiness,

			account most of the morphology parameters?	keeping in view the total number of experiment slots available in a semester.
2	Dr. Praveen Kumar	Professor, Civil Engineering Department, IIT Roorkee	In Transportation Engg Lab instead of crushing test, it is better to include 10% fine value test which is similar to it but nowadays this is included in various codes.	We have changed the crushing test to 10% fine value test.

6. The course is recommended by DFB/SFB after due deliberation.

   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.

Title	Transportation Engineering Lab	Number	CI2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	0-0-2 [1]
Offered for	B. Tech [CIE]	Type	Program Compulsory (PC)
Prerequisite	Transportation Engineering I		

Objectives

The instructor will:

1. Introduce students to laboratory testing procedures related to highway construction materials and pavement engineering.
2. Provide hands-on experience in evaluating the engineering properties of aggregates and bituminous materials used in transportation infrastructure.

Learning Outcomes

The students will have the ability to:

1. Perform standard laboratory tests on aggregates and bituminous materials as per relevant IS/IRC specifications.
2. Determine engineering properties of pavement materials and assess their suitability for highway construction.

Contents

1. 10% fine value test
2. Aggregate Impact Test
3. Los Angles Abrasion Test
4. Specific Gravity and Water Absorption Test on Aggregates
5. Shape Tests
6. Penetration Test and Softening Point Test
7. Viscosity Test
8. Ductility Test
9. Flash & Fire Test
10. Specific Gravity Tests on Bitumen
11. Marshall Stability Test

Text Books and References

1. Khanna, S.K., Justo, C.E.G. and A Veeraragavan, "Highway Material and Pavement Testing Manual", Nem Chand & Bros
2. IRC and IS Codes related to Highway Materials Testing

CHECKLIST FOR THE NEW COURSE APPROVAL

Fundamentals of Hydrology

1. Justification for the proposal in line with the philosophy of the program.
   This course introduces students to concepts of surface and subsurface hydrology. This is a core course as the understanding of hydrological principles is required for any Civil Engineer to work on irrigation projects, designing river structures and ascertaining water availability for the community.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Saran Aadhar, Dr. Deepak Swami

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   N.A.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Rajib Maity	Professor, Department of Civil Engineering, IIT Kharagpur	In brief, all three courses are good and nicely developed. Being said that, three minor changes can be made as follows: Fundamentals of Hydrology: One or two lectures may be included (incomplete indications are marked in the attached document in track change). Geoinformatics: All good. Expand LIDAR somewhere. Secondly, I am bit doubtful whether 12 lectures for Module 1 will be sufficient or not. Statistics and Probability in Civil Engineering: All good. One	All changes have been implemented. The lectures schedule is updated as per comments.

			minor edit shown in the attached file.	
2.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources. **I therefore recommend that the revised B.Tech. course syllabi in Water Resources be approved.**	Thank you for the review. No change required.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	Fundamentals of Hydrology	Number	CIL2XXX
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech.		
Prerequisite			

Objectives

- To provide an understanding of hydrologic processes, analysis methods, and design applications with emphasis on mathematical and statistical analysis in water resources management.

Learning outcomes

- Become familiar with the engineering aspects of surface and subsurface hydrology
- Utilize knowledge of hydrological processes in solving engineering problems within hydrology and water resources
- Develop an understanding of the major impact of changing climate on the water cycle.

Contents

Introduction: Engineering hydrology and its applications, Hydrological cycle, Concept of Catchment, Water balance and fluxes, World Water Balance, Residence time, Climate Change Impacts (Lectures 4).

Precipitation: Introduction, Forms of precipitation, Measurement of Precipitation, Mean Precipitation over an Area, Analysis of precipitation data, Precipitation variability and monsoon characteristics in India, Depth–Area–Duration Relationships, Maximum Intensity/Depth–Duration–Frequency Relationship, Probable Maximum Precipitation (PMP) (Lectures 6)

Abstractions from precipitation: Introduction, Evaporation and Evapotranspiration: factors affecting evaporation, measurement and estimation of evaporation, Reservoir Evaporation and Methods for Its Reduction, Transpiration, Measurement of Evapotranspiration, Potential Evapotranspiration and its estimation, Actual Evapotranspiration; Interception; Depression Storage; Infiltration: Infiltration Capacity and factors affecting Infiltration capacity, Measuring and modeling infiltration, Infiltration Indices (Lectures 7)

Runoff and streamflow: Types of runoff, Runoff characteristics of stream, Catchment factors affecting runoff, Flow–Duration and Flow–Mass Curve, Estimation of Reservoir Volume, Hydrograph, Factors affecting Storm Hydrograph, Components of a Hydrograph, Base Flow Separation, Effective Rainfall, Unit Hydrograph and its derivation, Unit Hydrographs of Different Durations, Synthetic Unit Hydrograph (Lectures 8)

Droughts, Floods and Flood Routing: Introduction to Hydrologic Extremes, Drought, Drought Indices, Flood Estimation: Rational Formula, Runoff Coefficient and Unit hydrograph method, Flood Frequency Analysis, Design Flood and Design Storm, Risk, reliability and Safety Factor in hydrologic design, Introduction to Flood routing, Level Pool Routing, Hydrological Channel Routing, Hydraulic Methods of Flood Routing, Flood Control, Flood in changing climate (Lectures 9)

Groundwater: Aquifer formations and their Properties, specific retention, specific yield, Storage Coefficient, Darcy's Law, Groundwater Movement, confined and unconfined aquifer, Well Hydraulics, Groundwater Resources of India (Lectures 5)

Text Books

1. Subramanya, K. (2024). Engineering hydrology (6th ed.). McGraw Hill Education (India) Pvt Ltd
2. Ojha, C. S. P., Berndtsson, R., & Bhunya, P. (2008). Engineering hydrology. Oxford University Press.
3. Viessman Jr., W., Lewis, G. (2002). Introduction to Hydrology. 5th Edition, Pearson

Reference Books

- Chow, V.T., Applied Hydrology, McGraw-Hill
- Todd, D.K., & Mays, W.L. (2004) Groundwater Hydrology, 3rd Edition, Wiley

Online Resources

- Margulis, S. (2016). Introduction to hydrology. https://margulis-group.github.io/textbook/
- Prof. Rajib Maity (2012). Advanced Hydrology. NPTEL. https://nptel.ac.in/courses/105105214

CHECKLIST FOR THE NEW COURSE APPROVAL

Air Pollution and Solid Waste Management

1. Justification for the proposal in line with the philosophy of the program.
   The course equips students with fundamental knowledge and practical understanding of challenges associated with air quality deterioration and waste generation. The course integrates scientific principles, engineering design, regulatory frameworks, and sustainable management practices to prepare students for addressing real-world pollution control and resource management issues.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core course.

3. Name of faculty members who will teach the proposed course.
   Dr. Deepika Bhattu, Dr. Amit Sharma, Dr. Monika Dubey, Dr. Trishikhi Raychoudhury

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   -

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Prof. Anurag Garg	Professor	1. Some edits and a new book recommended	1. All recommended changes are incorporated
2	Prof. Arun Kumar	Professor	1. Objectives are not properly balanced; ideally it should be 50%. 2. Textbook - No formal air pollution book is included 3. Reference book- At B.Tech level, regular air pollution treatment control type book should be included.	1. Objectives are revised and balanced between air pollution and solid waste management 2. Textbook- New air pollution book "Noel De Nevers, Air Pollution Control Engineering, 3rd Ed., Waveland Press, Inc., 2016" is added. 3. Reference book- B.Tech level reference book "Air Pollution Control: A Design

				Approach, C. David Cooper & F. C. Alley, 4th Ed., Waveland Press, Inc., 2011", is included

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   NA

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	Air pollution and solid waste management	Number	20XX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech [CIE]	Type	Program Core (PC)
Prerequisite	None		

Objectives

- To explain and classify sources, types, impacts of air pollutants, along with relevant standards and legislation.
- To analyze the role of meteorological parameters in air pollutant dispersion and evaluate control strategies.
- To explain and characterize solid waste sources, composition, and generation, along with regulatory and management frameworks.
- To apply and evaluate integrated solid waste management principles, including waste minimization, treatment, and disposal technologies.

Learning Outcomes

Upon successful completion of the course, students will be able to:

- Classify and assess air pollutants and interpret their impacts on health, environment, and regulatory compliance.
- Analyze and predict pollutant dispersion using meteorological concepts such as lapse rate, stability, and plume behavior.
- Evaluate and justify the selection of air pollution monitoring and control technologies for specific applications.
- Design and propose integrated solid waste management strategies including collection, processing, treatment, and disposal.

Contents

Air pollution

Sources, types and classification of air pollutants, air pollution episodes, effect of air pollutants on human health, vegetation and materials, air quality and emission standards and legislation **[7]**

Effect of meteorology: Lapse rate, Atmospheric Stability, Plume behaviour and dispersion, Stack height **[4]**

Engineering emission control concepts, Selection and basic design approach for air pollution control equipments, source control technologies, Indoor air pollution sources and control. **[9]**

Municipal Solid Waste Management

Fundamentals of Integrated Solid Waste Management, legislation and regulations **[4]**

Waste generation, collection, and transportation; Characteristics of solid waste. **[6]**

Transfer stations, material recovery facility; biological treatment methods (such as composting and anaerobic digestion), incineration, pyrolysis and gasification, and sanitary landfilling. **[9]**

Text books:

- Noel De Nevers, Air Pollution Control Engineering, 3rd Ed., Waveland Press, Inc., 2016
- Tchobanoglous, G., Theisen, H., and Vigil, S.A. (2014) Integrated Solid Waste Management: Engineering Principles and Management Issues, McGraw Hill Book Company, Singapore.
- Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.

Reference books:

- M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.
- Air Pollution Control: A Design Approach, C. David Cooper & F. C. Alley, 4th Ed., Waveland Press, Inc., 2011
- CPHEEO Manual of Solid Waste Management, GOI Publication, 2016

Comments received from reviewer 1:

Title	Air pollution and solid waste management	Number	20XX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech [CIE]	Type	Program Core (PC)
Prerequisite	None		

Objectives

- To explain and classify sources, types, and impacts of air pollutants, and relevant standards and legislation.
- To analyze the role of meteorological parameters in air pollutant dispersion and plume behavior.
- To evaluate and select appropriate monitoring techniques and control methods for air pollution mitigation.
- To apply principles of integrated solid waste management for waste minimization, treatment, and disposal.

Learning Outcomes

Upon successful completion of the course, students will be able to:

- Classify and assess air pollutants and interpret their impacts on health, environment, and regulatory compliance.
- Analyze and predict pollutant dispersion using meteorological concepts such as lapse rate, stability, and plume behavior.
- Evaluate and justify the selection of air pollution monitoring and control technologies for specific applications.
- Design and propose integrated solid waste management strategies including collection, processing, treatment, and disposal.

Contents

Air pollution

Sources, types and classification of air pollutants, air pollution episodes, effect of air pollutants on human health, vegetation and materials, air quality and emission standards and legislation **[7]**

Effect of meteorology: Lapse rate, Atmospheric Stability, Plume behaviour and dispersion, Stack height **[4]**

~~Methods for pollutant monitoring and control,~~ Engineering emission control concepts, Selection and basic design approach for air pollution control equipments, ~~pollutant removal and disposal~~, source control technologies, Indoor air pollution sources and control. **[9]**

Municipal Solid Waste Management

Fundamentals of Integrated Solid Waste Management, legislation and regulations, ~~resource management, waste minimization: 3Rs~~ **[4]**

Waste generation, collection, and transportation; Characteristics of solid waste.~~, waste fees processing~~ **[6]**

~~Material separation~~ Transfer stations, material recovery facility; biological treatment methods (such as composting and anaerobic digestion), incineration, pyrolysis and gasification, and sanitary landfilling. **[9]**

Text books:

- Stull, R. "Practical Meteorology: An algebra-based survey of atmospheric science" – version 1.02b. Univ. of British Columbia. 2017.
- Tchobanoglous, G., Theisen, H., and Vigil, S.A. (2014) Integrated Solid Waste Management: Engineering Principles and Management Issues, McGraw Hill Book Company, Singapore.
- Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.

- M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.
- Cooper - Alley - Air Pollution Control-A Design Approach

Reference books:

- Aerosol Measurement: Principles, Techniques and Applications, edited by K. Willeke, P. Baron, Van Nostrand Reinhold, 1993.
- ~~Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.~~
- CPHEEO Manual of Solid Waste Management, GOI Publication, 2001

Commented [a1]: Please add full reference.

Commented [a2]: Added above.

Commented [a3]: Please check. It is '2016'.

Comments received from reviewer 2:

Title	Air pollution and solid waste management	Number	20XX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech [CIE]	Type	Program Core (PC)
Prerequisite	None		

Objectives

- To explain and classify sources, types, and impacts of air pollutants, and relevant standards and legislation.
- To analyze the role of meteorological parameters in air pollutant dispersion and plume behavior.
- To evaluate and select appropriate monitoring techniques and control methods for air pollution mitigation.
- To apply principles of integrated solid waste management for waste minimization, treatment, and disposal.

Commented [A4]: Not properly balanced; ideally it should be 50%

Learning Outcomes

Upon successful completion of the course, students will be able to:

- Classify and assess air pollutants and interpret their impacts on health, environment, and regulatory compliance.
- Analyze and predict pollutant dispersion using meteorological concepts such as lapse rate, stability, and plume behavior.
- Evaluate and justify the selection of air pollution monitoring and control technologies for specific applications.
- Design and propose integrated solid waste management strategies including collection, processing, treatment, and disposal.

Contents

Air pollution

Sources, types and classification of air pollutants, air pollution episodes, effect of air pollutants on human health, vegetation and materials, air quality and emission standards and legislation **[7]**

Effect of meteorology: Lapse rate, Atmospheric Stability, Plume behaviour and dispersion, Stack height **[4]**

Methods for pollutant monitoring and control, design and selection of control equipment, engineering emission control concepts, pollutant removal and disposal, source control technologies, Indoor air pollution sources and control. **[9]**

Municipal Solid Waste Management

Fundamentals of Integrated Solid Waste Management, legislation and regulations, resource management, waste minimization: 3Rs **[4]**

Characteristics of solid waste, Waste generation, collection, transportation, waste fees processing **[6]**

Material separation, biological treatment, incineration, pyrolysis and gasification, disposal. **[9]**

Text books:

- Stull, R. "Practical Meteorology: An algebra-based survey of atmospheric science" – version 1.02b. Univ. of British Columbia. 2017.
- Tchobanoglous, G., Theisen, H., and Vigil, S.A. (2014) Integrated Solid Waste Management: Engineering Principles and Management Issues, McGraw Hill Book Company, Singapore.

Reference books:

- Aerosol Measurement: Principles, Techniques and Applications, edited by K. Willeke, P. Baron, Van Nostrand Reinhold, 1993.
- Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
- CPHEEO Manual of Solid Waste Management, GOI Publication, 2001

Commented [A5]: No formal air pollution book here

Commented [A6]: Btech level; so regular air pollution treatment control type book

CHECKLIST FOR THE NEW COURSE APPROVAL

Water and Wastewater Engineering (PC)

1. Justification for the proposal in line with the philosophy of the program.: This is a UG level core course (PC)

2. Total number of electives in the department/Center/School already approved by the Senate.

3. Name of faculty members who will teach the proposed course. Dr. Trishikhi Raychoudhury, Dr. Monika Dubey, Dr. Amit Sharma, Dr. Deepika Bhattu

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully : Yes

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Prof. Anurag Garg	Professor, IIT Bombay	Yes (a) For one semester, the content needs to be reduced (b) Air-quality related laboratory components: are not related with the course title so may be considered to remove (c) Some edits, language change and addition of three books as text books are suggested	Yes Content is reduced by eliminating (a) population forecast, (b) sewer line design, and (c) sludge disposal which are likely to be covered in other courses (i.e., Fluid Mechanics and Solid Waste Management) Two of three recommended books are included as reference book.
2	Prof. Arun Kumar	Professor, IIT Delhi	(a) Big scope for the 302 credit (b) sludge disposal may be covered in solid waste course (c) Air-quality related laboratory components: Why included in this course?	Yes (a) Content reduced as specified above (b) Air related lab components is removed and relevant instrumentation could be shown under advanced instrumentation demonstration lab (L10)

6. The course is recommended by DFB/SFB after due deliberation.: Yes
7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   NA. The existing course is splatted in two components

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   NA. The existing course is splatted in two components

Title	**Water and Wastewater Engineering**	Number	CIL3XXX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-2
Offered for	B.Tech (CIE)	Type	PC
Prerequisite	Not applicable		

Objectives

The Instructor will:

- Provide an overview on water demand, water quality and treatment unit operation
- Provide an overview on wastewater characteristics and treatment unit operation
- Provide knowledge on water distribution system

Learning Outcomes

The students are expected to be able to:

- understand water treatment unit operation and would be able to design it
- understand wastewater treatment unit operation and would be able to design it
- analyse water distribution system for further design

Contents

Water demand and Water quality assessment [4]

Water demand; Water quality parameters: Physical, chemical and biological parameters; Water quality standards.

Water treatment theory and operation [12]

Theory and design parameters for various water treatment unit processes such as screening; sedimentation tank; coagulation-flocculation; filtration; disinfection; hardness; Exposure to advanced water treatment processes.

Water distribution system [4]

Design of water supply pipeline; Pumping; Head loss and water distribution network analysis.

Wastewater characterization [5]

Wastewater characterization : physical, chemical and biological parameters.

Wastewater treatment and sludge disposal [14]

Introduction to Unit processes in wastewater treatment plant and design criteria; Focus on secondary treatment: suspended growth system (e.g., activated sludge process) and attached growth systems (e.g., trickling filter); nutrient removal; Introduction to natural system; Exposure to advanced treatment process and design software

<u>Laboratory Components</u>

1. Introduction to environmental engineering laboratory
2. Determination of alkalinity in water
3. Determination of hardness in water
4. Determination of TDS and TSS : Gravimetric method
5. Determination of pH, TDS or EC and turbidity
6. Determination of optimal coagulant dosing: Jar test
7. Determination of COD in wastewater
8. Determination of BOD in wastewater
9. Coliform test: demonstration and analysis
10. Demonstration of advanced instrumentation

Text books:

- Metcalf & Eddy, Wastewater Engineering- Treatment and Reuse (Revised by G. Tchobanoglous, F. L. Burton and H.D. Stensel), Tata McGraw Hill, 4th Edn., 2004.H.
- Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
- M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Reference books:

- Nathanson and Jerry, Basic Environmental Technology: Water Supply, Waste Management and Pollution Control, Prentice Hall India Learning Pvt. Ltd., India 2009
- Garg, Environmental Engineering: Water Supply Engineering, Khanna Publishers, 18th Edn, India 2004
- Manual on Sewerage and Sewage Treatment Systems, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 2013.
- Manual on Water Supply and Treatment, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 1999.

Online Course Material

- Guidelines for drinking-water quality, World Health Organization 2022, ISBN 978-92-4-004506-4, https://www.who.int/publications/i/item/9789240045064
- Drinking Water — Specification, Bureau of Indian Standard , IS 10500 : 2012, New Delhi, India, 110002

Original with revision for reference

Title	**Water and Wastewater Engineering**	Number	CIL3XXX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-2
Offered for	B.Tech (CIE)	Type	PC
Prerequisite	Not applicable		

Objectives

The Instructor will:

- Provide an overview on water demand, water quality and treatment unit operation
- Provide an overview on wastewater ~~generation,~~ characteristics and treatment unit operation
- Provide knowledge on water distribution system ~~network and sewage collection system~~

Learning Outcomes

The students are expected to be able to:

- ~~predict water demand for a city considering population growth and other factors~~
- understand water treatment unit operation and would be able to design it
- understand wastewater treatment unit operation and would be able to design it
- analyse water distribution ~~and sewage collection~~ system for further design

Contents

~~Population forecast, w~~Water demand and W~~w~~ater quality assessment [4]
~~Population forecast;~~ Water demand; Water quality parameters: Physical, chemical and biological parameters; Water quality standards

Water treatment theory and operation [12]
Theory and design parameters for various water treatment unit processes such as screening; sedimentation tank; coagulation-flocculation, filtration, disinfection, hardness, Exposure to advanced water treatment processes.

Water distribution system~~network, stormwater and sewage collection system~~ [8 to 4]
Design of water supply pipeline; Pumping; Head loss and water distribution network analysis; ~~Prediction of sewage and storm water and design of sewer lines.~~

Wastewater characterization [4 to 5]
Wastewater characterization : physical, chemical and biological parameters.

Wastewater treatment and sludge disposal [11 to 14]
Introduction to Unit processes in wastewater treatment plant and design criteria, Focusing on secondary treatment: suspended growth system (e.g., activated sludge process) and attached growth systems (e.g., trickling filter); nutrient removal; Introduction to natural system, ~~sludge handling and disposal techniques~~; Exposure to design software

Laboratory Components

1. Introduction to environmental engineering laboratory
2. Determination of alkalinity in water
3. Determination of hardness in water
4. Determination of TDS and TSS : Gravimetric method
5. Determination of pH, TDS or EC and turbidity
6. Determination of optimal coagulant dosing: Jar test
7. Determination of COD in wastewater
8. Determination of BOD in wastewater
9. Coliform test: demonstration and analysis
10. Demonstration of advanced instrumentation

~~9. Measurement of PM2.5 and PM10 in Air~~
~~10. Demonstration of Weather Monitoring Station: Sampling and analysis~~

Text books:

- Metcalf & Eddy, Wastewater Engineering- Treatment and Reuse (Revised by G. Tchobanoglous, F. L. Burton and H.D. Stensel), Tata McGraw Hill, 4th Edn., 2004.H.
- Peavy, D. R. Rowe and George Tchobanoglous, Environmental Engineering, McGraw-Hill International Ed., 1985.
- M. L. Davis and D. A. Cornwell, Introduction to Environmental Engineering, McGraw-Hill, Inc., 2014.

Reference books:

- Nathanson and Jerry, Basic Environmental Technology: Water Supply, Waste Management and Pollution Control, Prentice Hall India Learning Pvt. Ltd., India 2009
- Garg, Environmental Engineering: Water Supply Engineering, Khanna Publishers, 18th Edn, India 2004
- Manual on Sewerage and Sewage Treatment Systems, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 2013.
- Manual on Water Supply and Treatment, Central Public Health & Environmental Engineering Organisation, Ministry of Housing and Urban Development, Govt. of India, 1999.

Online Course Material

- Guidelines for drinking-water quality, World Health Organization 2022, ISBN 978-92-4-004506-4, https://www.who.int/publications/i/item/9789240045064

Drinking Water — Specification, Bureau of Indian Standard , IS 10500 : 2012, New Delhi, India, 110002

CHECKLIST FOR THE NEW COURSE APPROVAL

Course Title: Transportation Engineering II

1. Justification for the proposal in line with the philosophy of the program.

   The proposed course Transportation Engineering II is designed to strengthen the core competencies of undergraduate students in the domain of transportation systems while aligning with the department's vision of developing sustainable, smart, and digitally enabled infrastructure. Furthermore, the course supports in nurturing future-ready engineers by equipping students with practical and analytical skills relevant to modern transportation planning and operations.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core course.

3. Name of faculty members who will teach the proposed course.
   Dr. Ranju Mohan, Dr. Bhupendra Singh, Dr. Akshay Gupta, Dr. Subham Jain

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   Yes.

5. Experts' comments on the proposed courses:

S. No.	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Dr. Dharamveer Singh	Professor, Civil Engineering Department, IIT Bombay	The proposal sounds good. You may consider some soft tool along with theory course, and align lab with theory courses to appreciate concept.	The software tools along with AI/ML Applications are added in the revised syllabus.
2	Dr. Indrajit Ghosh	Professor, Civil Engineering Department, IIT Roorkee	Inclusion of multimodal mobility, data-driven mobility, ITS, simulation, surrogate safety	We have added multimodal mobility in Module IV and ITS in module III. Data driven mobility will be included within ITS. We appreciate the suggestion regarding surrogate safety measures. However, considering the limited course duration, it can be

				addressed in elective course.
			Addition of Transit-Oriented Development (TOD), NMT planning	We have added them in Module IV.
			Addition of Activity based models	This has been incorporated in Module V.
			Module VI: Emerging Trends in Transportation [Lectures - 2]: EV infra, CAV, AI/ML applications, digital twins What about including something on Noise pollution?	Addition of additional module will be difficult within the given 39 lectures. However, as per your valuable suggestions, we have added AI/ML applications in module III.
			Adding the reference: 4. Coleman A. O'Flaherty. Transport Planning and Traffic Engineering, CRC Press	We have added this reference in the revised course structure.

6. The course is recommended by DFB/SFB after due deliberation.

   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes.

Title	Transportation Engineering II	Number	CIL3XXX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech [CIE]	Type	Program Compulsory (PC)
Prerequisite	Not applicable		

Objectives:

The course will introduce students to the principles of traffic engineering, traffic operations, and transportation planning. It aims to develop an understanding of traffic flow characteristics, traffic control systems, travel demand estimation, and urban transportation planning for sustainable mobility systems.

Learning Outcomes:

At the end of the course, the student should be able to:

- Analyze traffic flow characteristics and interpret speed–flow–density relationships
- Conduct traffic engineering studies and evaluate traffic operations at different road facilities
- Understand transportation planning principles and carry out surveys involved in transport planning
- Apply four-step travel demand forecasting models

Course Content:

Introduction to Transportation Systems [Lectures - 2]: Overview of transportation systems and their role in economic and social development, Elements of traffic engineering and transport planning and Current transportation challenges

Traffic Engineering [Lectures - 9]: Transportation system, Types of facilities, Factors affecting traffic performance, Categories of traffic flow, Traffic stream components and characteristics, Analysis of speed, Flow and density relationships, Fundamental diagram of traffic flow, Trajectory diagrams

Traffic Operations [Lectures – 11]: Theories of traffic flow; Traffic studies (volume, speed, delay, parking, accident studies); Design of control strategies for simple systems like intersections, roundabouts, freeways, etc., Introduction to traffic signs and signals, Capacity and level of services of various transportation facilities, Introduction of Intelligent Transportation System (ITS), software tools, AI/ML applications

Transportation Planning [Lectures – 8]: Characteristics and objectives of planning, Urban Transportation Planning- Comprehensive planning process, multimodal mobility, Transit-Oriented Development (TOD), NMT planning, Transportation data needs, and sources, types of trips, Study area and analysis zone, Transportation surveys/data collection methods,

Travel demand estimation [Lectures - 9]: Four-step transportation planning process, Trip generation, Trip distribution, Modal split, Route assignment, Traffic impact assessment studies, Activity based models

Text Books:

1. N. J. Garber, L.A. Hoel. Traffic and Highway Engineering, 4th Edition
2. R. P. Roess, E.S. Prassas, W.R. McShane. Traffic Engineering, 4th edition, Pearson.
3. C. S. Papacostas, P. D. Prevedouros. Transportation Engineering and Planning, Third Edition,

Pearson

Reference Books:

1. Coleman A. O'Flaherty. Transport Planning and Traffic Engineering, CRC Press
2. K. Khanna, C. E. G. Justo, A. Veeraragavan. Highway Engineering, Tenth Edition, Nem Chand & Bros.
3. L. R. Kadiyali, Traffic Engineering and Transport Planning, Khanna Publishers.
4. James H. Banks, Introduction to Transportation Engineering, McGraw Hill
5. Saxena S. C. Textbook of Highway and Traffic Engineering, CBS Publishers.

Online Resources:

1. Introduction to Transportation Engineering by Prof K.S. Reddy and Prof Bhargab Maitra. NPTEL. https://nptel.ac.in/courses/105105107
2. Urban Transportation Planning by Prof. V. Thamizh Arasan. NPTEL. https://nptel.ac.in/courses/105106058

CHECKLIST FOR THE NEW COURSE APPROVAL

Geotechnical Engineering

1. Justification for the proposal in line with the philosophy of the program.

The course "Geotechnical Engineering" is proposed to give students a solid understanding of how soils behave and how this knowledge is applied in engineering practice. The course covers key topics such as site investigation, design of shallow and deep foundations, slope stability, and retaining wall structures. By the end of the course, students will be able to interpret site investigation data, assess the bearing capacity and settlement of foundations, evaluate earth pressures on retaining structures and conduct slope stability analysis. This course fits well within the program as it builds the technical skills students need to handle common geotechnical problems encountered in civil engineering practice.

2. Total number of electives in the department/Center/School already approved by the Senate.

3. Name of faculty members who will teach the proposed course.
   a. **Dr. Sarath Chandra Reddy Nallala**
   b. **Dr. Aali Pant**
   c. **Dr. Pradeep Kumar Dammala**
   d. **Dr. Susmita Panda**
   e. **Dr. Debanjan Guha Roy**

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   No Lab component

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Prof. G.V. Ramana	Professor, Department of Civil and Environmental Engineering, IIT Delhi	a. Learning outcomes: Editorial correction – apply b. Module 1: Remove plate load test from this module as it duplicated in other module where Bearing capacity is discussed c. Move Module 4 and Module 5 above Module 2 and Module 3 to suit conceptual flow and text book template structure d. Remove Earth pressure distribution concept from the module on "Lateral	a. Learning outcomes: Replaced the word "Analyse" with "Assess" and word "problem" is removed in point 2. Replaced the word "Determine" with the word

			Earth Pressure and Retaining Structures". e. In module on shallow foundations, discuss the concept of Bearing capacity factors before water table effects. f. Remove the word "safe" and replace it with the word "Design" in the module on shallow foundations. g. In the module on deep foundation, replace the word "skin friction" with the word "shaft resistance" h. Reorganize the references in the suggested order and add the following reference: 1. Ranjan, Gopal, and A. S. R. Rao. 2023. *Basic and Applied Soil Mechanics*. 5th ed. New Delhi: New Age International Private Limited	"Estimate" in point 4. b. Module 1: Removed plate load test from the module 1 c. Modules are reorganized. Now, Modules 4 and 5 are Module 2 and 3 and Module 3 is moved to Module 4 and Module 2 is moved to Module 5 respectively. d. Removed the Earth pressure distribution concept from Module 4 which discusses Lateral Earth Pressure and Retaining structures. e. In module on shallow foundation (at present module 2) the concept of Bearing capacity factor is moved to a position before effect of water table. f. The word "safe" is replaced by the word "Design" in module on Shallow

				foundations (Module 2) g. In the Module on deep foundations, the word "skin resistance" is replaced with the word "shaft resistance. h. References are reorganized as per the suggestion and the new suggested reference is added: 1. Das, Braja M. (2015). Principles of Foundation Engineering, SI (7th Edition). Cengage Learning. 2. Ranjan, Gopal, and A. S. R. Rao. 2023. *Basic and Applied Soil Mechanics*. 5th ed. New Delhi: New Age International Private Limited. 3. Budhu, Muni. (2011). Soil Mechanics and Foundations (3rd Edition). John Wiley & Sons, Hoboken, New Jersey. 4. Murthy, V. N. S. (2002). Geotechnical Engineering: Principles and

				Practices of Soil Mechanics and Foundation Engineering. New York: Marcel Dekker Inc. 5. Deb, D., Verma AK (2016) Fundamentals and Applications of Rock Mechanics. PHI Learning Pvt. Ltd.
2	Prof. Deepankar Choudhury	Professor, Department of Civil Engineering, Indian Institute of Technology Jodhpur	It is suggested to move the module on "Lateral Earth Pressure and Retaining Structures" before the module on "Stability of slopes" for conceptual flow.	In the revised syllabus, the module 4 deals with "Lateral Earth Pressure and Retaining structures" and module 5 deals with "Stability of slopes"
3	Prof. Ramanathan Ayothiraman	Professor, Department of Civil and Environmental Engineering, IIT Tirupati	The following are the comments: a. Design/stability of Reinforced soil walls can be added in retaining walls. You may also introduce geosynthetics and functions briefly. b. Kindly avoid Skin friction and end bearing; use Shaft resistance and Base resistance	a. The suggested two concepts are taught in the elective course "Ground Improvement methods" so it is not included in the present syllabus of this course. b. The word "skin friction" is replaced by "shaft resistance" and the word "end bearing" is replaced by "base resistance"

6. The course is recommended by DFB/SFB after due deliberation. - **Yes**

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas. - **Yes**

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts. - **No**

Title	**Geotechnical Engineering**	**Number**	CIE4XX0
Department	Civil and Infrastructure Engineering	L-T-P	3-1-0 [4]
Offered For	B. Tech	Type	PC
Prerequisite	**Soil Mechanics**		

Objective:

To introduce geotechnical investigation methods and the design principles of slopes, retaining structures, and shallow and deep foundations.

Learning outcomes: Students will be able to:

1. Plan and interpret geotechnical site investigations methods and data
2. Assess slope stability problems
3. Evaluate earth pressure on retaining structures
4. Evaluate bearing capacity and settlement of shallow foundations
5. Understand design principles of deep foundations

Contents:

Module 1: Site Investigation and Data Collection: Objectives of site investigation, Methods of subsurface exploration, Boring, drilling, and logging methods, Sampling techniques (disturbed and undisturbed), Rock Mass Characterization, Rock Mass Classification – RMR, Q-System, and GSI.; Standard Penetration Test (SPT), Cone Penetration Test (CPT), Geophysical exploration methods, Preparation of soil/engineering geology investigation reports **Lectures 10**

Module 2: Shallow Foundations: Types of shallow foundations, Bearing capacity concepts, Modes of shear failure, Terzaghi bearing capacity theory, Bearing capacity factors, Effect of water table, Plate load test, Settlement of foundations, Safe bearing capacity. **Lectures 8**

Module 3: Deep Foundations: Types of piles, Methods of pile installation, Load carrying capacity of piles, shaft resistance and base resistance, Pile load tests, Group action of piles, Settlement of pile groups, Introduction to drilled piers and well foundations. **Lectures 7**

Module 4: Lateral Earth Pressure and Retaining Structures: Types of earth pressure, Earth pressure at rest, Rankine earth pressure theory, Coulomb earth pressure theory, Retaining wall types, Stability of retaining walls. **Lectures 7**

Module 5: Stability of Slopes: Types of slopes and slope failures, Factor of safety concept, Stability of infinite slopes, Stability of finite slopes, Plane failure and circular failure, Method of slices, Bishop simplified method. **Lectures 7**

Textbook:

1. Das, Braja M. (2015). Principles of Foundation Engineering, SI (7th Edition). Cengage Learning.

2. Ranjan, Gopal, and A. S. R. Rao. 2023. *Basic and Applied Soil Mechanics*. 5th ed. New Delhi: New Age International Private Limited.
3. Budhu, Muni. (2011). Soil Mechanics and Foundations (3rd Edition). John Wiley & Sons, Hoboken, New Jersey.
4. Murthy, V. N. S. (2002). Geotechnical Engineering: Principles and Practices of Soil Mechanics and Foundation Engineering. New York: Marcel Dekker Inc.
5. Deb, D., Verma AK (2016) Fundamentals and Applications of Rock Mechanics. PHI Learning Pvt. Ltd.

CHECKLIST FOR THE NEW COURSE APPROVAL

AI/ML in Civil Engineering

1. Justification for the proposal in line with the philosophy of the program.
   This course introduces students to AI/ML techniques and tools being used in Civil and Infrastructure Engineering to solve field problems (such as structural damage assessment, traffic forecasting, forecasting of hydrological extremes and uncertainty quantification). Students also learn to implement end-to-end ML pipelines, from raw sensor data to deployable infrastructure models.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Aali Pant

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   Yes. The Computer Center facilities can be utilized as students need access to the computer lab.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Mark Girolami	Professor, Department of Engineering, University of Cambridge Chief Scientist, Alan Turing Institute	I have had a brief look at the course description - it looks very copmprhensive coverage of modern ML / AI methods is provided - the exercises seemed to start at quite a low level (XOR etc) and I would have expected to see more direct engineering applications - as well as examples such as the use of LLM's to develeop construction plans and reports, solving PDEs describing structures etc.	All changes have been implemented. The lab exercises are updated as the comment.
2.	Prof. C.S. Manohar	Professor, Department of Civil Engineering, IISc Bangalore	I have gone through the course outline. As I already mentioned, coverage wise, the course outline is already good. The following observations may assist you to	All comments are implemented. The lab exercises are designed in a way to introduce students to the field application of

			finalize the document and to plan the course delivery: 1. The course content is strongly biased towards ML tools and with inadequate emphasis on "infrastructure engineering". This could be remedied by planning a lecture that provides overview of nature of problems in infrastructure engineering where AI/ML are of value. A few case studies of successful application may also be provided. 2. Based on the course document on the first course, Introduction to Machine Learning, and the present document, I get a feeling that a few details of ANN, namely, universal approximation theorem, and details of back propagation theorem may not be getting adequate emphasis. 3. The book by C Bishop on deep learning could be suggested https://link.springer.com/book/10.1007/978-3-031-45468-4?utm_medium=referral&utm_source=sn&utm_content=RM&&utm_campaign=CONR_TBOOK_CON1_GL_PBOK_05YVV_CTB25-COMP Solution manual for the book is available at https://www.bishopbook.com/Concepts_in_Deep_Learning_Solutions_v1.0.pdf 4. The field of ML is rapidly advancing, and the course may include, in the concluding part, a review of current topics, like embodiment, edge computing, and agentic AI. This could be done in about one/two hours. 5. The course content seems bit too heavy. The discussion on reinforcement learning could be omitted and the time that becomes available could be	AI/ML in Civil and Infrastructure Engineering. As pointed out, the discussion of Universal Approximation Theorem and backpropagation is added to the revised course (Module 1). The book from C. Bishop on Deep Learning is added as a reference book and recent advances in the field are added in the Module 3. Yes, covering reinforcement learning in detail might make the course lecture heavy; hence, those details are dropped in the lecture component of the revised course.

			shifted to module 4. Topics like Markov Decision Processes may be difficult to cover.	

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	AI/ML in Civil Engineering	Number	CIL4XXX
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-2 [4]
Offered for	B.Tech.[Core], M. Tech, M. Tech – Ph.D., Ph.D		
Prerequisite			

Objectives

- Comprehensive understanding of the problem settings and application of AI/ML in civil and infrastructure engineering.
- Development of programming skills to execute AI/ML algorithms in civil and infrastructure engineering.
- The ability to bridge the gap between traditional engineering physics and data-driven discovery.

Learning outcomes

By the end of this course, students will have the ability to:

- Analyze engineering problems within both deterministic and random (uncertain) environments.
- Apply modern architectures (CNNs, Transformers, GNNs) to address challenges such as structural damage assessment, traffic forecasting, and uncertainty quantification.
- Implement end-to-end ML pipelines, from raw sensor data to deployable infrastructure models.

Contents

Foundations and Information Theory (Lectures: 12) Review of AI/ML: Fundamentals of data-driven infrastructure engineering; probability and information theory basics, including Shannon entropy and KL divergence; modern optimization algorithms, including stochastic gradient descent and Adam; support vector machines and feature embedding spaces; Universal Approximation Theorem; autograd and backpropagation.

Vision, Representation, and Sequential Intelligence (Lectures: 15) Autoencoder; Stacked Autoencoder; Computer vision in infrastructure: CNN architectures, and transfer learning; image classification and semantic segmentation for automated damage assessment; sequential sensor data and time series: Recurrent Neural Networks and LSTMs; attention mechanisms; Transformer architecture; Transformers for real-time structural health monitoring and traffic flow forecasting, Introduction to Graph Neural Network.

Surrogate Modeling and Active Learning (Lectures: 12): Introduction to surrogate modeling; dimensionality reduction of complex physical fields; Gaussian processes and neural network surrogates. Physics-aware ML: Brief conceptual introduction to Physics-Informed Neural Networks (PINNs) and embedding governing differential equations into AI loss functions. Active learning: Bayesian optimization, uncertainty quantification, and optimal data sampling strategies for data-scarce engineering environments, Recent advances in the field of AI relevant to Infrastructure Engineering (Sensor Network and Edge Computing, Agentic AI in Infrastructure Engineering).

Tentative list of laboratory exercises

- Exploratory Data Analysis and Data Preprocessing Pipelines
- Linear Regression and Logistic Regression using a single Perceptron
- Regularization and performance metrics for Regression and Classification
- Training Autoencoders on sensor data for unsupervised anomaly detection.
- Automated Visual Inspection and classification (or image segmentation) using CNNs and Transfer Learning
- Time Series Forecasting using LSTM & Transformers
- Segmentation of multispectral remote sensing images
- Grid Resilience Analysis using Graph Neural Networks
- Interpreting and Auditing Deep Learning Models using xAI for Structural Safety or other critical systems, such as drinking water supply.
- Building a fast surrogate model for flood routing, soil settlement under dynamic loading, or beam deflection, and making it Physics-Informed.

Text Books

- J.-A. Goulet (2020) Probabilistic Machine Learning for Civil Engineers, The MIT Press.
- Goodfellow, I., Bengio, Y., and Courville, A. (2016) Deep Learning, MIT Press.
- S. Shalev-Shwartz, and S. Ben-David (2014) Understanding Machine Learning From Theory to Algorithms, Cambridge University Press. ISBN 9781107057135.

Reference Books

- R. S. Sutton and A. G. Barto (2018) Reinforcement Learning: An Introduction (2nd Edition), MIT Press.
- A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola (2023) Dive into Deep Learning, Cambridge University Press.
- Y. Singh et al. (2023). *Machine Learning Applications in Civil Engineering*, Elsevier.
- E. Stevens, L. Antiga, and T. Viehmann (2020) Deep Learning with PyTorch, Manning Publications.
- Brunton, S. L., & Kutz, J. N. (2019). *Data-driven science and engineering: Machine learning, dynamical systems, and control*. Cambridge University Press.

Online Resources and Journal Articles

- Raissi, M., Perdikaris, P. and Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, pp.686-707. https://doi.org/10.1016/j.jcp.2018.10.045
- Torralba, A., Isola, P., & Freeman, W., Foundations of Computer Vision, The MIT Press, https://visionbook.mit.edu/

CHECKLIST FOR THE NEW COURSE APPROVAL

Water Resources Engineering

1. Justification for the proposal in line with the philosophy of the program.
   This course introduces students to assess the water demand and availability. The course also introduces hydraulic design of water storage, diversion and conveyance structures. The understanding of water resources is essential for designing a water distribution/collection system for communities in the field.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is a core subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Saran Aadhar, Dr. Deepak Swami

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   The lab component involves design exercises for different hydraulic structures. Adequate resources for running lab are already available.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources. **I therefore recommend that the revised B.Tech. course syllabi**	Thank you for the review. No change required.

			in Water Resources be approved.	
2.	Prof. C.S.P. Ojha	Professor, Department of Civil Engineering Indian Institute of Technology Roorkee	No specific comment on this course.	Thank you for the review. No change required.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	Water Resources Engineering	Number	CIL2XXX
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-2 [4]
Offered for	B. Tech.		
Prerequisite			

Objectives

- To quantify water demand for varied sectors and estimate available surface and groundwater resources.
- To master the hydraulic design of storage, diversion, and conveyance structures.
- To apply computational tools (like EPANET) for the analysis of complex water distribution networks.
- To integrate watershed management principles for sustainable water resource development.

Learning outcomes:

A graduate of this course will have a thorough idea of:

- Predict future water demands and estimate reservoir capacity.
- Design stable alluvial and non-alluvial canals using Lacey's and Kennedy's theories.
- Analyze and optimize pipe networks using the Hardy-Cross method and software.
- Design hydraulic profiles for spillways, energy dissipators, and cross-drainage works.
- Develop a River Basin Master Plan incorporating soil conservation and rainwater harvesting.

Contents:

Module I [Water Demand]: Introduction, Role of Water in Modern Lifestyle, Estimating Urban Water Demand, Irrigation requirement, Industrial Water Demand, Projecting future demand, Concept of Water Equivalent. (Lectures: 4)

Module II [Available Water Resources]: Introduction; Overview of Available Water Resources in India; Surface Water Resources: Stream flow Measurement Techniques, Hydrograph, Flow-Duration and Flow-mass Curve, Reservoir Capacity Estimation using Sequent Peak Algorithm; Groundwater Resources: Occurrence, Aquifer Properties, Hydraulics of Wells, Yield, Artificial Recharge. (Lectures: 8)

Module III [Water Retention]: Dam: types and their role in water retention, Suitable Site; Spillways: Design Considerations, Location of Spillway, Spillway Types; Reservoir: Different Storage in Reservoir, Reservoir Losses, Reservoir Sedimentation, Trap Efficiency, Methods to Control Silting, Safe Yield, Multi-Purpose Reservoir Operation. (Lectures: 10)

Module IV [Water Conveyance]: Intake Structures: Introduction; design considerations; Pipe Network: Design of Pipe Network, Turbulent Flow in Pipes, Minor Loss, Hardy-Cross Analysis, Software Analysis (for instance EPANET) of pipe network; Canal System: Alignment, Command Area, Time and Capacity Factors, Water losses in canal, Canal Falls, Canal Regulation, Alluvial vs Non-Alluvial Canal, Regime Theories, Design and economics of Canal Lining, Cross Drainage Works. (Lectures: 8)

Module V [Integrated Watershed Management (IWM)]: Watershed Concepts: Definition, types, and characterization, River Basin Master Plan; Soil Erosion and Conservation: Mechanics and estimation of water erosion, Structural and vegetative measures (contouring, terracing, gully plugs); Rainwater Harvesting: Rooftop harvesting, recharge pits, and check dams; Land Use Management: Impact of deforestation and urbanization on runoff, Concept of Low Impact Development; Participatory Management. (Lectures: 9)

Tentative list of laboratory exercises:

- Estimation of Available Water Resources in a basin
- Estimation of Safe Reservoir Capacity
- Analysis of a Water Supply Pipe Network
- Design of Irrigation Canal [lined and unlined]
- Design of Alluvial channels carrying clear and sediment laden water
- Design of Trapezoidal Notch Fall, SARDA fall and Glacis Fall
- Design of Canal Head Regulator, Distributory Head Regulator and Cross Regulator
- Design of canal outlets
- Design the section of hydraulic jump
- Design of profile of ogee spillway

- Design of cross drainage structures with typical plan and section
- Design of Roof Top rainwater harvesting system
- Design of drip and Sprinkler Irrigation and water delivery system

Text Books

- Mays, L.W. Water Resources Engineering, John Wiley and sons, Inc., Kundli
- Garg, S.K. Irrigation Engineering and Hydraulic Structures. Khanna Publishers, Delhi.

Reference Books:

- Chow, V.T., Maidment, D.R., and Mays L.W. Applied Hydrology. McGraw-Hill Inc., New York.
- James, L.D., and Lee, R.R. Economics of water resources planning. McGraw-Hill Book Inc., New York.
- Todd, D. K. and Mays, L. W. Ground Water Hydrology, John Wiley India Edition
- Linsley, R.K., Franzini, J.B., Freyberg, D.L., and Tchobanoglous, G. Water Resources Engineering, McGraw-Hill Inc., Singapore.
- Rohitashw Kumar, Vijay P. Singh, and Sabah Parvaze. Watershed Planning and Management, CRC Press
- Subramanya, K. (2024). Engineering hydrology (6th ed.). McGraw-Hill Education (India) Pvt Ltd
- Rossman, L. et al., (2000) EPANET 2.2 User Manual. U.S. Environmental Protection Agency

CHECKLIST FOR THE NEW COURSE APPROVAL

Railway and Airport Engineering

1. Justification for the proposal in line with the philosophy of the program.
   This course is widely taught in all IITs and NITs. It introduces students to the design and operation of railway and airport transportation systems, an important part of civil engineering. As the department strives for sustainability and to inculcate in students leadership qualities which drive social welfare, making efficient public transportation systems such as railways will help them make meaningful contributions to society in this regard, thus highlighting the importance of this course.

2. Total number of electives in the department/Center/School already approved by the Senate.

3. Name of faculty members who will teach the proposed course.
   Dr. Ranju Mohan, Dr. Bhupendra Singh, Dr. Akshay Gupta, Dr. Subham Jain

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   Not applicable

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
	Prof Surender Singh	Associate Professor, Department of Civil Engineering, IIT Madras	No comments	Not applicable
	Prof Praveen Kumar	Professor, Department of Civil Engineering, IIT Roorkee	In railway course an introduction to Bullet Train and Metro Trains may also be included.	"Introduction to Bullet Train and Metro Trains" added to Module I

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.

Title	Railway and Airport Engineering	Number	CI4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0
Offered for	B.Tech (CIE) 6th and 8th Semester students	Type	Program Elective (PE)
Prerequisite	Transportation Engineering-I		

Objectives

The course will help students to understand the fundamentals of railway and airport engineering, including the design and operation of their facilities.

Learning Outcomes

At the end of the course, the student should be able to:

- Identify and explain the functions of railway track components such as rails, sleepers, ballast, subgrade and fittings.
- Explain the principles of railway traction, train resistance, hauling capacity, tractive effort and track stresses.
- Design railway alignment and geometric elements, including gradients, cant, horizontal curves, transition curves and vertical curves.
- Identify and define the components of railway track components.
- Understand the requirements of airport planning including aircraft characteristics, site selection criteria and airport layout planning.
- Design airport airside facilities such as runways, taxiways, aprons and circulation areas based on operational and geometric requirements.

Course Content

Module I: Introduction to railways [Lectures – 4]:
History of railways - Indian and international perspective, gauge: factors affecting gauge choice, wheel and axle arrangement, Introduction to Bullet Train and Metro Trains.

Module II: Components and structural design of railway tracks [Lectures – 7]:
Rails, sleepers, ballast, subgrade, track fittings, railway cross-section, Traction and Resistances, hauling capacity, tractive effort, track stresses.

Module III: Alignment and geometric design [Lectures – 9]:
Track alignment – need and requirements, selection factors; speed on track – safe, permissible, equilibrium, cant excess, cant deficiency; various types of gradients, design of horizontal curves, transition curves and vertical curves, widening at curve.

Module IV: Points, Crossings, Signals and Introduction to [Lectures – 7]:
Turnouts – Types and Design; Crossings - Types and Design; Signals – Classification and functions, Types of signals, Signal placements; Interlocking on tracks: types.

Module V: Air transport [Lectures – 5]:
Overview, Aircraft characteristics, Airport planning, site selection, layout plan.

Module VI: Design of airside and landside facilities [Lectures – 7]
Runway orientation; geometric design of runways, taxiways, and aprons; circulation areas; hangars; terminal building functional areas and facilities; visual aids; drainage.

CHECKLIST FOR THE NEW COURSE APPROVAL

Statistics and Probability in Civil Engineering

1. Justification for the proposal in line with the philosophy of the program.
   This course provides a broad understanding of statistical methods and techniques utilized in different fields of Civil Engineering. In many civil engineering problems, historical observed data plays an important role. The course is designed to develop students' critical thinking and problem-solving abilities to interpret data accurately and communicate findings effectively.

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2. Total number of electives in the department/Center/School already approved by the Senate.

   18

3. Name of faculty members who will teach the proposed course.
   Dr. Mayank Suman, Dr. Saran Aadhar

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   N.A.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Rajib Maity	Professor, Department of Civil Engineering, IIT Kharagpur	In brief, all three courses are good and nicely developed. Being said that, three minor changes can be made as follows: Fundamentals of Hydrology: One or two lectures may be included (incomplete indications are marked in the attached document in track change). Geoinformatics: All good. Expand LIDAR somewhere. Secondly, I am bit doubtful whether 12 lectures for Module 1 will be sufficient or not.	All changes have been implemented. The lectures schedule is updated as per comments.

			Statistics and Probability in Civil Engineering: All good. One minor edit shown in the attached file.	
2.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources. **I therefore recommend that the revised B.Tech. course syllabi in Water Resources be approved.**	Thank you for the review. No change required.
3.	Prof. C.S.P. Ojha	Professor, Department of Civil Engineering Indian Institute of Technology Roorkee	No specific comment on this course.	Thank you for the review. No change required.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	**Statistics and Probability in Civil Engineering**	Number	CIL7XXX
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B.Tech., M. Tech, Ph.D.		
Prerequisite			

Objectives
The course aims to provide students with a broad understanding of statistical methods and techniques utilized in different fields of Civil Engineering.

Learning outcomes
1. Understand fundamental concepts such as joint, conditional probabilities, and their applications in assessing risks and inherent uncertainty in different fields of Civil Engineering
2. Gain knowledge about different probability distributions and statistical procedures (such as descriptive statistics or visualization) for data interpretation.
3. Designing and conducting experiments to support data related decisions in different fields of Civil Engineering.
4. Develop critical thinking and problem-solving abilities to interpret data accurately and communicate findings effectively.

Contents
Statistics and Probability: Descriptive Statistics; Probability concepts; Sample Space and Random Variable; Joint Probability; Conditional and Marginal Probability; Bayes theorem, Chebyshev's Theorem [5 Lectures]

Probability Distribution and Their Applications: Discrete and Continuous Probability Distribution; Civil Engineering Applications; Sampling Distribution; Central limit Theorem [5 Lectures].

Concepts of sampling and monitoring: Simple random, systematic, ranked, stratified, composite and cluster sampling; sampling from normal distribution, Ranked set sampling, sampling in wild (quadrant, Recapture, transect) Completely randomized design, randomized block design; Control charts [3 Lectures].

Frequency Analysis; Risk and Reliability in Civil Engineering Design; Concept of uncertainty [2 Lectures]

Hypothesis Testing and Non-parametric Test: Null and alternative hypothesis; Type-I and II errors; one-tail and two-tail tests for mean and proportion (one- and two-population), inferences about population

variance, goodness of fit (Kolmogorov-smirnov test, Anderson Darling test), Man Whitney test, Wald statistic [6 Lectures]

Uni- and Multi-variate Data Analysis: Simple and Multiple Linear Regression, Pearson Correlation and other measures of Association; Principal Component Analysis; ANOVA test (one- and two-way); Monte Carlo methods [7 Lectures]

Time Series Analysis: Stationary time series; Realization and Ensemble; Trend and Periodicity Analysis; Data Transformation; Time Series Modeling (auto-regressive, moving average, auto-regressive moving average model with Exogenous Inputs) [7 Lectures]

Copula: Sklar's Theorem; Properties of Copula Function; Survival Copula; Common Copula Function; Prediction using Copula [4 Lectures]

Programming assignments may be given to support the understanding of theory.

Text Books

1. Maity, R. (2022). Statistical methods in hydrology and hydroclimatology. 2nd Edition, Springer.
2. Manly, B.F.J. (2008), Statistics for Environmental science and Management, Chapman and Hall/CRC Press.
3. Johnson, R.A. (2020), Miller and Freund's Probability and Statistics for Engineering, Pearson

Reference Books

1. Ang, A. H-S. and W. H. Tang, Probability Concepts in Engineering: Emphasis on Applications to Civil and Environmental Engineering, John Wiley & Sons, Inc., USA, Second Edition, 2006.
2. Benjamin, J.R. and Cornell, C.A., Probability, Statistics and Decision for Civil Engineers, McGraw-Hill, New York, 2014.
3. Wilks, D S. (2011), Statistical Methods in Atmospheric Sciences, Academic Press.

CHECKLIST FOR THE NEW COURSE APPROVAL

Hydraulic Structures and Machines

1. Justification for the proposal in line with the philosophy of the program.
   This course introduces students to rigorous stability analysis and hydraulic design of different hydraulic structures such as dam, weir and barrages, spillway, energy dissipator etc. Additionally, the course also covers basic design considerations of different turbines and pumps used in hydraulic systems. Using these principles/theory, a Civil Engineering graduate can analyse the stability and check the designs of hydraulic structures.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is an elective subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Saran Aadhar, Dr. Mayank Suman

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   No lab component.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources.	Thank you for the review. No change required.

			I therefore recommend that the revised B.Tech. course syllabi in Water Resources be approved.	
2.	Prof. C.S.P. Ojha	Professor, Department of Civil Engineering Indian Institute of Technology Roorkee	No specific comment on this course.	Thank you for the review. No change required.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	**Hydraulic Structures and Machines**	Number	CIL7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D.		
Prerequisite	-		

Objectives

- To understand the subsurface flow physics under diversion structures and master the theories of seepage and uplift pressure.
- To perform rigorous stability analysis for gravity and embankment dams under various loading conditions.
- To design complex energy dissipation systems and spillway profiles to ensure structural safety.

- To analyze the performance and selection criteria for hydraulic turbines and pumps in water resource projects.

Learning outcomes

- Apply Khosla's Theory to design safe foundations against piping and uplift.
- Evaluate the safety of dams against overturning, sliding, and internal erosion (piping).
- Design river training works like Guide Banks and Groynes based on river morphology.
- Select and size Impulse and Reaction turbines based on specific speed and head requirements.
- Predict and mitigate cavitation in hydraulic machines.

Contents

Weirs and Barrages: Creep Theories (Bligh's, Lane's, Khosla's), Design of Weir/Barrages on Permeable Foundations, Critical Exit Gradient and Safe Exit Gradient. (Lectures: 8)

Dams, Spillways and Energy Dissipators: Dams: types, forces, failure types and causes, stability analysis of Gravity and Homogenous Embankment dams, dam outlet structures, dam terminal structures, Estimation of Reservoir Capacity, sediment flushing from reservoir; Spillway: Types, Design Considerations, Controlled and Uncontrolled Spillway Characteristics, Design of Uncontrolled Ogee Spillway, Design of Chute/Sid-Channel/Syphon/Piano-key/Morning-glory Spillway (any one is sufficient), Energy Dissipators, Stilling Basin, Design of Indian Standard Hydraulic Jump Type Stilling Basins, Spillway Crest Gates and their design considerations (Lectures: 16)

River Training Works: River Classification, Behavior of rivers (straight reaches, bends, meanders), Control and Training of Rivers: Objective, Classification, Methods (Leeves, Guide Banks, Spurs/Groynes, and Marginal Bunds), Stability Analysis of Leeves, Design of Guide Banks (Lectures: 8)

Hydraulic Machines: Hydraulic Turbines: Impulse and Reactive Turbines, Runaway Speed, Simplitude in turbines, draft tube, cavitation, Selection and Design of a Hydro-Turbine for a Specific Site Profile; Pumps: Centrifugal and Reciprocating (Lectures: 7)

Text Books

1. Garg, S.K (2023). Irrigation Engineering and Hydraulic Structures. Khanna Publishers.

2. Novak, P., Nalluri, C. & Narayanan, R. (2007). Hydraulic Structures (4th ed.). Taylor & Francis.
3. Subramanya. K. (2007). Hydraulic Machines. McGraw-Hill Education

Reference Books

1. Peterka, A. J. (1984). Hydraulic design of stilling basins and energy dissipators (USBR Engineering Monograph No. 25). U.S. Bureau of Reclamation.
2. Julien, P. Y. (1987). River mechanics. Cambridge University Press.
3. Mays, L.W. (2010), Water Resources Engineering, John Wiley & Sons, Inc.
4. U.S. Bureau of Reclamation. (1987). Design of Small Dams
5. Central Water Commission (CWC). Manual on River Training, Control and Management.

Additional Reading Materials:

1. Different IS Codes: IS 6512, IS 6966, IS 4997 and others

CHECKLIST FOR THE NEW COURSE APPROVAL

Quantity Estimation, Costing and Construction Management

1. Justification for the proposal in line with the philosophy of the program.

   The proposed course aligns with the philosophy of the Department of Civil and Infrastructure Engineering at Indian Institute of Technology Jodhpur by strengthening interdisciplinary, industry-relevant, and research-oriented learning in emerging areas of infrastructure engineering.

2. Total number of electives in the department/Center/School already approved by the Senate.

   18

3. Name of faculty members who will teach the proposed course.

   Dr. Anusha S. Basavaraj, Dr. Sreejith Krishnan, Dr. Ravi Prakash, Dr. Yogiraj Sargam

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
	Prof. Ashwin Mahalingam	Professor, IIT Madras	My quick view is that while the suggested curriculum is good, it is a bit old fashioned and may not meet the needs of today/tomorrow. My tweaks would be: 1. Engineering economics is very important for an engineer to learn today and requires far more than 2 hours. I would suggest upping that to at least 10 hours and getting into detail. 2. On estimation, I would reduce the focus on quantity analysis - most BIM tools do this very easily now. Instead focus a bit more on rate analysis, labour and equipment costs and bidding in general - contracts are best brought in here because once you have an estimate, this is the next stage. 3. On project management, I doubt you will have much time to get deep into contracts and that might also not be needed at this stage. Focusing on CPM and PERT but more importantly on	Added

			topics like Earned Value and Crashing may provide a toolkit that makes the students more employable..	
	Prof. K.N. Jha	Professor, IIT Delhi	The syllabus is suitable in principle, but I would recommend approval subject to revision. The most important corrections are: (1) Rename the course to Quantity Estimation, Costing and Construction Management. (2) Check the number of hours. At IITD we have 42 hours for 3 credit course. Your template shows total hours = 40. Please check. (3)Add standard methods of measurement and Bill of Quantities preparation. (4) Add Work Breakdown Structure before Critical Path Method and Programme Evaluation and Review Technique. (5) Add construction safety introduction (6) Add construction quality introduction. (7) Add spreadsheet/software-based estimation and scheduling exposure. (8) Correct the reference numbering and bibliographic errors.	Added all the comments

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.

Yes

Title	**Quantity Estimation, Costing and Construction Management**	Number	CILXXXX
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0
Offered for	B.Tech	Type	PE
Prerequisite	Not applicable		

Objectives:

1. To provide a foundational understanding of estimation, costing, and rate analysis in construction projects.
2. To introduce principles of engineering economics for effective financial decision-making.
3. To equip students with essential techniques of project planning, scheduling, and control using modern tools.
4. To familiarize students with bidding, tendering, and construction resource management practices.

Learning Outcome:

Following this course, the student will have the ability to:

1. Prepare quantity estimates for basic civil engineering works and buildings.
2. Perform rate analysis considering materials, labour, equipment, and overheads.
3. Understand and participate in bidding and tendering processes and interpret basic contract documents.
4. Apply engineering economics principles to evaluate project alternatives and financial decisions.
5. Develop project schedules using CPM and PERT techniques.
6. Monitor and control projects using tools like Earned Value Management and basic time–cost optimization.
7. Identify and manage construction resources including manpower and equipment effectively.

Course Contents:

Module 1 – Estimation, Costing and Engineering Economics (21 Hours)

1. Introduction to Estimation (2 hours)

Necessity of estimation, Units and measurements, Types and methods of estimate, Load bearing vs framed structures, Overview of Standard Methods of Measurement (SMM)

2. Quantity Estimation and BOQ (8 hours)

Application of Standard Methods of Measurement (SMM) in quantity take-off, Quantities of: Brickwork, RCC, PCC Plastering, painting, finishing works, Basic residential buildings (flat & pitched roofs), Arches (brickwork & RCC basics), Joineries (doors, windows, ventilators), Steel estimation for RCC (basics), Introduction to spreadsheet/software based estimation, Preparation of Bill of Quantities (BOQ)

3. Rate Analysis and Costing (4 hours)

Rate analysis (materials, labour, equipment), Labour productivity, Equipment costing basics, Overheads, contingencies, profit

4. Bidding, Tendering and Contracts (3 hours)

Tendering process, Types of tenders, Basics of contracts and essentials, Procurement overview

5. Construction Economics (4 hours)

Time value of money, Discounted cash flow basics, Present worth and comparison of alternatives, Basic cost-benefit analysis

Module 2 – Construction and Project Management (18 Hours)

1. Construction Resources, Safety and Quality (4 hours)

Resource requirements, Manpower planning basics, Equipment selection factors, Introduction to construction safety, Introduction to construction quality

2. Project Planning and Scheduling (8 hours)

Project scope and planning stages, Work Breakdown Structure (WBS), Bar charts, CPM and PERT, Time computation, Introduction to software-based scheduling

3. Project Control and Optimization (4 hours)

Schedule monitoring and control, Resource levelling and allocation, Crashing of networks Earned Value Management (EVM), Time-cost trade-off

4. Project Management Practices (2 hours)

Risk management basics, Leadership and teamwork, Stakeholder management, Project handover basics

Text book:

1.Jha, K. N. (2015). Construction Project Management: Theory and Practice. Pearson Education India.
2.Dutta, B. N. (2021). Estimating and Costing in Civil Engineering. UBS Publishers & Distributors Pvt. Ltd.

References:

1.Oberlender, G. D. (2014). Project Management for Engineering and Construction (2nd ed.). McGraw-Hill Education.
2.Harris, F., McCaffer, R., Baldwin, A., & Edum-Fotwe, F. (2021). Modern Construction Management (8th ed.). John Wiley & Sons.
3.Sawhney, A., Riley, M., & Irizarry, J. (Eds.). (2020). Construction 4.0: An Innovation Platform for the Built Environment. Routledge.
4.Sears, S. K., Sears, G. A., & Clough, R. H. (2010). Construction Project Management: A Practical Guide to Field Construction Management (5th ed.). John Wiley & Sons.
5.Walker, A. (2015). Project Management in Construction (6th ed.). John Wiley & Sons.
6.Kohli, D. D., & Kohli, R. C. (2013). A Textbook of Estimating, Costing and Accounts (Civil). S. Chand & Company Ltd.
7.Holm, L., & Schaufelberger, J. E. (2021). Construction Cost Estimating. Routledge.
8.Brook, M. (2016). Estimating and Tendering for Construction Work (5th ed.). Routledge.
9.Kumar, A. (2003). Value Engineering: Concept, Technique and Application. SAGE Publications

Online Courses:

Principles of Construction Management – Prof. Sudhir Misra, IIT Kanpur
(https://nptel.ac.in/courses/105104161)

CHECKLIST FOR THE NEW COURSE APPROVAL

Non-Destructive Evaluation of Structures

1. Justification for the proposal in line with the philosophy of the program

   It is estimated that billions of dollars are required annually for the repair, maintenance, and rehabilitation of concrete structures. Non-destructive evaluation of concrete structures is, therefore, a vital step in identifying the distress and developing repair strategies for critical infrastructure. The objective of this course will be to understand the underlying principles of the different non-destructive tests (NDTs) for assessing concrete quality, to understand the capabilities and limitations of the various NDTs for assessing concrete quality and to familiarise oneself with the relevant provisions in the standards and codes of practice relating to NDT of concrete.

   One of the missions of the Department of Civil and Infrastructure Engineering is to establish high-quality research facilities in the area of resilient and sustainable infrastructure. Non-destructive testing is a critical part of building resilient and sustainable infrastructure. Therefore, the proposed course aligns well with the philosophy of the program.

2. Total number of electives in the department/Center/School already approved by the Senate.

3. Name of faculty members who will teach the proposed course.

   Dr. Sreejith Krishnan, Dr. Yogiraj Sargam, Dr. Anusha Basavaraj, Dr. Amit Kumar Rathi

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.

   Not Applicable

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Prof. Shashank Bishnoi	Professor, Indian Institute of Technology Delhi	The course template looks good. Please see my specific comments: 1) The first module may increase a part on the philosophy of non-destructive evaluation. 2) GPR does not fit into the advanced techniques, it is a well-established method. 3) The word permeability may be replaced by mass-transport as permeability is only one type of flow. 4) While the course talks about "Evaluation" the content mainly focuses on testing. Evaluation and interpretation techniques are not included.	1) The philosophy of non-destructive evaluation has been added to module 1. 2) The GPR section has been removed as suggested. 3) The permeability has been replaced by mass transport as suggested. 4) A section on interpretation and evaluation has been added to Module 1 as suggested.

2.	Prof. Arunasis Chakraborty	Professor, IIT Guwahati	The proposal for the course on Non-Destructive Evaluation of Structures is well planned. It covers all major tests commonly carried out for the health monitoring of concrete structures. The students will really benefit from this course. Hence, the course in its present form may be considered for offering in you PG curriculum.	We thank the reviewer for the positive comments.

6. The course is recommended by DFB/SFB after due deliberation.

   Yes, the course was recommended

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.

   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.

   Yes.

Title	**Non-destructive Evaluation of Structures**	Number	CIL7xxx
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	BTech	Type	Program Elective
Prerequisite	-		

Objectives:

1. To understand the underlying principles of the different non-destructive tests (NDTs) for assessing concrete quality.
2. To understand the capabilities and limitations of the various NDTs for assessing concrete quality.
3. To familiarise with the relevant provisions in the standards and codes of practice relating to NDT of concrete.

Learning Outcomes:

Following this course, the student will have the ability to

1. Develop a testing plan for non-destructive evaluation of structures
2. Identify and perform the relevant NDT suitable for different scenarios
3. Analyse and interpret the NDT results obtained

Course Contents:

- **Module 1 (10 Lectures)**
  Philosophy, Importance and need of non-destructive testing, defects in concrete structures, deteriorations in concrete and steel, testing and quality control of concrete, Planning non-destructive evaluation of structures, Evaluation and Interpretation of NDT Results

- **Module 2 (10 Lectures)**
  Visual Inspection – Introduction, general procedures, and applications
  Fundamental principles, test procedures, applications, interpretation and limitations of Surface hardness test, penetration resistance test, pullout test, break-off test, core tests, and maturity methods.

  Stress wave methods for flaw detection: Ultrasonic pulse velocity, Impact echo, Impulse response, shear wave tomography**.**

- **Module 3 (10 Lectures)**
  NDT for evaluating corrosion - Surface resistivity, Half-cell potential, Polarisation resistance, Electromagnetic cover measurement, Direct measurement of corrosion rates.

Test methods for measuring mass transport, carbonation, and moisture content. Chemical testing and allied techniques for concrete, Concrete Petrography

- **Module 4 (9 Lectures)**
  Advanced Non-destructive techniques for concrete evaluation – Nuclear magnetic resonance, infrared thermography, electromagnetic tests for concrete, radioisotope gauge, Computer tomography, Magnetic-flux leakage

Textbooks

1. Carino, N.J., and Malhotra, V.M. "Handbook on Non-destructive Testing of Concrete (2nd Ed)" Taylor and Francis, 2003

2. Maierhofer, C., Reinhardt, H., and Dobmann, G. "Non-destructive evaluation of reinforced concrete structures Volume 2: Non- Non-destructive testing methods" Woodhead Publishing Series in Civil and Structural Engineering, 2010

References

1) Bungey, J.H, Millard, S.G., and Grantham, M. "Testing of concrete in structures (4th Ed)", Taylor and Francis, 2006

2) Breysse D. "Non-Destructive Assessment of Concrete Structures: Reliability and Limits of Single and Combined Techniques" Springer 2012

3) International Atomic Energy Agency, "Guidebook on Non-destructive Evaluation of Concrete Structures", 2002

4) American Concrete Institute "Report on Nondestructive Test Methods for Evaluation of Concrete in Structures" Report by ACI Committee 228

CHECKLIST FOR THE NEW COURSE APPROVAL

Flow in Open Channels

1. Justification for the proposal in line with the philosophy of the program.
   This course introduces students to fluid mechanics of steady and unsteady free surface flow in open channels such as rivers, artificial canals etc. As surface water bodies are covered, the understanding of these concepts are essential for studying flow in any open water conveyance system, which a working Civil Engineer may encounter in the field.

2. Total number of electives in the department/Center/School already approved by the Senate.
   This is an elective subject.

3. Name of faculty members who will teach the proposed course.
   Dr. Saran Aadhar, Dr. Mayank Suman

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   No lab component.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1.	Prof. Subhasish Dey	Visiting Professor, Department of Civil Engineering Indian Institute of Technology Gandhinagar	Thanks for sending the revised B.Tech. course syllabi in Water Resources (CIE Department, IIT Jodhpur). I have closely reviewed the course contents of the various subjects related to Water Resources. In my opinion, the revised courses have been well designed and cover all the important aspects of Water Resources. **I therefore recommend that the revised B.Tech. course syllabi**	Thank you for the review. No change required.

			in Water Resources be approved.	
2.	Prof. C.S.P. Ojha	Professor, Department of Civil Engineering Indian Institute of Technology Roorkee	No specific comment on this course.	Thank you for the review. No change required.

6. The course is recommended by DFB/SFB after due deliberation.
   Yes

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas.
   Yes

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts.
   Yes

Title	**Flow in Open Channels**	Number	CIL4XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B.Tech., Ph.D.		
Prerequisite			

Objectives

The course discusses the fluid mechanics of steady and unsteady free surface flow in open channels such as rivers, artificial canals etc.

Learning outcomes

1. Understand channel types, flow classifications, and basic equations governing flow and velocity distributions.
2. Learn about specific energy, critical flow conditions, methods for computing critical flow, and concepts like specific force.
3. Study shear stress distribution, apply Chezy's and Manning's equations, and understand uniform flow computation.
4. Learn the gradually varied flow (GVF), analyze flow profiles, and compute GVF for different conditions.
5. Understand flow dynamics over sharp-crested weirs, spillways, sluice gates, and hydraulic jumps.
6. Explore unsteady flow principles, Saint-Venant equations, surge propagation, and wave behavior in unsteady conditions.

Contents

Basic flow concepts: Types of channels, classification of flows, basic equations, velocity distribution, velocity coefficients, pressure distribution (Lectures: 3)

Energy and momentum principles: Specific energy, critical flow, section factor for critical flow computation, first hydraulic exponent, computation of critical flow, specific force, channel transitions, and chokes (Lectures 6)

Uniform flows: Shear stress distribution, Chezy's equation, Manning's equation, conveyance of a channel, section factor for uniform flow computation, second hydraulic exponent, computation of uniform flow, Hydraulically Efficient Channel Section, Critical Slope (Lectures 6)

Gradually varied flows: Differential equation of gradually varied flow (GVF), classification and analysis of flow profiles, Control Section, computation of GVF, Advanced Numerical Methods for solving GVF (Lectures 6)

Rapidly varied flows: Flow over sharp crested weir, spillways, flow under sluice gate, types of hydraulic jump, general equation for hydraulic jump in prismatic channels (Lectures 6)

Unsteady flows: Unsteady flow in open channels, derivation of Saint-Venant equations, surge propagation, negative and positive surges, different types of waves in River and Ocean systems (Lecture 6)

Hydraulics of Mobile Bed Channel: Sediment Properties, Initiation of motion, Bed Load, Sediment Load (Lectures 6)

Text Books

1. M. Hanif Chaudhry (2008) Open-channel flow. Springer
2. V. T. Chow (1959) Open channel hydraulics. Mc-Graw Hill
3. S. Dey (2024) Fluvial hydrodynamics. Springer

Reference Books

1. H. Chanson (2004) The hydraulics of open channel flow: an introduction. Elsevier
2. S. Montes (1998) The hydraulics of open channel flow. ASCE Press
3. K. Subramanya (2019) Flow in open channels. Mc-Graw Hill

CHECKLIST FOR THE NEW COURSE APPROVAL

Soil Mechanics

1. Justification for the proposal in line with the philosophy of the program.

The course titled "Soil Mechanics" provides the fundamental characteristics of soil and their mechanical relevance to substructure design supporting superstructure loads. The course delves into an understanding of index properties, soil phase relationships, soil constitutive response under typical loading scenario and determination of shear strength parameters. Overall, the course will enhance the understanding of readers on practical loading on soils and the parameters required for stable foundation designs, which is in line with the philosophy of Civil and Infrastructure Engineering.

2. Total number of electives in the department/Center/School already approved by the Senate.

3. Name of faculty members who will teach the proposed course.
   a. **Dr. Pradeep Kumar Dammala**
   b. **Dr. Aali Pant**
   c. **Dr. Sarath Chandra Reddy Nallala**
   d. **Dr. Susmita Panda**
   e. **Dr. Debanjan Guha Roy**

4. If the course involves a lab component, please confirm whether adequate infrastructure resources are available to run the laboratory successfully.
   The course has an independent laboratory course titled "Soil Mechanics Laboratory" (revised from Geotechnical Engineering Laboratory) and the department is equipped with all the mentioned experimental facilities in the laboratory course.

5. Experts' comments on the proposed courses:

	Name of Expert	Designation of Expert	Comments received	Action Taken
1	Prof. Murali Krishna Adapa	Professor, Department of Civil and Environmental Engineering, IIT Tirupati, India	a. Objectives: Editorial correction – properties b. Learning outcomes: Editorial correction – apply c. Module 1: Add engineering geology introduction, replaced unit weights, density relationships and relative density of sands with Index properties of soils d. Module 2: Field permeability to be added	a. Objectives: Replaced *properties* with *principles* b. Learning outcomes: Replaced *apply* with *assess* c. Module 1: Added *geological soil formation and classification*, and replaced unit weights, density relationships

			e. Reference Textbooks: To be added if allowed as per institute rules	and relative density of sands with *Index properties of soils* d. Module 2: *Field permeability of soils* added. e. Reference Textbooks: Following reference textbooks added ● Terzaghi, K., Peck, R. B., & Mesri, G. (1996). *Soil mechanics in engineering practice*. John wiley & sons. ● Lambe, T. W. (1951). *Soil testing for engineers* (Vol. 72, No. 5, p. 406). LWW.
2	Prof. GL Sivakumar Babu	Professor, Department of Civil Engineering, IISC Bangalore	Revised course content is fine	No comments

6. The course is recommended by DFB/SFB after due deliberation. - **Yes**

7. The course is designed after discussion with other relevant faculty in the institute working in same/or allied areas. - **Yes**

8. The course was circulated to all Heads/Coordinators for their comments before sending it to experts. - **Yes**

Title	**Soil Mechanics**	Number	CIL2XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B. Tech	Type	PC
Prerequisite	Not applicable		

Objectives

1. To introduce the fundamental engineering principles of soil mechanics including their formation, classification, hydraulic behavior, stress distribution, consolidation and shear strength.

Learning outcomes: Students will have the ability to:

1. Identify and classify soils based on index properties.
2. Assess soil phase relationships and engineering indices.
3. Analyze seepage and permeability problems.
4. Determine stress distribution in soil due to applied loads.
5. Evaluate consolidation and settlement behaviour.
6. Determine shear strength parameters of soils.

Course Content

Module 1: Soil Formation, Phase Relationships, Index Properties and Compaction: Geological soil formation and classification, Types of soils, Soil particle size and shape, Clay minerals and clay-water interaction, Three phase system of soils, Phase relationships, Index properties of Soils, Grain size distribution, Atterberg limits, Soil classification systems (USCS), Relative density of sands, Mechanism of compaction, Standard and modified Proctor tests **Lectures 9**

Module 2: Permeability and Seepage: Darcy's law, Hydraulic gradient and seepage velocity, Coefficient of permeability, Constant head permeability test, Falling head permeability test, Permeability of stratified soils, Field permeability of soils, Seepage through soils, Flow net construction and applications, Quick sand condition and piping. **Lectures 7**

Module 3: Effective Stress and Stress Distribution: Effective stress principle, Pore water pressure, Capillary rise in soils, Vertical stresses in soil mass, Stress distribution due to surface loads, Boussinesq's theory, Westergaard's theory, Influence charts. **Lectures 7**

Module 4: Compressibility and Consolidation: Compressibility of soils, One-dimensional consolidation, Terzaghi consolidation theory, Consolidation test (oedometer), Determination of coefficient of consolidation, Primary and secondary consolidation, Consolidation settlement. **Lectures 8**

Module 5: Shear Strength of Soils: Concept of shear strength, Mohr circle of stress, Mohr–Coulomb failure criterion, Stress conditions in soil mass, Direct shear test, Triaxial compression test, Unconfined compression test, Drained and undrained shear strength. **Lectures 8**

Text Books

1. Budhu, M. (2010). *Soil mechanics and foundations*. John Wiley and Sons.
2. Murthy, V. N. S. (2002). Principles and practices of soil mechanics and foundation engineering. *New York: Marcek Decker INC*.
3. Ranjan, G., & Rao, A. S. R. (2011). *Basic and applied soil mechanics*. New Age International.

Reference Books:

1. Terzaghi, K., Peck, R. B., & Mesri, G. (1996). *Soil mechanics in engineering practice*. John wiley & sons.
2. Lambe, T. W. (1951). *Soil testing for engineers* (Vol. 72, No. 5, p. 406). LWW.

Title	**Soil Mechanics Laboratory**	**Number**	CIE4XX0
Department	Civil and Infrastructure Engineering	L-T-P	0-0-3 [1.5]
Offered For	B. Tech	Type	PC
Prerequisite	**Soil Mechanics**		

Objective:

This course aims at:

- Familiarizing students with the physical and engineering properties of soil.
- Introducing the laboratory testing equipment and their functioning principles.
- Introducing basic field soil testing techniques.

Learning outcomes: Students will be able to:

1. Classify the soils roughly based on visual identification
2. Determine the index and engineering properties of soils
3. Use simple and complex field and laboratory testing techniques to determine the shear strength of soils

Contents:

1. Visual classification of soils
2. Specific gravity and moisture content determination
3. Dry and wet sieve analysis of soils
4. Sedimentation analysis: Hydrometer analysis
5. Atterberg limits: liquid limit; plastic limit and shrinkage limit
6. Permeability of soils through constant head and falling head tests
7. Field unit weight through sand replacement and core cutter tests
8. Laboratory compaction: light weight and heavy weight
9. Consolidation of soil
10. Direct shear test for sands
11. Unconfined compression test on cohesive specimens and CBR test
12. Triaxial compression test: Unconsolidated undrained; consolidated undrained
13. ~~Vane shear test~~
14. ~~14.~~13. Standard penetration test and other miscellaneous field tests

Textbook:

1. Das B.M. (2009). Soil mechanics laboratory manual (7th Ed). OUP USA International Edition. ISBN 10-0195367596.
2. Fratta D. Aguettant J. Smith L.R. (2007). Introduction to Soil Mechanics Laboratory Testing. CRC Press-Taylor & Francis Group (1st Ed). ISBN-10: 1138430242
3. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.

Title: Advanced Structural Analysis

Contents:

Module 1. Indeterminate structure: Static & Kinematic Indeterminacy; Force method of analysis of indeterminate structures: Fixed and continuous beams, frames, etc. - analysis by consistent deformation method; application of moment area and conjugate beam methods for fixed beams; theorem of three moments for continuous beams. (Lectures: 7)

Displacement method of analysis of indeterminate structures: Slope deflection method: No chord rotation and with chord rotation (derivation), Analysis of continuous beams, beams with overhang, rigid frames with sloping legs, settlement effects etc.~~;~~ moment distribution method (Lectures: 5)

Module 2. Matrix Method of Analysis: Flexibility Matrix Method of Analysis; Stiffness Matrix Method of Analysis; Stiffness Matrix for a beam-column element; Direct Stiffness Approach; Transformation; Static Condensation; Rigid end offset (Lectures 6) Introduction to FEM: Energy methods and variational principles; shape functions; ~~generation of stiffness matrix of 1D and 2D elements by FEM approach;~~ use of virtual work method for generating stiffness matrix of structure; ~~matrix formulation of structural analysis~~. (Lectures: 4)

Module-3: Stability of structure: Concept of structural stability; classical stability analysis of beams and frames; energy method of the stability analysis of beams, frames, and plates. (Lectures: 6)

Module 4. ~~Stress and strain tensors: Generalized strain-displacement relationship; compatibility condition and constitutive relationship; plane stress and plain strain problems; some boundary value problems in elasticity; yield criterion. (Lectures: 6)~~ Plastic analysis: Introduction to plastic theory; concept of plastic hinge; plastic modulus; shape factor; redistribution of moments; collapse mechanism; plastic analysis of beams and portal frames by equilibrium and mechanism methods; Moment-curvature for Arbitrary Composite Section. (Lectures: 6)

Module 5 Introduction to structural dynamics: Basic concepts of dynamic loading and response; types of mass-spring damper system; equation of motion using direct equilibration, Hamilton's principle. (Lectures: 2) Single-degree of freedom system: Development of equation of motion; statement of problem and solution; damped and undamped systems; free vibration response. (Lectures: 5)

Laboratory

- Introduction of computer applications in structural analysis.
- Create and analyze models to determine the shear force, bending moment, deflection diagrams for statically indeterminate beams.
- Create and analyze models to determine the shear force, bending moment, deflection diagrams for portal frames.
- Create and analyze models to determine the shear force, bending moment, deflection diagrams for multi storeyed buildings.
- Determine shear force, bending moment, deflection diagrams for multi-storey frames using approximate methods.
- Modelling and analysis of 3D structures.

Title	Green Building and Sustainable Materials	Number	CI7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech, M.Tech, M.Tech.-Ph.D. [CIE]	Type	Program Elective (PE)
Prerequisite -			

Objectives:

● To introduce elements of green building construction and use of sustainable materials.

● To familiarize students with different national and international green performance ratings.

● To teach students methods of carbon footprint reduction and increase in energy efficiency of a building.

Learning Outcomes:

A graduate of this course will have a thorough idea of:

● National and international green building assessment system and requirements.

● Green building design elements and use of sustainable materials.

● Techniques to reduce the carbon footprint of a green building.

● Management of hydrological cycle, treatment of waste, and maintenance of indoor air quality.

● Green building economics, business plan, and management.

Content

● Introduction to green building: Rationale for high-performance green building; green building progress and obstacles; trends in high-performance green building; the green building movement. ***(Lectures: 2)[[Removed 1 hour and the content highlighted]***

● Ecological design: design vs ecological design; contemporary ecological design; future ecological design; thermodynamics: limits on recycling and the dissipation of materials ***(Lectures: 2) [Removed 1 hour and the content highlighted]***

● Green building assessment: US green building assessment system; international green building assessment system; performance assessment. ***(Lectures: 2) [Removed 1 hour]***

● LEED building rating system: history of LEED; structure of the LEED suite of building assessment systems; LEED credentials; the LEED process; LEED categories. ***(Lectures: 2) [Removed 1 hour]***

● The Green Globes building assessment system: the Green Globes process; Green Globes verification and certification; structure of the ANSI/GBI 01-2010 standard; the Green Globes professional and Green Globes Assessor. ***(Lectures: 2) [Removed 1 hour]***

● GRIHA/IGBC: Overview of IGBC; IGBC Certification Process; Overview of GRIHA; GRIHA Certification Process; Variants and Professionals ***(Lectures: 2)[Added 2 hours newly]***

● Green building design process: conventional vs green building delivery systems; executing the green building project; the integrated design progress; role of Charrette in the design process; green building documentation requirements. ***(Lectures: 2) [Removed 1 hour]***

● Sustainable site and landscape: land and landscape approaches for green buildings; sustainable landscapes; enhancing ecosystems; stormwater management; low-impact development; heat island mitigation; light trespass and pollution reduction; sustainable site initiative. Case study – IIT Jodhpur ***(Lectures: 5) [Increased 1 hour by adding a relevant case study]***

● Life cycle assessment for assessing the energy and carbon footprint reduction in buildings: LCA as a tool of building materials and products; environmental product declaration; materials and product certification systems; emerging construction materials and products; distinguishing between green building products and green building materials; Closing materials loops: design for deconstruction. ***(Lectures: 4) [Changed the title of module from closing materials loops to LCA and emphasis on using the tool, increasing the number of hours to 4]***

● Energy and carbon footprint reduction – Embodied energy: Wall components and alternative systems for construction; Flooring systems and alternatives; Roofing systems and alternatives ***(Lectures: 4) – Added 4 hours newly***

● Energy and carbon footprint reduction – Operational energy: building energy issues; high-performance building energy design strategy; passive design strategy; building envelope; internal load reduction; active mechanical system; water-heating system; electrical power systems; innovative energy optimization strategies; renewable energy systems; fuel cells; smart buildings and energy management systems; ozone-depleting chemicals in HVAC&R systems; reducing carbon footprint of the built environment. ***(Lectures: 4)***

● Building Information Modelling (BIM)- Fundamentals of BIM, Various tools and their applications, BIM in building design, BIM Ecosystem, BIM aided green building design. ***(Lectures 2)***

● Built environment hydrological cycle: global water resource depletion; hydrological cycle terminology; high-performance building hydrological cycle strategy; designing the high-performance building hydrological cycle; water budget rules of thumb; sustainable stormwater management; landscaping water efficiency. ***(Lectures: 2) [Removed 1 hour]***

● Indoor environmental quality: IEQ issues; design; main components of integrated IEQ design; emission from building materials; economic benefits of good indoor environmental quality. ***(Lectures: 2) [Removed 1 hour]***

● Green building economics: the business case for high-performance green building; the economics of green building; quantifying green building benefits. ***(Lectures:2) [Removed 1 hour]***

Text Books

● Charles J. Kibert (2013). Sustainable Construction. John Wiley & Sons. ISBN 978-1- 118-33013-5

● J. Newman and Ban Sang Choo (2003) Advanced Concrete Technology-Constituent Materials (1st Ed), Elsevier. ISBN 9780080489988.

● S. Kubba (2010) LEED Practices, Certification, and Accreditation Handbook (1st Ed) Elsevier. ISBN 9780080958590.

Additional reference:

● Ministry of Power, Energy Conservation Building Code 2018, Revised Version, Bureau of Energy Efficiency, 2018,

● Indian Building Congress, Practical Handbook on Energy Conservation in Buildings, 1st ed. Nabhi Publication, 2008.

● TERI-Griha's Green Design practices (www.teriin.org/bcsd/griha/griha.htm)

● Sartori, I., and Hestnes, A. G. "Energy use in the life cycle of conventional and low-energy buildings: A review article", Energy and Buildings., 20, 249-257.2007

● Green Building Basics, California Integrated Waste Management Board

(www.ciwmb.ca.gov/GREENBUILDING/Basics.htm#What)

● Catarina Thormark. "A low energy building in a life cycle—its embodied energy, energy need for operation and recycling potential", Building and Environment., 37,429-435.2001.

Online course Material

"NOC:Energy Efficiency, Acoustics and Daylighting in Building" by Prof. B. Bhattacharjee on NPTEL. https://nptel.ac.in/courses/105/102/105102175/

Title	Ground Improvement Techniques	Number	CI4XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech [CIE]	Type	Program Elective (PE)
Prerequisite			

Objectives:

This course aims at :

1. Introducing the need for ground improvement and available techniques.
2. Introducing the aspects of soil shear strength modification for effective design of foundations.
3. Introducing recent soil modification techniques.

Students will have the ability to

1. Apply the ground modification techniques for efficient design of foundations
2. Utilize the concepts of preloading and dewatering in soils of cohesive nature
3. Stabilize the soils using different mechanical and chemical approaches

Contents

Module 1 [Lectures: 9]
Classical examples of soil failure, need of ground improvement, In-situ densification methods, In-situ densification of granular and cohesive soils; vibration at ground surface and at depth

Module 2 [Lectures: 10]
Preloading and dewatering, prefabricated vertical drains (PVDs), sand drains, geo-drains, filter requirements, spacing effects, stone columns, sumps and interceptor ditches, single and multi-stage well points, vacuum well points, horizontal wells, electro-osmosis.

Module 3 [Lectures: 10]
Stabilization of soils, methods of soil stabilization- mechanical, cement, lime, bitumen and polymer stabilization, use of industrial wastes like fly ash and granulated blast furnace slag, stabilization of expansive soils; stabilization with cement, lime and chemicals, Grouting techniques, types of grouts, grouting equipment and machinery, injection methods, grout monitoring.

Module 4 [Lectures: 10]
Geosynthetics, type of geosynthetics, geotextiles, geomembranes and gabions, Reinforced Earth Technology, concept of soil reinforcement; reinforcing materials; backfill criteria; design of reinforcement for internal stability; applications of reinforced earth structures; miscellaneous methods-soil nailing, thermal methods.

Text Books

1. Hausmann, M.R. (1990). Engineering Principles of Ground Modification, McGraw-Hill International Editions. ISBN: 9780070272798.
2. Dr. Purushottama Raj (2005). Ground Improvement Techniques. Laxmi

Commented [AP1]: Moved from Module 4 to module 3, considering the relevance of the topics with soil stabilisation than with geosynthetics

Commented [AP2]: Moved from Module 3 to Module 4, considering its relevance with geosynthetics

Publications. ISBN: 9788170088370.

3. Budhu. M (2012). Soil Mechanics and Foundations (3rd Ed). John Wiley and Sons. ISBN : 978-0-470-55684-9.

Online References

Ground Improvement Techniques by Prof. G.L. Sivakumar Babu on NTPEL.
https://nptel.ac.in/courses/105/108/105108075/
Ground Improvement Techniques by Prof. Dilip Kumar Baidya on NTPEL.
https://nptel.ac.in/courses/105105210

Commented [AP3]: Added one additional Online reference

Title	**Stability of Slope**	Number	CI7XX0
Department	Civil and Infrastructure Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, Ph.D.	Type	Elective
Prerequisite			

Objectives:

1. To understand the influence of geology, rockmass properties, and hydrology on the stability of slopes.
2. To present basic types of analytical and numerical methods of slope stability analysis.
3. To introduce methods and instruments of slope monitoring and stabilization.
4. To enable planning and design of preventive measures, and redesign of failed slopes

Learning Outcomes:

1. Knowledge of data collection from field-work, aerial photography, geophysics, and geological map.
2. Knowledge of kinematic analysis of slopes using geological data and stereographic projection.
3. Knowledge of kinetic analysis of slopes and influence of critical geometrical and hydrological parameters.
4. Knowledge of available methods of slope stabilization by reinforcements and rock removal.
5. Knowledge of instrumental facilities to monitor slope movement and subsequent classification.

Contents:

Structural geology for slope analysis: Geological structures; Orientation of discontinuities; Introduction to geological maps; Identification of modes of slope instability [Lecture: 3]

Site investigation and geological data collection: Site reconnaissance; Geological mapping; Spacing, persistence, and roughness of discontinuities; Probabilistic analysis of structural geology; Drilling and logging. [Lecture: 4]

Rock strength properties: Scale effect and rock strength; Shear strength of discontinuities; Laboratory testing of discontinuities; Hoek-Brown strength criterion for fractured rock mass; Rock durability and compressive strength [Lecture: 4]

Plane failure: Conditions of plane failure; Influence of groundwater on stability; Critical tension crack: identification and influence; Influence of critical slide plane inclination; Analysis of failure on a rough plane; Reinforcement of a slope; Seismic analysis of slopes [Lecture: 5]

Wedge failure: Conditions of wedge failure; Wedge analysis including cohesion, friction, and water pressure; Wedge stability chart for friction; Compression wedge analysis [Lecture: 5]

Circular failure: Conditions of circular failure; Methods of analysis; Derivation and use of circular failure charts; Location of critical slide surface and tension crack; Bishop's and Janbu's method of failure analysis. [Lecture: 5]

Toppling failure: Types of toppling failure; Kinematic analysis of block toppling failure; Limit equilibrium analysis of toppling [Lecture: 5]

Stabilization of slopes: Stablization programs: Rock slope Inventory, Hazard rating analysis; Stabilization by rock reinforcements; Stabilization by rock removal; Measures against rock fall [Lecture: 4]

Movement monitoring: Types of slope movement: Initial response, Regressive and progressive movements, Long-term creep; Surface monitoring methods; Data interpretation [Lecture: 4]

Books

1. Duncan C Wyllie, Christopher W. Mah (2005) Rock Slope Engineering, 4th Ed, Spon Press (Taylor & Francis Group), ISBN 0–415–28001–X (pbk).
2. Robin Chowdhury, Phil Flentje, Gautam Bhattacharya (2010) Geotechnical Slope Analysis. CRC Press (Taylor & Francis Group), ISBN: 978-0-415-46974-6 (Hbk).